



**GREEN
CLIMATE
FUND**

Meeting of the Board

4 – 7 October 2021

Virtual meeting

Provisional agenda item 15

GCF/B.30/02/Add.02

14 September 2021

Consideration of funding proposals - Addendum II

Funding proposal package for FP170

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Enhancing climate resilience in Thailand through effective water management and sustainable agriculture";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Enhancing climate resilience in Thailand through effective water management and sustainable agriculture
Country(ies):	Thailand
Accredited Entity:	United Nations Development Programme (UNDP)
Date of first submission:	<u>2019/06/17</u>
Date of current submission	<u>2021/09/09</u>
Version number	V.013



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Project	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of <u>GCF budget</u> devoted to it. The total of the percentages when summed should be 100%.		
	Mitigation: Reduced emissions from: ☐ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☐ Forestry and land use: Adaptation: Increased resilience of: ☒ Most vulnerable people, communities and regions: ☒ Health and well-being, and food and water security: ☒ Infrastructure and built environment: ☒ Ecosystem and ecosystem services:		GCF contribution: <u>Enter number</u>% <u>Enter number</u>% <u>Enter number</u>% <u>Enter number</u>% 33% 10% 35% 22%
A.5. Expected mitigation impact	N/A	A.6. Expected adaptation impact	<u>Direct beneficiaries:</u> 62,000 people <u>Indirect beneficiaries:</u> 471,561 (population of project districts) 25,000,000 (population of greater Chao Phraya River Basin) Respectively: 0.1% of population of Thailand 1% of population of Thailand 39% of population of Thailand
A.7. Total financing (GCF + co-finance)	33,911,323 USD	A.9. Project size	Small (Upto USD 50 million)
A.8. Total GCF funding requested	17,533,500 USD		
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. ☒ Grant 17,533,500 ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	5 years	A.12. Total lifespan	20 years

A.13. Expected date of AE internal approval	17 June 2019	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information	The Executing Entity is the Kingdom of Thailand, represented by the Ministry of Agriculture and Cooperatives (MOAC) (The MOAC is not a nationally designated authority)		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
1. Thailand is one of the 16 countries identified globally as being in the 'extreme risk' category of those most vulnerable to future climate change impacts over the next 30 years.¹ An extensive coastline, rural communities highly dependent on agriculture, and its extensive populous urban areas, located on flood prone plains, shape Thailand's vulnerability. Severe drought and flood events² are likely to be experienced increasingly in the near and longer-term future, as a result of the combined effects of a more vigorous hydrological cycle and enhanced surface drying.³ The IPCC Fifth Assessment Report (AR5), indicates that in Southeast Asia temperatures overall have been increasing at a rate of 0.14°C to 0.20°C per decade since the 1960s, and are projected to increase from 0.8°C to 3.2°C above pre-industrial levels by the end of the century⁴, with differences between regions and microclimates within and across Southeast Asia. A positive trend in the occurrence of heavy (top 10 percent by rain amount) rain events is also predicted⁵ with key determinants such as the amount and intensity of rainfall, and the rate of evaporation set to affect surface water flow and water availability.⁶ This results in a significantly greater frequency and intensity of flooding during the wet season, and extended drought periods during the dry season, presenting a significant challenge to effective water management in Thailand. 2. The frequency of extreme drought has been rising⁷ due to increases in evapotranspiration, combined with increases in rainfall variability (less frequent rain of higher intensity)⁸ adversely impacts economic activities and agricultural livelihoods. Until a few years ago Thailand was the largest exporter of rice in the world with below-normal rainfall potentially crippling the agricultural sector and slowing economic growth.⁹ Thailand's agricultural sector in particular will be impacted by changing precipitation patterns with increasingly common extreme rain events and floods (an increase of precipitation of 8 percent is expected by 2050)¹⁰, as well as more extensive drought periods.¹¹ In 2011, 66 out of 77 provinces in the country were affected by flooding, with over 20,000 sq. km of agricultural land damaged, and nearly 900 lives lost.¹² In 2012, the World Bank reported Thailand suffered USD46.5 billion in total damages and loss and required an estimated USD14 billion in loans for rehabilitation and reconstruction from the 2011 flooding alone.¹³ The recent drought in 2015-2016 is estimated to have resulted in losses of USD3.4 billion. Future climate change projections indicate an increase in both the magnitude and frequency of drought and extreme rainfall leading			

¹ Thailand's Nationally Determined Contribution (NDC), 2015

http://www4.unfccc.int/ndcregistry/PublishedDocuments/Thailand%20First/Thailand_INDC.pdf

² Thampanishvong, K. 2014, *Farmers' Adaptation to Extreme Weather Events in the Chao Phraya River Basin*, Thailand Development Research Institute, October 15 2014

³ Limsakul, A., Chidthaisong, A. and Boonpragob, K. 2011, Thailand's First Assessment Report on Climate Change: Working Group I-Scientific Basis of Climate Change. Thailand Research Fund.

⁴ IPCC. 2013, Climate Change 2013: The Physical Science Basis, *Intergovernmental Panel on Climate Change*, 2013

⁵ IPCC. 2013, Climate Change 2013: The Physical Science Basis, *Intergovernmental Panel on Climate Change*, 2013

⁶ Chaowiwat, W. Boonya-aroonnet, S. & Weesakul, S. 2016, *Impact of Climate Change Assessment on Agriculture Water Demand in Thailand*

⁷ Annex II: feasibility study, figure 32.

⁸ Annex II, feasibility study, figure 25.

⁹ USDA. 2015, 'Thailand: Irrigation shortage reduces 2015/16 rice production', *Commodity Intelligence Report*, United States Department of Agriculture (USDA) Foreign Agricultural Service, October 29, 2015: <http://www.pecad.fas.usda.gov/highlights/2015/10/th/index.htm>

¹⁰ SNC

¹¹ NAP-Ag. Inception Report Thailand, FAO and UNDP, 2016

¹² Germanwatch (2017). Global Climate Risk Index 2017. Bonn: Germanwatch

¹³ World Bank, 2012.

<http://documents.worldbank.org/curated/en/677841468335414861/pdf/698220WP0v10P106011020120Box370022B.pdf>

to these events. Projected negative impacts are estimated to induce losses in Thailand's agriculture nationally during 2040 – 2049 of between USD24 billion and USD94 billion.¹⁴

3. Climate impacts on agriculture will particularly affect poor households. Poverty in Thailand has a predominately rural-profile, which fluctuates according to vulnerabilities in the agricultural sector, such as faltering economic growth, falling agricultural prices, and droughts.¹⁵ In 2013, for example, 80 percent of the country's poor 7.3 million live in rural areas, and a further 6.7 million were living within 20 percent above the national poverty line.¹⁶ Proportionally, the Central and Northern Regions of Thailand have the highest levels of poverty. Sukhothai, Phitsanulok, and Uttaradit Provinces, included as part of the Northern Central region, have higher poverty levels when compared with other parts of the country.

4. With climate extremes expected to increase, climate-informed water management and climate-resilient water infrastructure will be critical in order to prepare for and respond to floods and droughts. In Thailand's National Adaptation Plan 2018, Flood Control and drought management are both highlighted as key priorities, with a focus on Chao Phraya River Basin. Given the cost of upgrading existing water infrastructure across the country, the RTG is seeking to complement its grey infrastructure with ecosystems-based adaptation measures. As agriculture households are the most vulnerable to changing climatic conditions, an integrated solution which brings together water management and agriculture is proposed. The project therefore seeks **to adapt water management and agricultural livelihoods in the Yom and Nan river basins to climate change induced extreme weather events (droughts and floods)**, through interventions across three Outputs:

- Output 1: Enhance climate and risk informed planning in the water and agricultural sectors through improved climate information and cross sectoral coordination
- Output 2: Improve water management through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts
- Output 3: Reduce volatility of agriculture livelihoods in drought and flood prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets

5. The project would directly benefit 62,000 people in the project site area, with indirect benefits reaching 471,561 people in the project districts, and ultimately 25,000,000 of the Greater Chao Phraya River Basin. Further, better integration of EbA measures would have environmental benefits, while capacity building interventions of the project would support cost-efficient and effective water and agriculture planning. Project design, which includes artificial intelligence to support climate-informed planning, precision agriculture for efficient water use and applies the internet of things (IoT) concept for sharing and applying data, has been guided by Thailand 4.0, which aims to shift Thailand's agriculture sector towards an innovation-driven and interconnected sector. At the same time, the project also supports low tech interventions to help farmers respond to changing rainfall patterns. These include on-farm EbA measures (e.g. farm ponds), small scale equipment to support water saving farming practices (e.g. system for rice intensification (SRI)) and community nurseries. Training will be provided to ensure that extension services can support farmers with adaptation measures, and the project will provide support to market access for products resulting from climate resilient practices.

6. UNDP and FAO currently support (through a BMU funded project) support the Ministry of Agriculture and Cooperatives (MoAC) to enhance their agriculture sector adaptation planning, e.g. through an updated climate change strategy and action plan to align with the CCMP and as sectoral input to Thailand's National Adaptation Plan (NAP). The MoAC is implementing the the Agricultural Strategic Plan on Climate Change (ASPC) 2017-2021 whereby RID takes lead for the Strategy 2 (Adaptation Actions). Furthermore, the Office of National Water Resource (ONWR) functions as the regulating agency in proposing policies, formulating master plan on water resources management, responsible for management and supervision as well as integration on the implementation plan of water related-agencies in accordance with the Water Resource Management Act (2018), the ONWR has developed the 20-year Master Plan on Water Management (2018-2037), aimed at solving Thailand's chronic drought, flood and wastewater problems. The Master Plan also stresses the importance of the need to bring in new ideas and technologies to address water related challenges which are exacerbated by climate change.

¹⁴ Attavanich, W. 2012, *The effect of climate change on Thailand's agriculture*, https://www.researchgate.net/profile/Witsanu_Attavanich/publication/262067789_The_Effect_of_Climate_Change_on_Thailand's_Agriculture/links/566a311a08ae1a797e379c9b.pdf

¹⁵ Overview, Thailand: <http://www.worldbank.org/en/country/thailand/overview>

¹⁶ Overview, Thailand: <http://www.worldbank.org/en/country/thailand/overview>

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate rationale and context (max. 1000 words, approximately 2 pages)

Climate vulnerability baseline

7. Pressure from climate change induced variations in precipitation patterns (rainfall volume and intensities) challenge the functionality of existing water management practices through the use of existing canals and flood gates, which are increasingly inadequate to contain increased rainfall and consequent discharges during the wet season. Extreme rain events enhance flood risk during wet seasons, while increasing evapotranspiration reduces water availability – both present serious challenges for effective water management and a productive agriculture sector.

8. Thailand has 25 river basin systems (including 254 sub-basins). Of these river basins – there are two principal systems, the Mekong and Chao Phraya River systems. The Greater Chao Phraya River Basin Area consists of eight sub-river basins including Ping, Wang, Yom, Nan, Chao Phraya, Sakaekrung, Pa Sak and Tha Chin, covering an area of 158,586 km² with a population of approximately 25 million people. The average annual rainfall is 1,100 mm while the average annual runoff is 34,600 million m³. In the northern part of the basin, the entire storage capacity is approximately 27,446 million m³, while the central area can store only 1,808 million m³. The Chao Phraya River will not overflow its banks if the discharge does not exceed 2,800 m³ per second, which was exceeded during the devastating 2011 floods due to higher discharges from the Yom-Nan basin.

9. The topography of the Yom-Nan river basin in particular, and the related challenges it presents in managing changes in precipitation and related water flow, make the area an adaptation priority for Thailand. The Yom River (watershed area 23,600 km²), and Nan River (watershed area 34,300 km²) flow from the northern mountain system and join together at Nakhon Sawan. Upstream along the Nan River, the Sirikit Dam reservoir (capacity 9.5 billion m³, catchment area 13,000 km², built in 1974) was constructed for multiple purposes, including flood and drought management. There is, however, no major dam or reservoir that can regulate water throughout the year in the Yom River basin. Plans to build the Kaeng Sua Ten River Dam (1.15 billion m³) for flood and drought protection were not approved by CSOs or communities. In the lower part of Yom and Nan Rivers, the river gradients are around 1/10,000 to 1/15,000 which is relatively flat and the width of the Yom River at the downstream section is narrow with low capacity for water retention. Specific to the Yom-Nan area, RID has designed for enhancements to the existing canal and regulator network in the Yom and Nan river basins at 13 critical schemes. The designs consider the increased capacities needed to retain water (so called 'Monkey Cheek' concept) in the upper reaches of the watersheds, as well as the policy of government to not relocate the area's population. Planned under RID, the estimate cost of upgrading all 13 schemes is approximately USD 42 million over the next 20 years.

Key climate change risks and hazards

Increased temperatures and rainfall variability

10. Thailand has experienced significant country-wide warming over the last four decades, as well as associated warm extremes.¹⁷ There is evidence that maximum, mean and minimum temperatures in Thailand as a whole have increased significantly by 0.86°C, 0.95°C and 1.45°C respectively over the 1955-2009 period (see Figure 1). For temperature extreme indices, a significant upward trend is seen over the annual number of warm nights and days, the annual occurrence of warm spells, tropical nights, summer days, and the highest and lowest values of daily minimum temperatures – indicating a significant shift in the temperature distribution toward warmer conditions over the recent decades. In contrast, significant downward trends are observed in the annual number of cold nights and days, the annual occurrence of cold spells, diurnal temperature range and annual extreme temperature range. These impacts are consistent with the fifth assessment report (AR5) from the Intergovernmental Panel on Climate Change (IPCC), which indicates that temperatures in Southeast Asia have been increasing at a rate of 0.14°C to 0.20°C per decade since the 1960s and predicts increases from 0.8°C to 3.2°C for the overall region by the end of this century.

¹⁷ e.g. Limsakul, A. and Singhruck, P. (2016). Long-term trends and variability of total and extreme precipitation in Thailand. Atmospheric Research, 169, 301-317; Limjirakan, S. and Limsakul, A. (2012). Observed trends in surface air temperatures and their extremes in Thailand from 1970 to 2009. Journal of the Meteorological Society of Japan, 90(5), 647-662.; Limsakul, A., Chidthaisong, A. and Boonpragob, K. (2011). Thailand's First Assessment Report on Climate Change: Working Group I-Scientific Basis of Climate Change. Thailand Research Fund.

Limsakul, A., et al. (2016). Thailand's Second Assessment Report on Climate Change: Working Group I- Updated climate change knowledge and information of Thailand.; Limsakul A, Limjirakan S. Suthamanuswong B. (2010). Asian summer monsoon and its associated rainfall variability in Thailand. EnvironmentAsia, 3, 79-89; Limsakul, A., and S. Limjirakan, (2011). Observation of surface and atmospheric climate change in Thailand (in Thai). Thailand's First Assessment Report on Climate Change-Synthesis Knowledge on Scientific Basis (Eds: Boonpragob K, Limsakul A, Chidthaisong A). Thailand Research Fund, Thailand, 39-62.

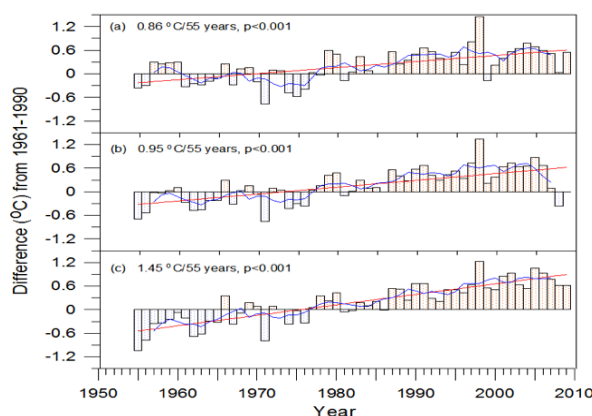


Figure 1: Maximum, mean and minimum temperature¹⁸

11. Nationwide averages of rainfall show little change in annual totals since 1950 (see Annex 2, section 1.7). However, the IPCC highlights the positive trend in the occurrence of heavy (top 10% by rain amount) rain events, and the influence of climate change on several large-scale phenomena affecting the region¹⁹. This is consistent with simple daily intensity index (SDII) observations in Thailand which show a consistent increase between 1955 and 2014 both nationally and over the northern and central areas of the Chao Phraya basin (see Figure 2) and with recent analyses of 44 long record quality controlled first order surface weather stations across Thailand for the period from 1955 to 2011, which indicate that precipitation has become less frequent throughout most areas, though precipitation events have become more intense.²⁰ These results are further consistent with reduced numbers of rainy days and increases in mean daily rainfall intensity demonstrated for stations in the Chao Phraya basin (see FS, section 1.7).

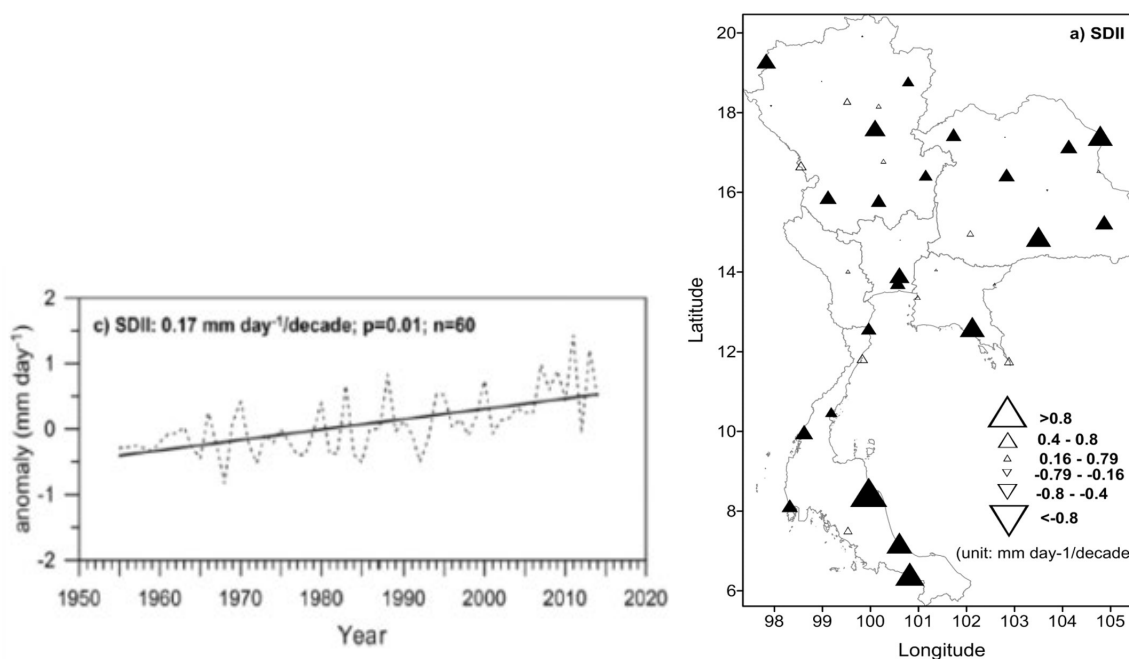


Figure 2: Average trends for Simple Daily Intensity Index (SDII) for the period from 1955 to 2014.²¹ Left – national average, right individual stations (solid triangle trend >95% significance)

Extreme events exacerbated by climate change

12. Downscaling of the ECHAM4 GCM using PRECIS, projects precipitation increases over the whole country under both A2 and B2 scenarios (see Annex 2, section 1.5.2). The length of rainy season are simulated to be similar

¹⁸ Thailand's Nationally Determined Contribution (NDC), 2015

¹⁹ Climate Change 2013: The Physical Science Basis, IPCC, 2013

²⁰ Limsakul, A. and Singhruck, P. 2016, 'Long-term trends and variability of total and extreme precipitation in Thailand', *Atmospheric Research*, 169, Pp 301-317.

²¹ Limsakul, A. and Singhruck, P. 2016, 'Long-term trends and variability of total and extreme precipitation in Thailand', *Atmospheric Research*, 169, Pp 301-317.

to the present, with increases in rainfall intensity and an increase of total precipitation by 2050.²² These results are confirmed through an analysis of the wider ensemble of models from the CMIP5 (37 GCMs), CMIP6 (24 GCMs) and CORDEX (7 RCMs) archives. These model ensembles demonstrate an expected increase in total annual rainfall of the order of 10-15% (Annex 2, section 1.7.4)

13. Increasing temperatures and potential evapotranspiration have resulted in increased incidences of severe droughts (as measured through the standardised precipitation evapotranspiration index, SPEI), which are likely to intensify in their magnitude due to climate change (see Annex 2, section 1.7.3 and 1.7.4). These increases in potential evapotranspiration have been shown to lead to increases in rice crop water requirements during all months, and these increases in requirements are greater than increases in rainfall, leading to increases in irrigation water requirements under climate change (see Annex 2, section 1.7)

14. Given the above evidence and that detailed further in Annex 2 (particularly section 1.7), the following key climate change impacts on the Chao Phraya basin have been identified: a) Increases in potential evapotranspiration, crop and irrigation water requirements; b) increases in annual rainfall which will manifest mostly through increases in rainfall intensity; c) increases in river discharges beyond current levels (including the devastating 2011 floods).

Key impacts of temperature changes, rainfall variability and extreme events on water management and agriculture
Impact on water management in the Chao Phraya River basin

15. The Greater Chao Phraya River Basin Area consists of eight sub-river basins including Ping, Wang, Yom, Nan, Chao Phraya, Sakaekrungs, Pa Sak and Tha Chin, covering an area of 158,586 km² with a population of approximately 25 million people. The average annual rainfall is 1,100 mm while the average annual runoff is 34,600 million m³. In the northern part of the basin, the entire storage capacity is approximately 27,446 million m³, while the central area can store only 1,808 million m³. The Chao Phraya River will not overflow its banks, if the discharge does not exceed 2,800 m³ per second. Several studies have projected the future river discharge in the Chao Phraya River Basin, using future climate projections indicate that peak discharges are expected to increase during the next 30 years.^{23 24 25}

16. Annex 2, section 1.7.4 shows that the projected increases in annual rainfall result from a relatively strong increase in rainfall in the peak of the rainy season (Aug-Oct). An earlier study²⁶ of hydrological impacts of climate change demonstrated that even a weak rainfall increase (relative to increases projected using the full range of models in the CMIP5/CMIP6/CORDEX ensembles) is strongly amplified in the hydrological system to create a considerable impact on streamflow in the lower parts of the basin. This and the increases in runoff in the Chao Phraya simulated using 6 GCMs²⁷ suggests that a wetter future will result in a considerable increase in flood discharges beyond current levels. This is confirmed in other data; the majority of CORDEX models (>75%) combined with a rainfall-runoff model simulate increases of approximately 50% (median estimate) in discharges of the Chao Phraya basin (see figure 38, Annex 2).

17. Overall, the following is the summary of the analysis for climate change for the project:

- Whilst station trends are mostly <95% significance, there are consistent trends between stations showing: increases in daily rainfall intensity, decreases in number of rainy days (this is consistent with future projections). This is consistent with existing peer-reviewed literature;
- Historically, there has been an increasing risk in drought as measured by SPEI (measures rainfall-evapotranspiration) but not SPI, implying that increases in evapotranspiration are mostly responsible and are greater than any increases in rainfall;
- Ensemble projections [CMIP5 (37 GCMs), CMIP6 (24 GCMs) and downscaled CORDEX (4-7 RCMs)] clearly show future projected increases of 10-15% in annual rainfall. The MRI model is at the lower end of

²² Naruchaikusol, S., 2016. Climate change and its impact in Thailand (TransRe Fact Sheet No. 2), Science, Results, Information. University of Bonn, Bonn.

²³ Ogata, T., Valeriano, O., Yoshimura, C., Liengcharennsit, W., Hirabayashi, Y. 2012, 'Past and future hydrological simulations of Chao Phraya River basin', Journal of Japan Society of Civil Engineers Ser. B1, 68, Pp 97-102

²⁴ Champathong, A., Komori, D., Kiguchi, M., Sukhaphunnaphan, T., Oki, T. 2013, 'Future projection of mean river discharge climatology for the Chao Phraya River Basin', Hydrological Research Letters, 7, Pp 36-41.

²⁵ Watanabe, S., Hirabayashi, Y., Kotsuki, S., Hanasaki, N., Tanaka, K., Mateo, C., Kiguchi, M., Ikoma, E., Kanae, S. & Oki, T. 2014, 'Application of performance metrics to climate models for projecting future river discharge in the Chao Phraya River Basin', Hydrological Research Letters, 8:1, Pp 33-38

²⁶ Wichakul, S., Tachikawa, Y., Shiiba, M., Yoroze, K., 2014. Prediction of water resources in the Chao Phraya River Basin, Thailand. Proceeding from Hydrology in Hydrology in a Changing World: Environmental and Human Dimensions, 7th Global FRIEND-Water Conference (IAHS). Publ. 363. Montpellier, France.

²⁷ Kotsuki, S., Tanaka, K., & Watanabe, S. (2014). Projected hydrological changes and their consistency under future climate in the Chao Phraya River basin using multi-model and multi-scenario of CMIP5 dataset. *Hydrological Research Letters*, 8(1), 27–32. <https://doi.org/10.3178/hrl.8.27>

these estimates, but still shows increases in rainfall intensities which lead to increases in discharges during the main rainfall season when combined with a calibrated hydrological model;

- Future temperature increases of about 2°C under RCP4.5 are virtually certain and consistent with station and gridded datasets. This leads to increases in evapotranspiration;
- Crop water requirements (due to increases in evapotranspiration) and irrigation water requirements (accounting for changes in rainfall) both increase in all months; growing both rainfed and irrigated rice will require more water and more efficient farming methods in the future; and
- Significant increases (approximately 50%) in modelled maximum discharges (using CORDEX models) are consistent with the +10% in annual precipitation since the catchments amplify the rainfall signal and the increases occur during the rainy season. This is also consistent with simulated increases in runoff for the basin found in peer-reviewed literature.

18. RID already practices water management in the critical Chao Phraya River basin to stave off excess flooding during the wet season and to ensure water availability during the dry season, using canals and regulators, including reservoir operations. In the upper section (upland area), dams and reservoirs are used to store and regulate the flow of water. In the middle section (low lying area), water is drawn into designated low-lying areas to store the water, for release when the rain intensifies to prevent flooding in the lower (urban) part. The Action Plan of Integrated and Sustainable Flood Mitigation in Chao Phraya River Basin includes a) soil improvement and conservation in the upper river basin by reforestation and rehabilitation of forest areas in the river basin in Ping, Wang, Yom, Nan, Sakae Krung, Tha-Chin and Pa Sak, as well as reservoir construction in Yom, Sakae Krung, Nan, and Pa Sak water basin. And in the lower section of the basin (urban and industrial area) flood diversion canals are used to divert excess water to the Gulf of Thailand, and dams are used to further regulate the distribution of water for industrial and agricultural use.

19. However, existing infrastructure, particularly in the proposed project area (Yom and Nan river basin, in the Uttaradit, Phitsanulok, Sukhothai provinces, middle section, low lying area), are not adequate to cope with the increasing pressures of climate change. With more intense wet seasons, this basin will be increasingly unable to slow runoff to the central plains, and the greater Bangkok area. And with drier dry seasons, the water that the infrastructure is able to retain is insufficient to cover the irrigation needs of farmers for the extended period. The middle section (the proposed project area) serves a critical function along the broader Chao Phraya River, as it has historically been able to slow or keep excess water from the downstream areas that cannot manage the water flow due to limited natural capacity. The channel capacity of the Yom River at Si Satchanalai district is 2,397cms⁻¹, which then reduces to just 565cms⁻¹ by the time it reaches downtown Sukhothai. In addition, the longitudinal river slope decreases from 1:2,300 to 1:11,000 at this point, and creates a point of overflow in the area of Sukhothai that connects Phitsanulok and Phichit provinces.

20. With rainwater as the country's most important fresh water source, increases in temperature and evaporation will lead to water stresses as well as water scarcity during certain periods in the Chao Phraya River Basin area. Annex 2, section 1.7.4 further showed a strong impact of the projected increase in air temperature and PET on agricultural water demand in the near future: both crop water demand and irrigation water requirement increase by up to 30%.

21. Increased runoff during the wet season and drier dry seasons will also accelerate soil erosion and inhibit groundwater resources, as well as increasing downstream flooding. The central plain area of Thailand (including the provinces of Phitsanulok, Sukhothai and Uttaradit) is a large plain, formed through sediment deposits as a result of fluvial processes into flood plains, terraces and swamp areas. In terms of hydrology, the central plain is also the main reservoir of groundwater for Thailand. It holds eight separate aquifers. Effective water management pursuing water retention during wet periods, has dual-value in not only slowing water flow during periods of flood, and reducing the volumes of water that continue downstream during wet periods, but also serves to secure water to be saved for use during the dry periods, assisting to maintain soil fertility and reduce sediment loss from agricultural lands, providing environmental flows and retaining ecological system functions, as well as allowing recharge and permeation to groundwater sources.

22. It is therefore crucial that water management provides: (i) flood protection, particularly given the risk to several cities located along the Yom River downstream; and (ii) adequate water storage through irrigation canals, weirs, backwater areas, diversion structures and embankments for farmers during the drier dry season.

Impact on agriculture and agriculture livelihoods

23. While flooding can damage crops and infrastructure across the country, drought affects crop production.²⁸ Thailand has been experiencing in recent years its worst periods of drought in decades, with significant impacts on overall food production and water availability. In 2008, for example, 10 million people across 55 of Thailand's provinces, experienced water shortages due to severe drought.²⁹ Drought in 2015 disproportionately impacted the agricultural sector, particularly subsistence farmers. The 2015 drought affected more than 250,000 people and cost USD400 million in government spending. In 2016, MoAC provided USD2 billion in soft loans to farmers in order to mitigate the impact of water shortages (drought) to the agricultural sector.³⁰ Estimates of total losses resulting from the 2016 drought are estimated at USD3.4 billion or 0.85 percent of GDP.³¹

24. Thailand has, in general, two annual rice-growing periods of wet season and dry season rice crops. Although called wet season rice, a portion is also irrigated. Wet season rice accounts for around 72 percent of Thailand's annual rice production³² and is heavily dependent on monsoonal rain with up to 70 percent being directly and entirely rainfed crop.³³ The other 30 percent of wet season rice is grown in the Chao Phraya River Basin and is irrigated from the upper catchment reservoirs and areas of water retention and controlled water management. Rice is the main agricultural crop of the Chao Phraya River Basin, as the basin has the greatest production advantages in terms of highly productive land, and irrigated areas.³⁴ In the Chao Phraya River basin region, climate projections show trends of increasing temperature with fluctuating precipitation³⁵ resulting in increases of crop and irrigation water requirements. Calculations for the region, found that on average (with variations between the provinces) projections for the periods during 2040-2049, farmland values per rai (0.16 hectares), are projected to decrease from USD2,703 to USD2,068. In terms of climate change adaptation activities, rice farmers currently try to reschedule their crop planning in order to adjust to the season, however they often lack reliable climate forecasts.

25. Farmer households and farmer communities are the most vulnerable groups to climate change in Thailand. An agriculture survey conducted during the development phase of the proposed GCF project (see Annex 23), showed that most respondents in the Yom and Nan River basin were dependent on rice farming as a source for income (95.8 percent of households surveyed). Household incomes in the area are split on average as THB 174,345 (USD 5,050) per year income generated from crop revenues, THB 31,652 (USD 918) from fishing revenue, and THB 2,895 (USD 84) generated from livestock revenues annually. In some of the areas where interviews took place for focus group discussions, rice farmers attempt to do three rounds of rice cropping per year. This is not consistent, however, and farmers are facing increasing volatility due to rainfall variability. For instance, cropping was drastically changed during the 2015-2016 drought due to constrained water availability.

Baseline efforts and investments

Climate information for the water and agricultural sectors

26. Given the cross-sectoral impacts of challenges related to water management, RID established the Smart Water Operation Center (SWOC) to link data with all relevant agencies for water resource management in a systematic manner during drought and flood crises. This center brings together information from the Thai Meteorological Department (TMD), the Hydro and Agro Informatics Institute, the Geo-Informatics and Space Technology Development Agency, the Department of Water Resources, and the Hydrographic Department of the Royal Thai Navy, to be compiled as central information for water analysis and forecasts. SWOC serves as a command center for water resource management for Thailand.

27. While there is a hydro-meteorological network in place, only recently has the data from the network started to get systematically analyzed, tailored for use, and disseminated for agricultural and water management purposes. SWOC and Thailand's research institutes have worked with the academic-King Mongkut Institute of Technology North Bangkok (KMUTNB) for developing a water forecast decision making tool, namely the Intelligence-based Real-time Decision-Making Modeling (IRDM) tool. This tool already serves RID's decision makers in managing water in Chao

²⁸ Kranz N, Menniken T, Hinkel J. 2010, 'Climate change adaptation strategies in the Mekong and Orange-Senqu basins: what determines the state of play?', *Environ Sci Policy*, 13:648–659

²⁹ Secretariat of the Cabinet of Thailand, 2008

³⁰ Thailand's drought weakens agricultural outlook, Oxford Business Group, May 19, 2016: <http://www.thailand-business-news.com/economics/53473-thailands-drought-weakens-agricultural-outlook.html>

³¹ 'Relief as rain falls, but drought's aftermath to linger', Bangkok Post, <http://www.bangkokpost.com/news/general/994313/relief-as-rain-falls-but-droughts-aftermath-to-linger>

³² USDA. 2015, 'Thailand: Irrigation shortage reduces 2015/16 rice production', *Commodity Intelligence Report*, United States Department of Agriculture (USDA) Foreign Agricultural Service, October 29, 2015: <http://www.pecad.fas.usda.gov/highlights/2015/10/th/index.htm>

³³ USDA. 2015, 'Thailand: Irrigation shortage reduces 2015/16 rice production', *Commodity Intelligence Report*, United States Department of Agriculture (USDA) Foreign Agricultural Service, October 29, 2015: <http://www.pecad.fas.usda.gov/highlights/2015/10/th/index.htm>

³⁴ Nara, P. Mao, G. & Yen, T. 2014, 'Climate change impacts on agricultural products in Thailand: a case study of Thai rice at the Chao Phraya River Basin', *APCBEE Procedia*, 8, Pp 136-140

³⁵ Attavanich, 2013

Phraya river basin. IRDM can calculate the amount of rainfall and inflow of water into reservoirs during the wet season and design water release schedules for 12-month periods, with a relatively high level of accuracy. Its mathematical models can be used to forecast inflows into different water sources, support water management plans for individual irrigation zones and estimate agricultural yield and revenue.

28. IRDM is a collaborative effort between 5 organisations - RID, the Agricultural Research Development Agency (ARDA), the National Research Council of Thailand (NRCT), the King Mongkut's University of Technology North Bangkok (KMUTNB), and the Thailand Research Organizations Network (TRON). IRDM is a milestone in enhancing the institutional coordination between the agricultural and irrigation departments, as well as to develop the public forecasting and modelling capacity³⁶. However, continuous support is needed to refine and upgrade the IRDM tool in order to also provide forecasts for shorter time frames (e.g. monthly or weekly) and tailor and distribute risk information to the needs of subnational decision makers and farmers/end-users. This can, ultimately, enable farmers to enhance their resilience against climate change induced droughts and floods through better water management planning.

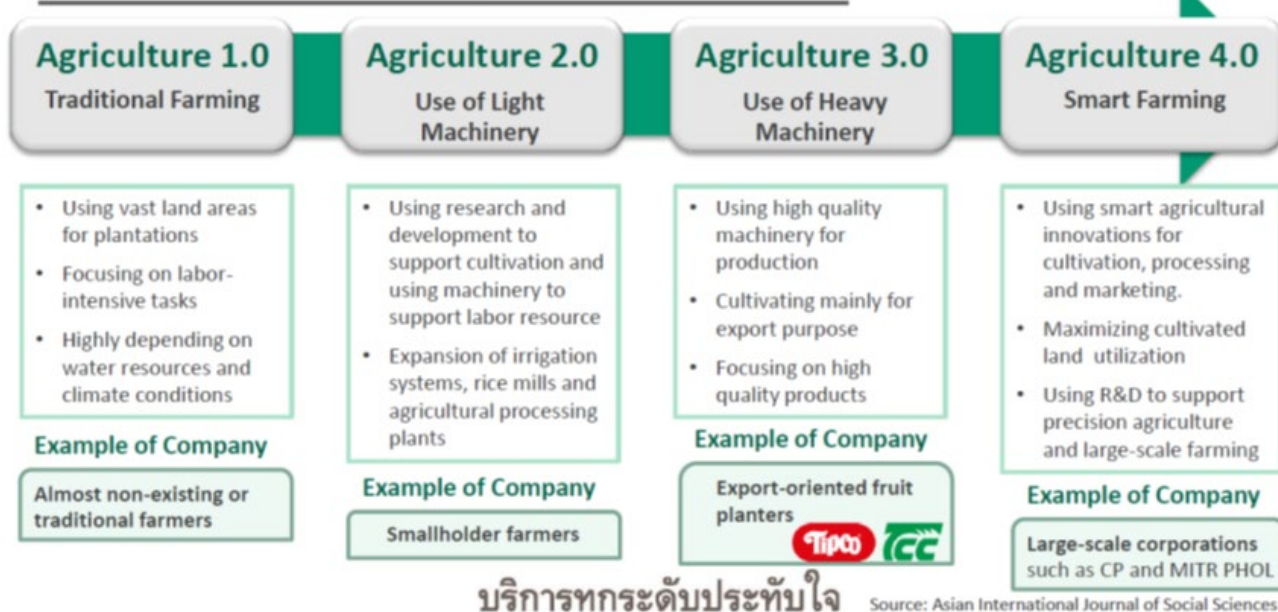
29. The concept of integrated water resources management (IWRM) based on the river basin model, is contained within Thailand's water policies and management. However, as found by a study by the World Bank (2012), Thailand's water resource management framework would benefit from a number of improvements – particularly in regard to strengthening and streamlining the IWRM approach through sectoral planning and programming. Given the dependence of the agriculture sector on water availability and sustainable water management, there is a need to bring climate change data into the respective strategies and plans, and to strengthen collaboration between the water and agriculture sectors. Work is already ongoing which can be built upon for more integrated planning. A broader (national) expert review ideally should take place, reviewing and analyzing the extreme events to provide further lessons and recommendations to the Strategic Committee on Water Management for integration into any future strategic plans. This assessment also found that reviewing and passing the water bill, would benefit from high level championing of the IWRM model in instituting the political clout required to institute IWRM throughout sectoral development planning long term. Institutions need to strengthen their roles in coordination to ensure that IWRM can take place, with a focus on management instruments and an enabling environment being established to engage inter-agency and widely with stakeholders.

30. With the growing challenges and particular vulnerabilities related to the agriculture sector, the RTG also seeks transformational change to the agriculture sector. The National Strategy - Thailand 4.0 aims to shift Thailand's agriculture sector towards an innovative-driven and interconnected sector through the integration of the concepts and approaches from sufficiency economy, smart farm, precision agriculture, intensive agriculture, sustainable agriculture, green economy, and agricultural biotechnology. Thailand 4.0 is an economic model that aims to unlock the country from several challenges (i.e. middle income trap, growing inequality and imbalanced development) resulting from past economic development models which place emphasis on agriculture (Thailand 1.0), light industry (Thailand 2.0), and advanced industry (Thailand 3.0). The four objectives of Thailand 4.0 are economic prosperity, social well-being, raising human values and environmental protection.

31. While significant progress has been made related to data collection, analysis, and sharing of information, there are opportunities to improve coordination between the water and agriculture sectors, building on IRDM to ensure policy makers are equipped with the tailored information needed for climate-informed planning and in line with the ambitions of Thailand 4.0.

³⁶ Thailand 'Technology needs assessments report for climate change adaptation'

Key Characteristics of Thailand's Agricultural Industry



Water Management in Chao Phraya River Basin

32. To cope with these additional climate-induced flood and drought risks, RID developed a 20-year roadmap towards Intelligent Organisation: Achieving water security and rendering value-added service within 2036 (see Annex 2). Investment in the Yom and Nan river basin area is part of a series of interventions ongoing and planned by RID to enhance water management in the Greater Chao Phraya River Basin Area.

33. RID already practices water management in the critical Chao Phraya River basin to stave off excess flooding during the wet season and to ensure water availability during the dry season, using canals and regulators, including reservoir operations. In the upper section (upland area), dams and reservoirs are used to store and regulate the flow of water. In the middle section (low lying area), water is drawn into designated low-lying areas to store the water for certain periods for release when the rain intensifies to prevent flooding in the lower (urban) part. And in the lower section of the basin (urban and industrial area) flood diversion canals are used to divert excess water to the Gulf of Thailand, and dams are used to further regulate the distribution of water for industrial and agricultural use.

34. Existing infrastructure, particularly in the proposed project area (Yom and Nan river basin, in the Uttaradit, Phitsanulok, Sukhothai provinces, middle section, low lying area), are no longer adequate to cope with the increasing pressures of climate change. With more intense wet seasons, this basin is increasingly unable to slow runoff to the central plains and the greater Bangkok area resulting in flooding. And with drier dry seasons, the water that the infrastructure is able to retain is insufficient to cover the irrigation needs of farmers for the extended period, resulting in maladaptive responses by farmers. The middle section (the GCF project area) serves a critical function along the broader Chao Phraya River, as it has historically been able to slow or keep excess water from the downstream areas that cannot manage the water flow due to limited natural capacity. The channel capacity of the Yom River at Si Satchanalai district is 2,397cms, which then reduces to just 565cms by the time it reaches downtown Sukhothai. In addition, the longitudinal river slope decreases from 1:2,300 to 1:11,000 at this point, and creates a point of overflow in the area of Sukhothai that connects Phitsanulok and Phichit provinces.

35. The central plain area of Thailand (including the provinces of Phitsanulok, Sukhothai and Uttaradit) is a large plain, formed through sediment deposits as a result of fluvial processes into flood plains, terraces and swamp areas. In terms of hydrology, the central plain is also the main reservoir of groundwater for Thailand, it holds eight separate aquifers. The increased runoff during the wet season is resulting in rapid soil erosion and inhibiting groundwater recharge.

36. The RTG seeks to find greener solutions to this challenge. Investment towards effective water management in the Yom-Nan river basin would have multiple benefits – i) slowing water flow to downstream areas during periods of flood risk, ii) retaining water for use during the dry season by farmers in the immediate area, and iii) assisting to maintain soil fertility and reduce sediment loss from agricultural lands, providing environmental flows and retaining ecological system functions, as well as allowing recharge and permeation to groundwater sources.

37. To inform needed enhancements to the broader Chao Phraya River system, RID commissioned a feasibility study, recommendations include:

- The capacity of drainage system should be improved to enable a drainage of water from low-lying areas at the beginning of the rainy season. Canals and waterways are needed to be dredged to the depth that can drain these low-lying areas. These waterways should be managed to connect to each other.
- The capacity to store water volume within the areas during the late rainy season and the beginning of the dry season should be increased to allow a sufficient storage of water to irrigate the area during the dry period. The dredging of waterways through the constructions of additional water regulators should be implemented for the purpose especially at the end of waterways connecting to the Yom River and Khlong Bang Kaeo Canal.
- The capacity to divert water into the area should be improved. Additional infrastructure could enable the inflow of water from the Yom River (possibly to be drawn from the Ping River where a project is underway) and from the Nan Basin through the feeder canal of the Nan River- Khlong Mem and the ditch of Khlong Wang Khon. The construction of water regulator facilities (Khlong Muang Chang and Khlong Ta Kay) in the Yom River and Khlong Bang Kaeo Canal are needed.
- An expansion of the irrigation zone in the area can be developed through improved water management planning in the Yom River Basin. This can also help in the preparation for the potential development of a large additional reservoir and would benefit the water co-management plan with the neighboring basins.
- Specific to the Yom-Nan area, RID has developed designs for enhancements to the existing canal and regulator network in the Yom and Nan river basins at 13 critical schemes. The designs consider the increased capacity needs of the area, as well as the policy of government to not relocate the area's population. Planned under RID, the estimate cost of upgrading all 13 schemes is approximately USD 42 million over the next 20 years.

Agriculture livelihoods in flood and drought prone areas

38. Rain variability and drier dry periods are affecting agriculture livelihoods in the proposed project area. Support is needed to both increase resilience of vulnerable agriculture livelihoods as well as to prevent maladaptation, which could inadvertently worsen conditions.

39. Through consultations with communities undertaken during project development, farmers were asked about measures employed to respond to drought conditions. A disproportionate number resorted to groundwater resources for irrigation. Constructing artesian wells to access groundwater was the most widely employed on-farm response for dealing with drought, despite the significant costs involved.

40. Overall, groundwater extraction in Thailand has dropped significantly since the introduction of the 2004 Groundwater Act, and a comprehensive monitoring system is in place, using 156 monitoring stations and 474 observation wells operates as an extensive network around greater Bangkok and its vicinity, and across the country 1,351 monitoring wells. There is, however, extraction of groundwater at a rate higher than can be recharged naturally – which has caused a decline in groundwater levels, land subsidence and seawater intrusion in certain areas. The Central Plain area of Thailand, which holds the Provinces of Phitsanulok, Sukhothai and Uttaradit – is a large plain, formed through sediment deposits as a result of fluvial processes into flood plains, terraces and swamp areas. In terms of hydrology, the Central Plain is also the main reservoir of groundwater for Thailand. It holds eight separate aquifers. The groundwater that farmers have increasingly come to rely on during dry periods, particularly in and around Phitsanulok, has faced continuous level decreases due to over-extraction for irrigation³⁷. Within the northeastern area, groundwater too is affected by drought.³⁸

41. Farmers have also been adapting autonomously to climate change by shifting to other crops such as fruit trees, and/or adopting and/or returning to use of aquaculture or aquaculture-diversified livelihoods. Similarly, fast growing and/or water resistant crops are used in the areas that are submerged by flood for long periods. Fishing and fish products are also utilized when rice and agricultural livelihood productions are inhibited by shorter grow periods and the early onset of rainy season or reduced rainfall and drought periods. In response to drought, farmers interviewed during project design, employed autonomous adaptation strategies that are wide-ranging. However, resorting to groundwater, and constructing artesian wells to access groundwater was widely employed as an on-farm response. Second was diversifying income streams with external employment.

42. The ability of rice farmers to adopt climate-resilient practices successfully is dependent on access to knowledge on climate change and its impacts on agriculture and natural resources, and on alternative practices, e.g. growing more

³⁷ Uppasit, S., Silaratana, N., Fuangswasdi, A., Chusanatus, S., Pavelic, P. & Kriengsak, S. 2013, 'Managed aquifer recharge using infiltration pond: Case study of Ban Nong Na, Phitsanulok, Thailand', *Journal of Earth Science and Engineering*, 1, Pp 14-22

³⁸ Fornes, J. & Pirarai, K. 2013, 'Groundwater in Thailand', *Journal of Environmental Science and Engineering B*, 3, Pp 304-315.
<http://www.davidpublisher.org/Public/uploads/Contribute/55078898cb5e0.pdf>

climate resilient crops and applying innovative agriculture and water saving approaches and technologies. Per consultations, farmers have been dependent upon their past experiences in gauging the water conditions to determine cropping seasons, while flood forecasting tends to rely upon information passed on by farmers further upstream. To reduce their risk to conditions limiting their ability to complete the cultivation season cycle – in terms of flood forecasting errors, or impacts of water insufficiency, most farmers in the area have settled on using short growth duration varieties of rice. These varieties are considered as being of lower quality than long duration rice, and the prices that farmers receive for this variety is less.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

43. The project targets to improve the water management capacity of the RID and local communities in the critical area between the Yom and Nan Rivers (Phitsanulok, Sukhothai and Uttaradit provinces) to address the core problem that climate-induced droughts and floods threatening agriculture livelihoods and infrastructure in Yom and Nan river basins, and increases flood risks downstream' (see problem statement in Theory of Change below). The targeted river basins are key to Thailand's flood risk management and irrigation system, and host some of the most vulnerable groups of society. Therefore, this project will address the following root causes and barriers to mitigate flood and drought risks, and enhance the resilience of local communities:

Lack of reliable data and risk information on localized climate change impacts available

44. The knowledge base on risk information and climate change impact projections is growing (e.g. the first version of IRDM³⁹ tool has been developed), however, little reliable and localized projections for changing temperature regimes and precipitation patterns for shorter time periods exists. This reduces the ability to predict droughts, precipitation patterns and other weather-related information to enable a climate change responsive water and irrigation management, as well as to plan adaptation strategies and climate-responsive cropping patterns. With this data, climate information can be further tailored for local level, as well as household level, planning.

Insufficient coordination between water and agriculture sectors for effective integrated water resources management

45. As detailed in its NDC, a top adaptation priority for Thailand is to promote and strengthen of IWRM practices to achieve water security, effective water resource management to mitigate flood and drought. To achieve this, greater coordination is needed across sectors, in particular water and agriculture as the most vulnerable to climate change.

Limited technical and financial capacity for the climate-informed upgrade existing infrastructure

46. As an extreme risk country⁴⁰, RTG faces daunting costs related to both response to extreme events exacerbated by climate change, as well as investments in adaptation. Existing water management infrastructure must be upgraded, to better prepare for drier dry seasons and to better manage increased rainfall during shorter periods. Current engineering designs for the planned upgrade to water management infrastructure (i.e. regulators) do not adequately take into account climate change, and could result in higher than expected repair costs or a short life of the investments (e.g. infrastructure needing replacement within a short period).

47. A review of existing engineering designs is needed, and enhancement as necessary, to ensure resilience to climate change – ensuring efficiency and effective use of public resources. Further, due to the high upfront costs of water management infrastructure, and long-term operations and maintenance (O&M) costs, financial support is needed to complement government investment.

Limited use of EbA measures to complement grey infrastructure for water management

48. Thailand generally uses grey infrastructure (e.g. regulators) for water management. Existing plans and designs for the Yom-Nan project area and the Greater Chao Phraya River Basin Area do not include EbA measures. RID seeks to complement grey infrastructure with ecosystem-based adaptation (EbA) measures however RID experience in this area is limited. Combining grey infrastructure with EbA measures would not only support critical ecosystem services, but also reduce overall costs (i.e. lower upfront investment and long-term O&M, compared to a purely grey infrastructure investment).

49. The project area is heavily altered by small settlements, farms, and cultivation; villages are scattered throughout the catchment area with many irrigation canals cutting through the area. While the area is significantly utilized, there are still different ecological habitats present. Lines of bushes and trees can be found quite often in between paddy fields or other farmland areas. As existing water management systems must be enhanced to be more climate-resilient, there is an opportunity to also introduce EbA measures to complement planned grey infrastructure investments by RID. An assessment on the possibility to integrate EbA measures was conducted during the development of this proposal

³⁹ Intelligent Based Real-time Decision-Making

⁴⁰ Thailand's Nationally Determined Contribution (NDC), 2015

(see Annex 22). The assessment concluded that there is a high potential to integrate EbA approaches in the project area, with specific recommendations include:

- Application of modeling comprising of three different methodologies each of which build on each other: hydrological river basin and water allocation modelling, hydraulic modelling, and erosion and sediment transport;
- River restoration, with duration curves and depth-discharge curves informed by the above models; and
- And the conduct of a fish survey to inform fish passes.

50. With targeted support to RID in planning and implementing EbA measures in the GCF project site area, RID can then replicate or expand on use of EbA, to complement grey infrastructure, in their plans for the Greater Chao Phraya River Basin Area.

Farmers lack of access to climate information and support to adapt to climate change

51. Closely related to the lack of localized climate information, there is currently a constrained access to information and support about climate change and adaptation measures to communities and farmers. While communities are already experiencing flood and drought related impacts, without information and support there is a risk for maladaptive measures (e.g. such as ground water extraction for water intensive crops). Diversification to climate-resilient agricultural practices (e.g. adapting drought tolerant crop varieties, techniques using short cycle crops) and water saving technologies (e.g. on farm ponds) requires specific knowledge, skills, and technology transfer that is limited, as communities lack the technical assistance and support owing to limited capacities among peers and local extension staff.

52. Importantly, with adaptation or change comes uncertainty. Farmers lack sufficient information on markets, and training on accessing new markets, to make informed decisions on changes in crops, agriculture practices and related risk mitigation measures.

53. Below is the theory of change for the project:

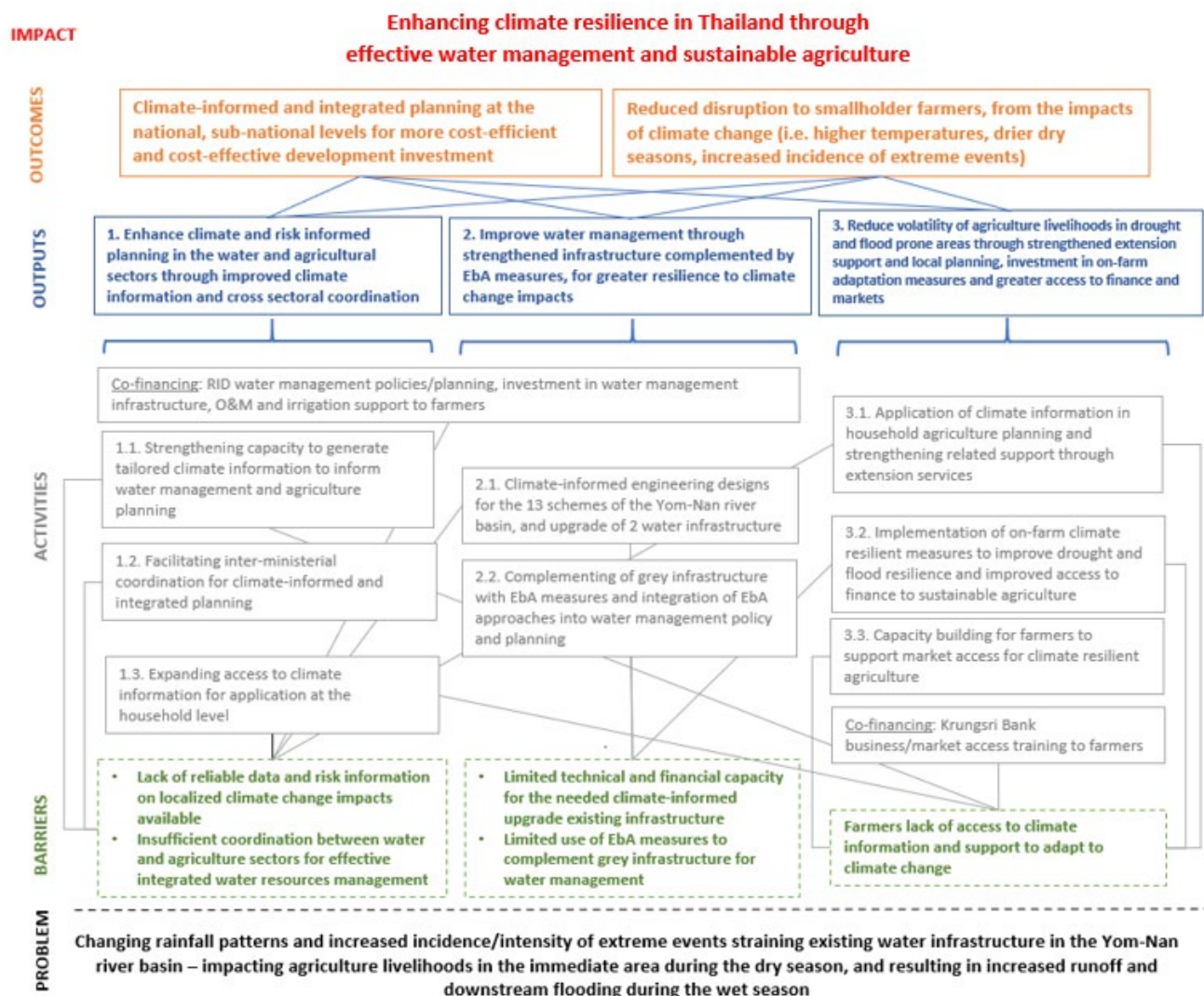


Figure 5: Theory of Change

Project objective, outcomes, and impacts

54. The target communities of the proposed project are in three provinces of the northern region of Thailand of the Greater Chao Phraya River Basin, in the confluence of the Yom and Nan Rivers: **Phitsanulok, Sukhothai, and Uttaradit**. These provinces were selected by RID under MoAC, in consultation with the Office of Natural Resources and Environmental Policy and Planning (ONEP), given the impacts of climate change to the area's critical function in water management in Thailand. The objective of the proposed project is to **adapt water management and agricultural livelihoods in the Yom and Nan river basins to climate change induced extreme weather events (droughts and floods)**. The Fund level impacts of the project are an increased resilience of: i) infrastructure and the built environment to climate change threats; iii) ecosystems and ecosystem services; iii) most vulnerable people and communities; and iv) health and well-being, and food and water security. The GCF-related outcome of the project is Strengthened adaptive capacity and reduced exposure to climate risks of green and grey water infrastructure, and agricultural livelihoods in the Yom and Nan river basins. The paradigm shift is to move towards climate risk informed water and livelihood management through tailored climate information, technical support, and the adoption of innovative technologies and practices, such as ecosystem-based adaptation.

55. First the project will address the barriers related to the low availability of reliable, localized, and easily accessible climate and risk information, as well as the improve the inter-ministerial coordination and distribution of risk

information to end-users. To do this the project will: (i) strengthen institutional capacities for generating climate risk information, which will directly support the continuous water management and irrigation planning; (ii) enable to mainstream climate change risk information into public planning processes through improved inter-ministerial coordination; and (iii) disseminate climate change information to the community and household levels. This includes upgrading the IRDM tool to provide forecasts for shorter time frames (e.g. monthly or weekly) and tailor and distribute risk information to the needs of subnational decision makers and farmers/ end-users. This can, ultimately, enable farmers to enhance their resilience against climate change induced droughts and floods through better water management planning.

56. Second, the project will address the barriers of limited technical and financial capacities to cope with intensifying climate change induced risks for water management and irrigation infrastructure in northern Chao Phraya river basin. GCF resources will be used to match public finance to: (i) strengthen water infrastructures and its management in the Yom and Nan river basins, to reduce flood risks and increase resilience of agriculture dependent communities to droughts; and (ii) enable RID to expand their capacities to implement ecosystem-based adaptation (EbA) in their planning and implementation approaches, which has the potential to be transformational to Thailand's traditional planning approaches that solely focussed on hard structure interventions.

57. Finally, GCF resources will help to overcome barriers related to a low awareness of and access to resilient livelihood practices, lack of technical and financial capacities, and limited adoption and scalability hindering farmers to diversify to climate-resilient livelihoods and implement adaptive livelihood strategies. GCF resources, combined with leveraged RGT co-financing, will ensure that: (i) generate localized climate assessments and adaptation options analysis to capacitate local decision makers and agriculture based communities to integrate climate change risk information into climate change resilient livelihood and adaptation action planning; (ii) promoting EbA on-farm solutions to enhance drought and flood resilience of local farmer communities; and (iii) support farmer household capacity to access markets for climate resilient agriculture products. A particular focus will be given on creating linkages of local producer groups and farmers with buyers, including the promotion of cooperatives and organic products. Complementing this effort, Krungsri Bank will provide training to farmers on financial literacy, so that they are more aware, and can better avail themselves, of market research and financial mechanisms.

58. Through these inter-related activities, the project will strengthen the resilience of water infrastructure and agriculture-reliant communities to cope with increasing climate change induced floods and droughts in the Yom and Nan river basins – benefiting communities and private sector through reduced damages or disruption, and relieving pressure on public resources related to flood and drought relief programmes. The project will also enhance the capacity of the RID to incorporate climate change risk information into water management planning processes and integrate EbA approaches into their portfolio of adaptive solutions, alongside traditionally used hard infrastructure interventions. Furthermore, the project will improve the quality of and access to climate change risk information available for decision makers at different levels and local communities. This will lead to the overall GCF, project-level outcome of strengthened adaptive capacity and reduced exposure to climate risks of green and grey infrastructure, and agricultural livelihoods in the Yom and Nan river basins.

59. This greener, integrated approach is new for Thailand. Best practices of the project will inform water and agriculture planning in the broader Chao Phraya river basin, as well as other river basins in the country. As poverty in Thailand has a largely rural profile, the project can have wide impacts on addressing the climate challenges faced by the most vulnerable in the country. Through partnerships and innovative approach of the project, there is significant potential for enhanced engagement with the private and finance sectors. This is already happening through with the new parallel financing from the Bank of Agriculture and Agriculture Cooperatives (BAAC) – a line of credit of USD 16 million through financial products with incentives for farmers to practice climate resilience. A letter of commitment is included in Annex 13. GCF resources will support the design of the financial product.

B.3. Project/programme description (max. 2000 words, approximately 4 pages)

Project objective, outcomes and impacts

60. The target communities and schemes of this proposed project are in three Thai provinces of the northern region of the Greater Chao Phraya River Basin, in the confluence of the Yom and Nan Rivers (see Figure 6): **Phitsanulok, Sukhothai and Uttaradit.**

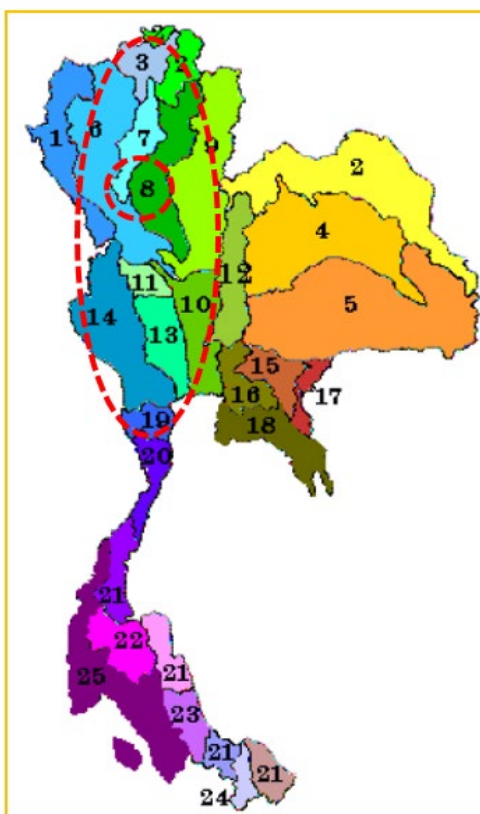


Figure 6: Provincial map of Thailand. The larger area highlighted by a red dashed line is the larger Chao Phraya river basin. The smaller circle within that area is the site of the proposed GCF project, between the Yom and Nan rivers.

61. These provinces were selected by RID under MoAC, in consultation with the Office of Natural Resources and Environmental Policy and Planning (ONEP), given the provinces' high exposure to climate change induced floods and droughts, and the area's critical function in water management in Thailand. The project site is situated in the lower area between two of the four main rivers that form the Chao Phraya River system - the Yom and the Nan rivers. Their basins provide a water retention function, slowing the flow of water before it gets to the country's central plain, and ultimately the Bangkok metropolitan area and the Gulf of Thailand.

62. With co-financing and GCF financing, the project seeks to a) support climate informed water management, planning and investment, and b) support vulnerable farmers in reducing volatility to changing climatic conditions. and planning project seeks to enhance climate-informed and integrated planning as well as reduce disruption to smallholder farmers. will achieve the objective through three complementary outputs:

- Output 1: Enhance climate and risk informed planning in the water and agricultural sectors through improved climate information and cross sectoral coordination
- Output 2: Improve water management through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts
- Output 3: Reduce volatility of agriculture livelihoods in drought and flood prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets

63. The project will complement ongoing efforts from RTG to enhance the resilience of the country and its citizens against climate change induced extreme weather events, such as floods and droughts. More specifically it will contribute to climate-risk informed decision making in the fields of water management and livelihoods planning processes through enhancing the knowledge-base on climate change, making information easily accessible for a spectrum of stakeholders and improve inter-ministerial coordination and information sharing. The project, furthermore, will reduce the flood risks within and downstream of the project site through green and grey infrastructure developments/ enhancements (including through ecosystem-based adaptation approaches), and increases the amount of water available for irrigation for local farmers during dry seasons, which are projected to increase in length due to climate change. The agriculture-based communities in the project areas will be supported in shifting towards more climate change resilient livelihoods

through the adoption of agriculture technologies and practices (e.g. using drought resilient crops), and integration of climate risk information in their decision-making processes.

Rationale and methodology for targeting of the project areas and populations

64. The project targets to enhance the water access and flood resilience of, particularly agricultural-based, households in the Yom and Nan river basins in light of intensifying climate change impacts. The targeting for the total direct beneficiaries is based on the entirety of people directly residing in the districts that form the Yom and Nan river basins, due to their high vulnerability to drought and flood impacts. The population of the targeted districts (solely based on population figures of each sub-districts within the proposed project site), which leads to a total of 62,000 people as direct beneficiaries (for output 2 and 3). The targeting for the direct beneficiaries for livelihood interventions is anchored in identifying the people most exposed to observed and projected drought and flood impacts, which are those household members currently engaged in agricultural income generating activities. The indirect beneficiaries, benefiting from improved water management and flood control, will comprise of the entire population in the proposed districts (471,561 people)⁴¹, as well as the 25,000,000 people of the greater Chao Phraya river basin area.

Output 1: Enhance climate and risk informed planning in the water and agricultural sectors through improved climate information and cross sectoral coordination

(GCF Grant: USD 1.683 million, Co-financing USD 1.532 million)

65. GCF resources will support the improvement of climate risk-responsive decision making in Thailand through the enhanced quality and availability of tailored climate information for both water management and agriculture livelihoods planning. This can be reached through improved climate projections and the dissemination of tailored to relevant stakeholders to support the integration into policies and planning across sectors (especially the water and agriculture sectors) – comparing supply to use and demand, and making adjustments in line with the philosophy of sufficiency economy⁴². This Output will also support inter-ministerial coordination, enabling stronger collaboration between key decision-making bodies in Thailand's water and agriculture sectors to incorporate climate risks and adaptation strategies in their sectoral planning processes. Importantly, tailored climate information will be generated and disseminated to agriculture households for application in household level planning.

Activity 1.1. Strengthening capacity to generate tailored climate information to inform water management and agriculture planning

66. This Activity aims to strengthen the climate risk-informed planning and implementation of water and agriculture sector interventions. The work under this Output will build upon the IRDM tool, which can forecast the amount of rainfall and inflow of water into reservoirs on the rainy season and optimize water release schedules for a 12 months ahead (a tool jointly developed by integrated data sources from several Thai organisations to enhance the utilization of weather information and projections to inform water management and irrigation practices throughout the Chao Phraya River Basin. Under this output the IRDM tool will be updated to increase the level of forecasting precision as well as generate projections for shorter time frames than annually (current standard of the tool). The climate analysis will also inform the necessary upgrades to existing infrastructure, namely the canal and regulator network in the Yom and Nan river basins. Per the recently-conducted feasibility study undertaken by the Royal Irrigation Department, in order to adequately regulate water during wet and dry season, water gates need to be upgraded and existing canals need to be enlarged to ensure greater climate resilience. Other plans to be supported by Activity 1.1 include Agri-Map (a mandatory and key zoning policy of MoAC, endorsed by the National Economic and Social Development Council (NESDC)), as well as support agriculture planning at the farmer level. This will require procurement of computing and measurement equipment and related training to technical staff in its use. Training will also be provided to RID and TMD to tailor products with information relevant for water management, with consideration of agriculture needs, including rainfall, runoff, temperature, and evapotranspiration. The Thai Meteorological Department and RID will develop the national scope. They will engage the national and sub-national researchers on climate change fields (policy, technical modelling) to the activity.

67. Sub-activities include:

⁴¹ Considering a total of 124,095 households in Bang Rakam, Prom Piram, Pichai, Kong Krailas and an average of 3.8 people per household.

⁴² The Sufficiency Economy Philosophy (SEP) originates from the development philosophy of the late King Bhumibol Adulyadej (Rama 9) and builds on three pillars: (i) Moderation; (ii) Reasonableness; and (iii) Risk Management. In addition to the SEP, the New Theory of Agriculture (NTA) was developed to help Thai farmers to build resilience against economic crisis and natural disasters. Proposed adaptation actions under the NTA include multi-cropping systems, diversified land-use, focus on households rather than commercialisation (mass production), and promotion of resilient seed varieties.

- 1.1.1. Development of Terms of Reference (ToR) for the climate information processing unit mandate
- 1.1.2. Procurement of hardware and software for enhanced generation of climate and environmental information, and related training
- 1.1.3. Technical training to 30 staff from TMD, RID and relevant agencies on integration of climate change into water management planning, including consideration of agriculture needs
- 1.1.4. Development of tailored climate information products to inform planning, and one-to-two week, monthly, seasonal, annual and long-term climate forecast models

Activity 1.2. Facilitating inter-ministerial coordination for climate-informed and integrated planning

68. In order to create an enabling environment to mainstream climate change considerations into planning processes in public entities of relevance to water management and livelihoods development, the project aims to promote the inter-ministerial coordination and flow of information. This Activity will support coordination between the Thai Meteorological Department, the Hydro and Agro Informatics Institute, the Geo-Informatics and Space Technology Development Agency, the Department of Water Resources, and the Hydrographic Department of the Royal Thai Navy for improved information sharing. The project will provide technical support in developing standard operating procedures (SOPs) for the related committee for information sharing across government stakeholders, established under the TMD mechanism/forum. This will entail reviewing existing practices for dissemination of climate information, revising in consultation with partners and related sensitization.

69. Support from the project will ensure consideration of existing committees to ensure complementarity, including integration with ongoing national adaptation plan (NAP) processes, which include capacity building and coordination efforts. As this project is an adaptation priority for the RTG, the project activities feed into national and international commitment on climate action. The Government has established the National Committee on Climate Change Policy (NCCC), chaired by the Prime Minister. The NCCC is responsible for (i) national climate change policy and strategy; (ii) determination of national positions in international negotiations under the UNFCCC and any relevant international agreements; and (iii) monitoring and evaluating implementation results of government agencies, as stated in the national policy and strategy. There are four sub-committees under the NCCC, namely, the Sub-Committee on Climate Change Policy and Planning Integration, the Sub-Committee on Climate Change Knowledge and Database, the Sub-Committee on Climate Change Negotiation and International Cooperation, and the Sub-Committee on Action for Climate Empowerment and Public Relations. Each Ministry has issued an order for the appointment of a Climate Change Focal Officer to ensure inter-ministerial coordination. According to the reporting line, the Director General of RID reports to the Permanent Secretary of the Ministry of Agriculture and Cooperatives (MoAC). Senior Management at RID is part of the Sub-Committees, and through this role can help ensure knowledge sharing, lessons learned from the project. The Sub-Committee members will be invited to the project seminar/workshop bi-annually under the Activity 1.2. In addition, Thailand's National Adaptation Plan (NAP) is under the Cabinet approval process and will then submit to the UNFCCC. UNDP/GCF is also supporting ONEP to update and implementation of the NAP. ONEP is expected to get an approval by the end of this year.

70. Sub-activities under this activity thus include:

- 1.2.1. Development of SOPs for sharing of climate information across relevant departments
- 1.2.2. Workshops and meetings for sensitization and collaboration in sharing of information and identifying opportunity for collaboration and complementarity

Activity 1.3. Expanding access to climate information for application at the household level

71. Activity 1.3. focuses on the last mile of dissemination of climate information, expanding access of information to the public, and in particular tailored climate information for agriculture households. Various communication means will be developed and employed to ensure climate information reaches a wide audience. The website will be registered under the RID; RID will own and manage the website. KMUTNB would proceed with copyright registration of the app under their institution, potentially together with RID. This point to be further consulted and finalized during the project inception. Intellectual property rights related to the app and website will be captured under the related clauses of cooperation agreements with partners.

72. Importantly, consideration will be given to what information is needed and how it is communicated, so that it is understood and applicable. To that end, an communications strategy will be developed, followed by the generation of communications products. RID will disseminate the tailored climate change information. Target groups include departments with MoAC (RID and agriculture, the Ministry of Natural Resources and Environment, Office of Water Management Policy, Thai Meteorological Department, the Hydro and Agro Informatics Institute, the Geo-Informatics

and Space Technology Development Agency, the Department of Water Resources, the Hydrographic Department of the Royal Thai Navy, financial institutions such as Krungsri Bank and the Bank of Agriculture and Agricultural Cooperative (BAAC), as well as agriculture households.

73. Sub-activities include:

- 1.3.1. Development of communication strategy for dissemination of information, including timing and media
- 1.3.2. Development of information briefs and reports released for circulation in local and social media (dissemination informed by communications strategy)
- 1.3.3. Development of smartphone and Line app with climate information that can be applied at the agriculture household level
- 1.3.4. Development of complementary website for interactive information in line with information apps or households

Output 2: Improve water management through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts

(GCF Grant: USD 9.640 million, Co-financing USD 13.485 million)

74. The second Output focuses on complementing RID investment to improve water management in Yom-Nan river basin area. GCF resources will support the necessary assessments to ensure that existing designs for the canal regulators/flood gates at 13 schemes (see Figure 7 below) are adequate given future climate projections. Given the high upfront costs of the basin-wide investment required to adapt to climate change, this will ensure long term sustainability of government investment. GCF will also complement government co-financing and planned long term investment by RID in the entire Yom-Nan river basin by supporting the upgrade of infrastructure at two of the planned schemes (13 schemes in total).

75. Water management in Thailand is largely dependent on grey infrastructure, although there are best practices available where use of reservoirs has been established (i.e. Monkey Cheek). GCF resources will support RID in identifying and implementing ecosystems-based approaches to complement the planned grey infrastructure. While ecosystems-based measures alone cannot provide address all climate risk challenges, complementing grey infrastructure with ecosystems-based solutions can be cost-efficient solution while also generating environmental co-benefits. This integration of EbA approaches has the potential to be transformational to Thailand's traditional planning concepts for water management that focus widely on hard structure interventions, through upscaling across the 13 schemes in the Yom-Nan river basin area and the greater Chao Phraya River Basin.

Activity 2.1. Climate-informed engineering designs for the 13 schemes of the Yom-Nan river basin, and upgrade of 2 water infrastructure

76. The climate information and enhanced coordination of Output 1 will inform the review of existing designs for water infrastructure, ensuring they are adequate given climate change and importantly that they are both responsive to the needs of people in the immediate Yom and Nan river basin area, as well as adequate to perform the function of water management as part of the greater Chao Phraya River Basin under projected climate change. Based on the review, enhancements will be made to the existing designs as necessary, including those to be supported through co-financing under this project.

77. While the review of designs will be for the all 13 schemes, GCF resources will only support the actual upgrades at one scheme at Khlong Muang Chang – which is considered as a critical scheme by RID as part of the canal and regulatory network in the Yom and Nan river basins. GCF resources will cover the upgrade construction cost of the two floodgates, size 6.00 * 5.00 meters, two-drainage type - increasing drainage capacity from 50 cubic meters/second to 100 cubic meters/second for flood mitigation and water storage for agriculture and consumption in Kong Kailash district in Sukhothai, and the Bang Rakam district in Phitsanulok. The cost is estimated at USD 5 million and includes contractual services and equipment for the canal head regulator and related appurtenant works. Ownership of related immovable objects will ultimately be with RID.

78. These upgrades alone however will not be sufficient, therefore the RTG through RID co-financing will support dredging canal and strengthening embankment at the Klong Muang Chang canal (USD 3 million), and operation and maintenance in Yom-Nan river basin (USD 4.24 million) as well as reconstruction of the Khlong Ta Kay culvert located in the Phrom Pira district, Phitsanulok province (USD 4 million). Further, RTG through RID has allocated resources for operations and maintenance of canals and regulators, including: maintenance of irrigation canals, headwork and

road; weed eradication; canal dredging; ecosystem improvement; and related administrative costs. The dredging of canals to a depth that can drain these low-lying areas, will improve connection of waterways to each other.

Schemes for Proposed Infrastructure Investments

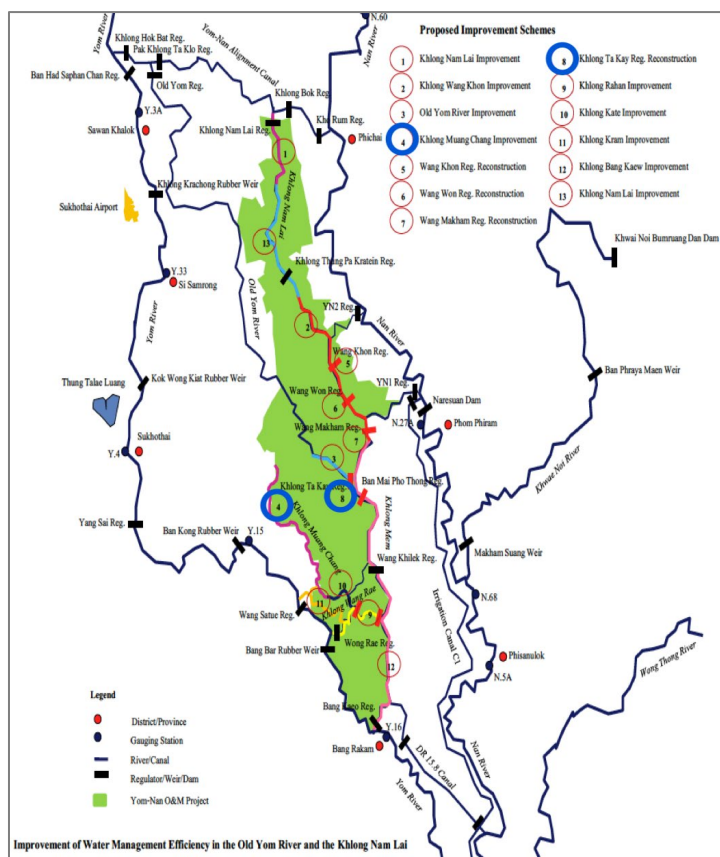


Figure 7: Planned upgrade by RID of water infrastructure in the Yom-Nan River basin

79. Together, GCF and co-financing will improve the drainage capacity for these low-lying areas – protecting the areas during the wet season, and increasing water storage capacity in the canals for the dry season. With the review of engineering designs, informed by the climate information under Output 1, there will be assurance that water management investments by government or other resources will be climate resilient and therefore cost-effective.

80. Sub-activities include:

- 2.1.1. Technical support to review and enhance existing engineering designs for the 13 schemes of the Yom-Nan river basin area, to ensure adequate integration of climate change
- 2.1.2. Construction activities to upgrade of 2 canal head regulators along Khlong Muang Chang and Khlong Ta Kay Reg in the Yom-Nan river basin area

Activity 2.2. Complementing of grey infrastructure with EbA measures and Integration of EbA approaches into water management policy and planning

81. Activity 2.2 focuses on the promotion of EbA approaches in Thailand in order to widen their range of adaptation options in the field of water management to cope with climate change induced pressures. The project seeks to retrofit ecosystem-based adaptation processes into the area, through the re-introduction and/or strengthening of integrated farming practices (such as rice-fish management), as well as the re-introduction and strengthening of vegetation and waterflow systems, within the reinforcement of the existing canal channels and large irrigation system structures already in place. Given that much of the project is focused on the retention of water in the upper tracts of the project area – some areas will have water areas increased (e.g. monkey cheeks⁴³) whereby wetlands management techniques can be employed, and the re-introduction of some of the original fauna and flora can take place. The 'greening' of reservoirs

⁴³ Monkey Cheek is a flood-control concept initiated by His Majesty the King of Thailand, who observed that most monkeys, when they obtain bananas, usually store them in their mouths, and then gradually chew and swallow them. The concept of water detention is to temporarily store excessive water during flooding period and afterwards gradually drain it, once the floods recede.

and canals is likely to take place through use in part of riparian vegetation sections – to act in combination with sediment filtration systems for more effective soil management and anti-topsoil runoff loss solutions. Ecosystem-based Adaptation (EbA) measures, in terms of improvements to water infrastructure through restoration principles include vegetating river banks, using erosion management techniques combined with vegetation buffers to conserve soil and maintain water quality, the slowing down of water flow, and the recharge of groundwater sources during periods high precipitation for water access during dry periods; soil conservation measures, among others. Appropriate measures will be introduced to complement the existing water management to slow water flows during flood periods and enable water to permeate to the ground, and to mitigate potential environmental impacts from the construction of two proposed hard infrastructures. These may include: wetland / floodplain / oxbow lake restoration; living weir; riparian zone rehabilitation / bank stabilization by vetiver grass, buffer zones, sedimentation traps and/or an appropriate combination of measures (see Figure below). Support to national partners will also include capacity building for EbA, integration of agriculture and biodiversity into EbA measures, floodplain development and conservation, river restoration, fish migration structures, and longitudinal connectivity of the Yom River.

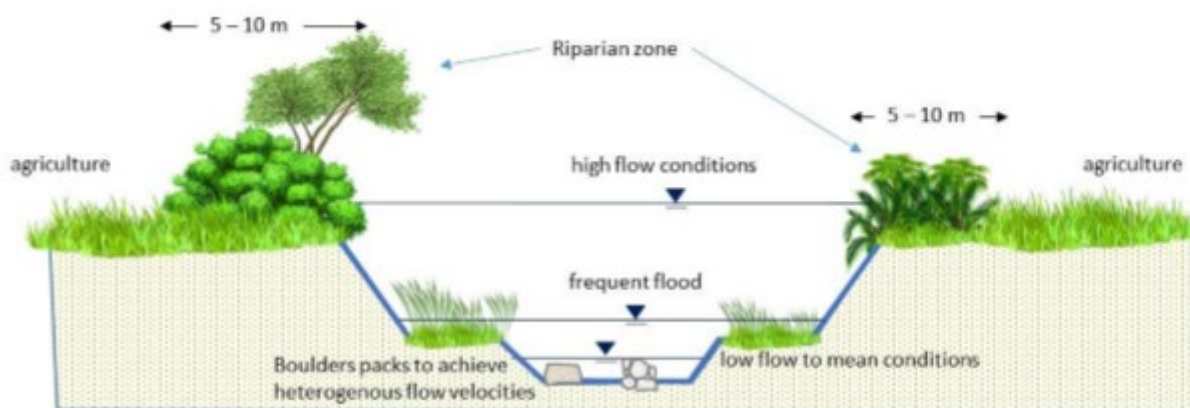


Figure 8: Example of a typical cross-section with berms and vegetation

82. Indicative EbA measures used for proposal development are based on the variable factors of the potential EbA features and the physical profile of each location, and the related costs per site for budgeting are based on previous EbA interventions in Thailand. Assessments and analyses preceding selection will consider interventions which are effective while benefiting the most, in terms of both area and beneficiaries. Since the EbA measures will be undertaken in and around the cultivated areas, the spatial coverage is indeed estimated based on these areas. The estimation will be vetted and refined during the participatory process with the communities whereby the final determination of the EbA measures will be made. A comprehensive Annex 18(b) - Lessons Learned from Baseline Projects has been included in the FP package detailing the process by which EbA measures will be identified/selected. In addition, the proposal package includes the safeguards measures to be put in place to protect farmers in the area. Please refer to Annex 6(b) – Environmental and Social Management Framework.

Project location	Target area	Local Administration Units	EbA feature	Cost estimation	Cultivated area
Yom-Nan Operation and Maintenance Project under the Royal Irrigation Department, Ministry of Agriculture and Agricultural Cooperation (MoAC)	Klong Mueang Chang	Sukhothai province, Khlongkairas District, Kainok Sub-district, Dongdueai Sub-district	- wetland / floodplain / oxbow lake restoration - living weir	Cost of the 3 EbA measures is based on different EbA features and geography of final locations	27,936 rai
	Klong Takae	Phisanulok province, Sukhothai province, Phromphiram District, Khlongkairas District, Wangwoon Sub-district, Promphiram Sub-district, Kokrat Sub-district	- riparian zone rehabilitation / bank stabilization by vetiver grass - buffer zones - sedimentation trap => and/or an appropriate combination of those		58,964 rai

83. Following best practices and lessons learned, final selection of the EbA measures will follow a consultative process, informed by vulnerability and economic assessments. GIZ is a Responsible Party for implementing this activity. GIZ's framework for mainstreaming EbA employs an iterative six-step approach (known as the adaptation mainstreaming cycle, see below). It also provides tools and methods that for each step. This guidance is in line with the Voluntary guidelines for the design and effective implementation of ecosystem-based approaches to climate change adaptation and disaster risk reduction upon request by the CBD Conference of the Parties (COP), in its decision XIII/4, to which GIZ contributed⁴⁴. Further, this approach is consistent with national regulations and policies. On 2 February 2021, the NWRC has announced a new regulation that 'In any area of water resource management initiatives, information must be provided sufficiently to the private sector, relevant people and communities to create awareness that will be useful in participating in the management of water resources in the area'. The consultative selection process will also formalize responsibility and accountability among district authorities, the Department of Water Resources and RID for maintenance of the investments. Should identified measures exceed the budget available in the GCF project, additional resources will be requested from government partners.

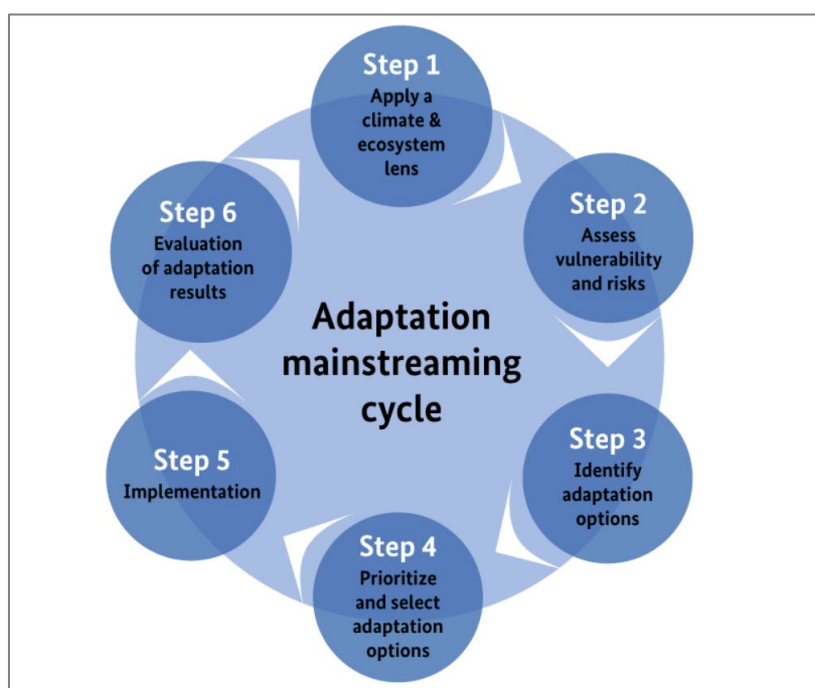


Figure 9: Adaptation mainstreaming cycle

84. A climate-sensitive IWRM approach needs to be applied in order to improve the framework conditions for sustainable water management and allow for better mainstreaming of adaptation targets, incl. EbA, into the water sector. The concept of climate-sensitive IWRM focusses on strengthening capacities in the planning process of water related bodies while integrating climate change adaptation needs in policy frameworks (national level) as well as in the planning and design of water management measures at river basin scale (sub-national level). This includes a comprehensive capacity development strategy using improved diagnostic analysis as well as emphasizing the prioritization and budgeting of most suitable adaptation actions. It will be essential to raise the awareness of the direct and co-benefits of ecosystems, their economic value and critical role for adaptation to climate change. To contribute to the overall monitoring and evaluation of the impact of the different measures in terms of water security, on the flood situation and on socio-economic developments digital tools (incl. UAVs/ drones) shall be applied. The digital solution based M&E methodology that will enable monitoring quantitative and qualitative changes at river basin scale is established and operated in collaboration with academia and linked to local and national monitoring system(s) of institutions like HII for the documentation of and reporting on climate change adaptation actions in the water sector.

85. The EbA measures are to be integrated in a climate-sensitive River Basin Management approach (reflected in the River Basin Master Plan – RBMP) that builds on climate risk and vulnerability information and an integrated water resources management (IWRM) concept. Quantity & quality of societal benefits are to be reflected as well (in terms of impact on livelihoods e.g. source of income, reduced risks as well as in terms of ecosystem services e.g. reduced

⁴⁴ <https://www.adaptationcommunity.net/ecosystem-based-adaptation/eba-mainstreaming-cycle/>

erosion, extreme event buffering for flood and drought situations). In close cooperation with the Thai water sector GIZ is currently in the process of developing an EbA Guidebook as a national standard for the systematic identification, prioritization and selection of EbA measures at river basin scale. The EbA Guidebook is to be complemented by a Code of Practice (CoP) for selected EbA measures that will cover specific design issues. The EbA Guidebook and CoP are to be finalized by end of 2021 and are planned to be published under the Thai Office of the National Water Resources (ONWR). RID and DWR are participating in the development of the EbA Guidebook and CoP. The national standard builds besides others on the Guidebook on Climate Risk Assessment for Ecosystem-based Adaptation and joint cause-impact chain analysis as key elements of a joint prioritization and selection process for effective EbA measures. <https://www.adaptationcommunity.net/wp-content/uploads/2018/06/giz-eurac-unu-2018-en-guidebook-climate-risk-assessment-eba.pdf>.

86. Sub-activities include:

- 2.2.1. Conduct of diagnostic analysis incl. vulnerability and risk assessment, modelling, cause-impact chains and economic evaluation
- 2.2.2. Based on the diagnostic analysis conduct joint prioritization of EbA solutions, finalization of design for three selected EbA measures, including related training and guidance materials
- 2.2.3. Implementation of identified ecosystem-based adaptation measures
- 2.2.4. Integration of ecosystem-based adaptation measures into water management rehabilitation plans: soil conservation measures, floodplain development and conservation, river restoration, fish migration passes, and longitudinal connection of the Yom River
- 2.2.5. Documentation of lessons learned in terms of linking local (river basin) level planning and implementation experience with national level policy developments [contribution to the implementation of the NAP/NDC]

Output 3: Reduce volatility of agriculture livelihoods in drought and flood prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets

(GCF Grant: USD 5.617 million, Co-financing USD 0.361 million)

87. The project area is dominated by agriculture activities. Farmers in the area are particularly vulnerable to climatic conditions and would be directly impacted by any interventions in the area. It is critical therefore that interventions in the project area engage farmers, and the project is informed by their experiences and needs.

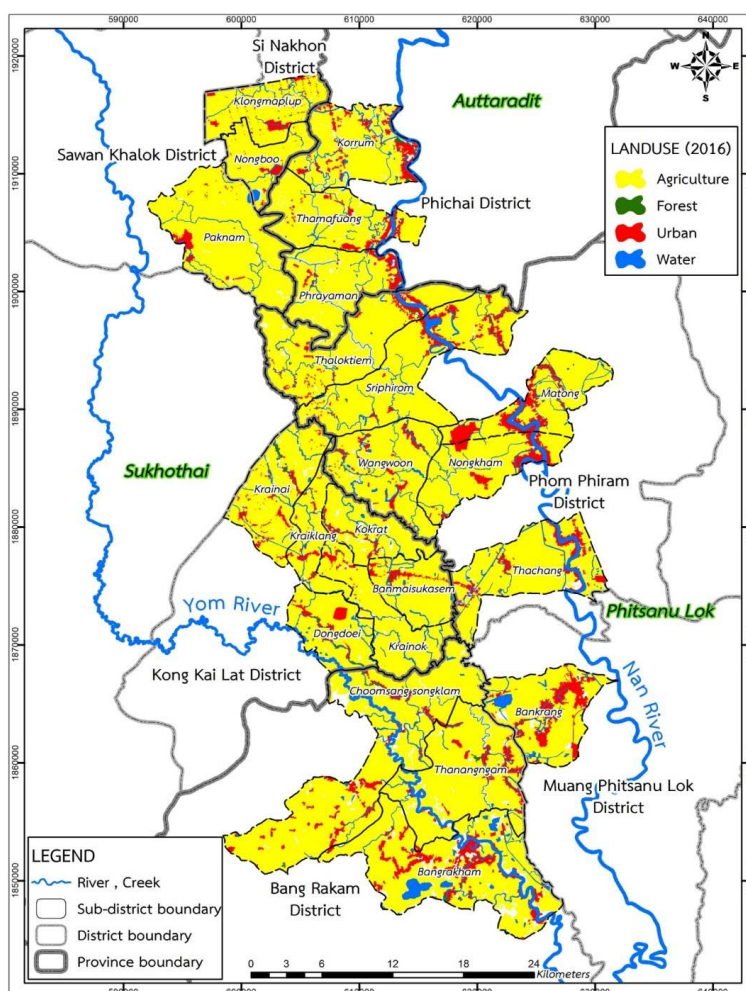


Figure 10: Land use map of project area

88. During consultations, farmers expressed the need for specific support to improve their climate resilience, (i.e. not just information on the design, but tailored support on implementation of measures to better cope with changing climatic conditions). Farmers requested guidance for on-farm measures for water storage during the dry season, support on drought resilient species and any related changes to agriculture practices to ensure viability of crops and help with business skills for better market access. To that end, under Output 3, GCF will support training to inform climate-informed decision-making related to agriculture practices, technical expertise and implementation of on-farm interventions for farmers in the immediate project site, with related training materials that can reach across the agriculture sector in Thailand through Farmer Field Training (FFT).

89. While the majority of farmers in the project sites grow rice, there are different types of agriculture and agriculture land in the area. Interventions and training will be tailored to what is suitable and proven effective in those conditions.

Province	Total Land	Paddy land	Upland field	Orchard and perennial crop	Vegetable and ornamental plants	Other agricultural land use
Sukhothai	1,823,937	1,215,015	393,530	128,331	9,548	77,514
Phitsanulok	3,057,529	1,777,793	727,441	345,998	14,718	191,579
Uttaradit	1,248,199	680,842	314,762	153,599	12,663	86,332
Total	6,129,665	3,673,650	1,435,733	627,928	36,929	355,425
Percentage	100.0	59.93	23.42	10.24	0.6	5.8

90. The Output will be implemented in partnership and coordination with King Mongkut's University of Technology North Bangkok (KMUTNB) and Krungsri Bank. KMUTNB will help to deliver training to farmers on applying water forecast based on climate information to household level agriculture practices; and Krungsri Bank, through co-financing

activities, will support farmers in developing their business skills for greater market access. Trainings under this Output will be coordinated for cost-savings purposes, but importantly to also ensure that extension officers and farmers get most of the collaborative approach of the project – bringing together expertise from the water and agriculture departments, academia and the finance/private sector, and of course the farmers themselves.

Activity 3.1. Application of climate information in household agriculture planning and strengthening related support through extension services

91. Effective application of climate information into household agriculture planning will require shifting the crop choice and crop calendar to correspond to the changes in temperatures and rainfall patterns. For this, inter-agency coordination at the local, provincial, and central levels is needed to adequately support agriculture households to make the necessary changes in order to successfully adapt to climate change.

92. Based on the tailored climate information of Output 1 and a close collaboration with the IRDM programme and with KMUTNB, GCF resources will support knowledge sharing and training to provincial extension officers from the Department of Agricultural Promotion under MoAC and agriculture households on climate change impacts on agriculture in the Yom and Nan river basins. GCF resources will support awareness raising, informed adaptation measures and training to communities, such as related to appropriate timing for agriculture practices and agriculture crops or technologies which are less water-intensive (e.g. precision farming, application of Unmanned Aerial Vehicle (UAV)). Conservation agriculture, on-farm water storage and aquaculture, linkages to market for diversified crop options are some of the key supportive measures to improving agricultural livelihood resilience.

93. Informed by vulnerability and risk assessment under Output 2 and the climate information of Output 1, district level action plans will be developed. The action plans will also provide a basis for the type of support households in the project area will require. For instance, where assessments indicate water scarcity issues for the village regions of the project target area, action plans may include the constructing of on-farm ponds. Agriculture extension officers will receive training to support the changing needs of farmers in the area.

94. In partnership with King Mongkut's University of Technology North Bangkok (KMUTNB), training will be provided to farmers in 22 sub-districts on using and applying the information from the app developed under Output 1. Training will include adaptation measures such as, changes to crop choice and crop calendar which may be required due to climatic changes, and that result in both reliable productivity as well as more sustainable use of natural resources by farmers.

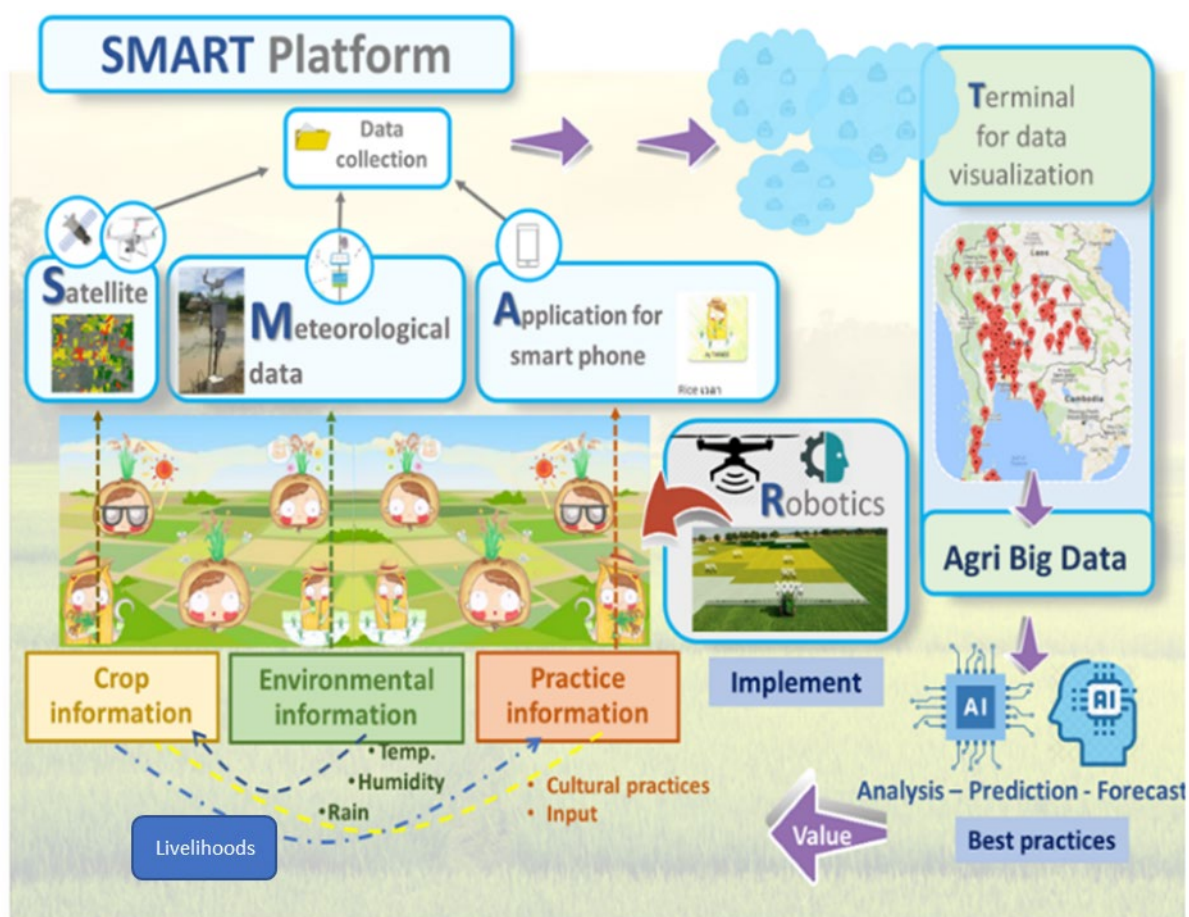
95. The project will provide technical support and bring in local experience to share lessons learned and best practices related to principles and application of, for example, conservation agriculture, organic agriculture, agroecology, integrated farming, as well as water saving approaches such as water saving techniques related to rice, such as system of rice intensification (SRI) to respond to drier conditions and changing rainfall patterns⁴⁵.

96. Technical support applies to both extension services and Farmer Field Schools. The Farmer Field School model is established as a platform for farmers to provide practical learning approaches for livelihoods and in replicating good practice techniques. Whilst being complementary to the Agricultural Extension Services approach and provision, the Farmer Field School differs in that it uses participatory teaching methods and is specific to adults and farmer needs – providing opportunity for farmer experimentation with different farming techniques and crop varieties, allowing demonstration, and generating observation and analysis of the outcomes. GCF resources will support climate-resilient agriculture trainings to farmers through existing farmer field schools, which will serve a long-term mandate of facilitating a transformation in agricultural practice through the promotion of ecologically supportive climate smart practices. In consultations during project development, farmers expressed training fatigue, asking that training be accompanied by on-farm support. The project will support on-farm and community-based activities, including demonstrations and regular follow ups.

97. Thailand has recognised the potential for technology to provide positive results for improved production, quality and agricultural products, and importantly their ability to be resilient to climate change. The National Strategy -Thailand 4.0 includes ambitious targets for agriculture, aiming to transit traditional agriculture to smart agriculture and increase farmer income by fivefold in 20 years; leveraging innovation and digital transformation. The advancement of technology in agriculture is also in line with Thailand's 20 year Agricultural Development Plan (2017-2036) and "MOAC's Digital

⁴⁵ It should be noted that alternate wetting and drying (AWD) in Thailand has had mixed results. Contributing factors included low lying areas and shallow groundwater levels, which are both present in the project area. Roll out of such approaches will be informed by on farm conditions and suitability to the location, based on best practices and lessons learned.

Agriculture Strategic Plan (2017-2021)" (For further details, refer to paragraph 153). Therefore, the project aims to pilot precision agriculture with IoT system and the usage of drones (see illustrative figure below).



98. Precision agriculture will be employed at five sites, covering 50 farms. IoT sensors will be installed at the farm and drones with multispectral camera will be used for agricultural area survey. Data will be collected and processed on the cloud server by using the AI from commercial software. The data from IoT sensors and the drones will be linked with the result of climate change data forecasting for decision making. Related information could help inform farmers to divide their fields into smaller and homogeneous zones, assess water quantity needs and even predict crop yield. Equipment will be owned and maintained by the district, and data incorporated into activities under Outputs 1 and 2. Results will be documented and disseminated to farmers in the area as well as provinces/districts with similar landscape and climatic conditions. As data can inform district planning, further upscaling can be done by district authorities through public resources – consistent with Thailand 4.0 aims to shift Thailand's agriculture sector towards an innovative-driven and interconnected sector through the integration of the concepts and approaches from sufficiency economy, smart farm, precision agriculture, intensive agriculture, sustainable agriculture, green economy, and agricultural biotechnology. A more tailored application of the technology may be upscaled by farmers through available loan instruments (for larger farms) or through the credit to be established by BAAC (see Sub-activity 3.2.5). Trainings will be combined where appropriate for efficiency purposes. . It is to be noted that in the project, district authorities would be responsible for O&M for this intervention and related resource commitments (e.g., personnel/finance) would be confirmed before roll out of interventions.

It is not expected that individual small farmers invest in IoT/drone technology, though groups of farmers may find it feasible – bridging the technology gap between small and larger farms; in additional, data collected from IoT can help enable policy and planning therefore benefiting smallholder farms. The use of IoT is meant not only to improve farm outcomes (e.g., increased yields), but also to provide information that can help with local planning and resource allocation (this is in line with National Strategy-Agriculture 4.0, which aims to shift Thailand's agriculture sector towards an innovation-driven and interconnected sector). By focusing on smallholder farmers for the intervention, data is collected from farms that need the greatest support. IoT is an innovative technology (approx. \$1M of the total GCF

funding request) that is being adopted by the government across its many investments (government and donor funded) and there is a demonstrated commitment for O&M and upscaling that will ensure that these technologies become cost-effective over time. The proposed project will undertake a technical assessment as part of the implementation of this activity to fully inform the design and roll-out, with a view towards replication and scale. The precision agriculture interventions will be based on a successful research conducted by the Kasetsart University Thailand in collaboration with the Ministry of Agriculture and Cooperatives⁴⁶ and international best practices, such as [International Rice Research Institute](#)⁴⁷. These research indicated the IoT system and drones are especially effectively on:

- Assist timing for planting;
- Land preparation;
- Drone usage for remote sensing (crop health data collection) and efficient herbicide / pesticide application if this is needed;
- Enabling farmers to effectively manage water levels through better management of soil moisture levels (as monitored by IoT sensors);
- Data collection for enabling policy and planning with local government;
- Climate data forecasting for better decision making.

99. Sub-activities include:

- 3.1.1. Assist in the development of district level adaptation action plans through technical assistance
- 3.1.2. Training of 20 provincial agricultural extension and land development officers in the target provinces to undertake climate-risk informed agricultural extension services
- 3.1.3. Training of agricultural extension and land development officers for the project area in climate-smart agricultural practices and management; including appropriate crops and inclusion of climate risk responsive farm planning
- 3.1.4. Precision agriculture demonstration and related dissemination of data
- 3.1.5. Technical assistance to support farmers to apply cause-effect analysis and climate information to agriculture decisions at the household level (e.g. crop choice and crop calendar)
- 3.1.6. Training through farmer fields schools in each sub-district to complement and support the activities of agricultural services (extension)

Activity 3.2. Implementation of on-farm climate resilient measures to improve drought and flood resilience and improved access to finance for sustainable agriculture

100. Based on results of cause-effect analysis and related action plans, on-farm water storage and groundwater recharge mechanisms will also be explored, including on farm storage/rainwater infrastructure, installation of underground water traps⁴⁸, vegetation management to promote recharge, canal and waterway management constructed and natural wetlands, and so forth⁴⁹. Techniques assist to prevent the dropping of the watershed. Groundwater recharge collects the water during the wet season for use during the dry season when the demand is highest. In the cases where farmers are using chemical fertilizers and pesticides – artificial recharge must only be done using filtration/percolation techniques as to prevent contamination. The project will, however, promote the usage of organic alternatives to chemical fertilizers and pesticides.

101. To support the EbA measures of Output 2, community nurseries will be established. The preparation for the seedling nursery will be in small pieces of land equipped with water pumps to drain those submerged pieces of land. In other cases, the project will support farmers to cooperate and establish community nurseries on higher ground. Importantly, this can provide means of longer term support to vulnerable farmers during periods of flooding. Ownership of the community nurseries will be with the farmer cooperatives in each sub-district. These cooperatives operate under the legal framework, managed by the Cooperative Promotion Department, under the MoCA. GCF will also support investment in small scale equipment to farmers to support EbA through water conservation and better soil management (e.g. alternate wetting and drying technologies that measure more precisely the amount of water needed for rice). Selection criteria for beneficiaries will be guided by vulnerability and priority, and economic needs of beneficiaries. Economic needs include a) elderly farmers, b) newly graduated (unemployed) youth, households with savings not higher than 50,000 THB and household income not higher than 100,000 THB, and c) households with usable area not

⁴⁶ SMART PLATFORM FOR PRECISION AGRICULTURE IN THAILAND, Tanee Sreewongchai and Sutkhet Nakasathien, Kasetsart University, Thailand, 2019

⁴⁷ A comprehensive solution for landscape-scale sustainable water management in rice Synthesis report (v 1.0), International Rice Research Institute, Philippine Rice Research Institute

⁴⁸ See USAID Mojokerto project.

⁴⁹ See www.unep.org/ipet/publications/techpublications/techpub-8e/artificial.asp

larger than 50 square meters and total residential area not larger than 5 rai. In the context of COVID-19 and the related economic impact, selection criteria may be revisited.

102. To support farmers with access to finance for investments, GCF resources will also support the co-design of a credit instrument for farmers practicing climate resilient agriculture. BAAC has committed USD 16 million for this effort. It has been counted as parallel finance for the proposal and the commitment letter is part of Annex 13. The BAAC credit will be available to small-scale farmers, community farmers/cooperatives, agriculture SMEs, as well as larger agricultural businesses. The range of credit available is expected to be 9,000 THB (USD 300) – 10 million THB (USD 320,000). The co-design of the credit instrument will follow a consultative process engaging relevant ministries and communities, to ensure that the product is one that both supports climate resilient agriculture investments and on-farm EbA measures, and suits the needs of farmers in the project area.

103. Sub-activities include:

- 3.2.1. Technical expertise for development of on-farm EbA guidelines for application by farmers
- 3.2.2. Technical assistance and investment to support to farmers related to on-farm ponds and enhancing on-farm measures to protect irrigation channels (e.g. vegetating on farm canal buffer zones, planting shade cover areas to minimise evaporation of water along canal ways, and vegetating irrigation infrastructure for flood protection)
- 3.2.3. Procurement of small scale equipment for farmers to support water conservation and soil management
- 3.2.4. Establishment of community nurseries to provide alternative livelihood income generation during drought-flood/non-farming periods, and to cultivate and supply climate smart crops, vegetables, and vegetation species for EbA measures implemented in Output 2
- 3.2.5. Co-design of financial instrument with BAAC to support farmers with adaptation investments towards climate resilient agriculture

Activity 3.3. Capacity building for farmers to support market access for climate resilient agriculture products

104. Uncertain yields due to climate change and falling prices for traditional rice varieties create volatility to farmer livelihoods. During consultations, farmers shared challenges faced by increased temperatures and changing rainfall patterns (particularly drought periods), but also expressed concerns over implementing changing due to the lack of knowledge about and access to markets. Knowledge sharing workshops will be conducted, so that farmers can learn from each other and benefit from each other's best practices and lessons learned. To support market access, the project will establish an online market platform, providing an additional means for farmers to reach consumers.

105. GCF resources will also support farmers with market access through training and support with identifying links to buyers and markets for climate-resilient agriculture products, including development of contract farming (incl. hospitals, schools, restaurants), linkages through rice cooperatives, linkages to local gatherers and processors, linkages to private sector and community support agriculture (CSA) for larger markets at Phitsanulok province, and comprising of larger market buyers, processors, and cooperatives. With support from Krungsri Bank, farmers will also receive training on business skills to help convert networking opportunities into buyer-seller partnerships.

106. Sub-activities include:

- 3.3.1. Farmer to farmer knowledge sharing workshops, including reciprocal learning processes (e.g. site visits by farmer field schools to project schemes)
- 3.3.2. Establishment of online market platform to increase visibility and marketability of farmers' products
- 3.3.3. Training to farmers to support market access for farmers, including capacity building related to financial literacy, linkages to buyers and markets, requirements for contract farming (incl. hospitals, schools, restaurants), linkages through rice cooperatives, linkages to local gatherer and processors, linkages to private sector and community support agriculture (CSA) for larger markets at Phitsanulok province, and comprising of larger market buyers, processor, cooperative

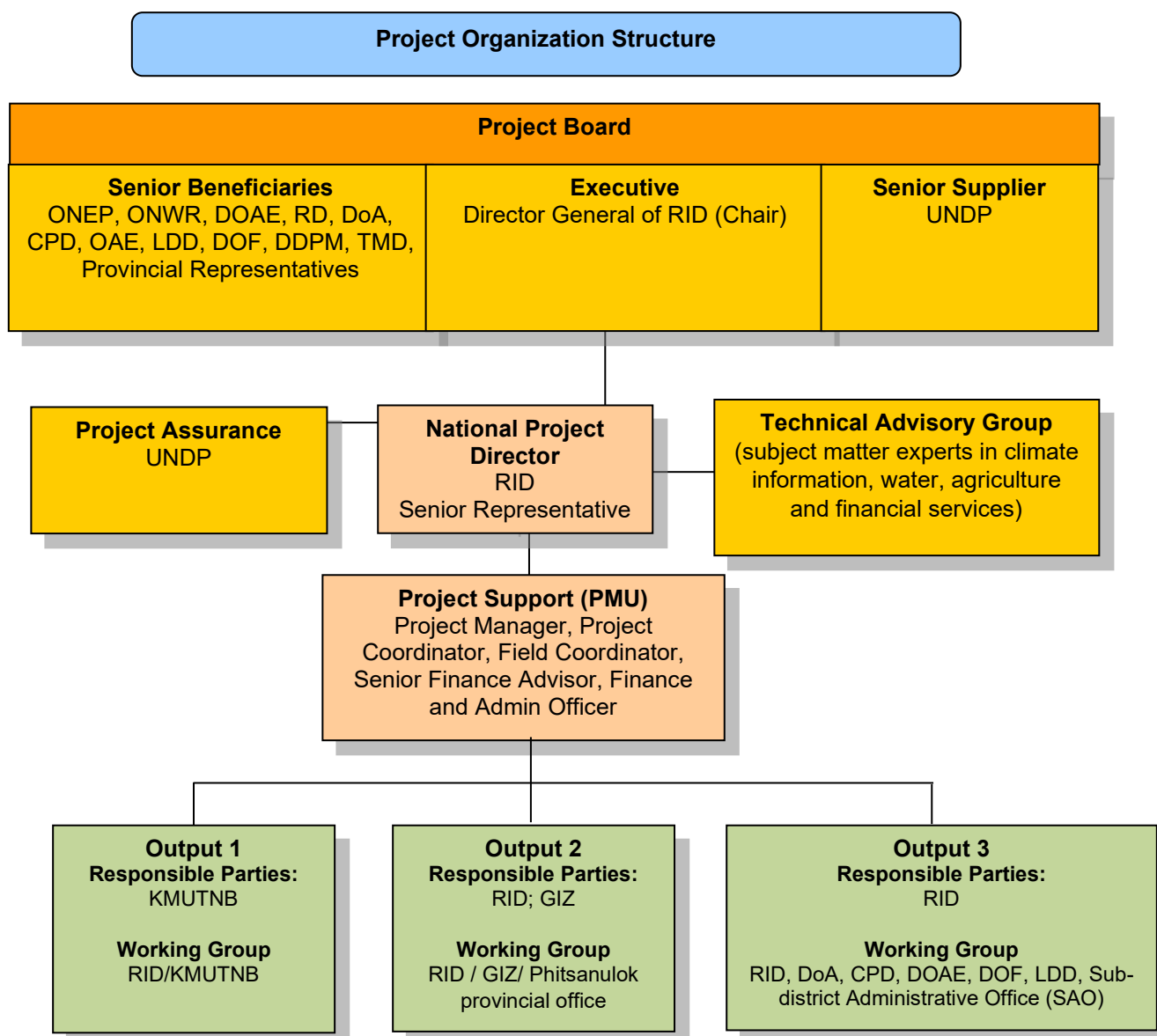
B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

107. The project will be implemented following UNDP's National Implementation Modality (NIM), according to the **Special Fund to Thailand**⁵⁰ entered into between the United Nations Special Fund and the Royal Thai Government, the

⁵⁰ The Government and UNDP have entered into an agreement to govern UNDP's assistance to the country (Agreement between the United Nations Special Fund and the Royal Thai Government (RTG) concerning Assistance from the Special Fund), which was signed by both parties on 4 June 1960. Based on such Agreement, UNDP's assistance to the RTG shall be made available to the RTG and shall be furnished and received in accordance with the relevant and applicable resolutions and decisions of the competent UNDP organs, and

Country Programme Action Plan (CPAP), and as policies and procedures outlined in the UNDP POPP (see: <https://popp.undp.org/SitePages/POPPSubject.aspx?SBJID=245&Menu=BusinessUnit>).

108. The national executing entity - also referred to as the national 'Implementing Partner' in UNDP terminology - is required to implement the project in compliance with UNDP rules and regulations, policies and procedures, including the NIM Guidelines. These include relevant requirements on fiduciary, procurement, environmental and social safeguards, and other performance standards. In legal terms, this is ensured through the national government's signature of the Special Fund Agreement, together with a UNDP project document which will be signed by the Implementing Partner to govern the use of the funds. **The (national) Implementing Partner** for this project is Ministry of Agriculture and Cooperatives (MOAC). MOAC will assign RID to implement the project outputs and will be accountable to UNDP for managing the project, including monitoring and evaluation of project interventions, achieving project outcomes, and for the effective use of UNDP resources.



Indicative Schematic of the Management Arrangements for the Proposed Project

subject to the availability of the necessary funds to the UNDP. The Agreement on Special Fund to Thailand is in lieu of the Standard Basic Assistance Agreements that UNDP entered into with the governments of other countries.

109. The Project Board is responsible for making, by consensus, management decisions when guidance is required by the National Project Director. Project Board decisions will be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency, and effective international competition. In case a consensus cannot be reached within the Board, final decision shall rest with the UNDP Programme Country Director. The Project Board will meet every six months.

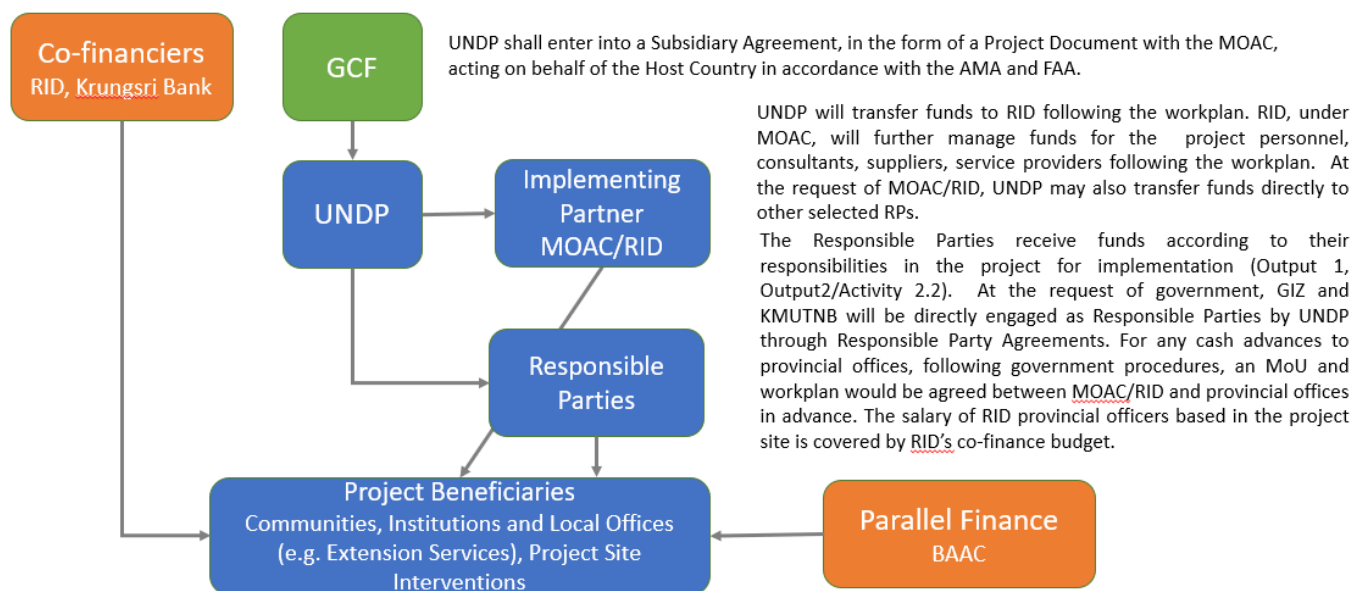
110. Project Board will consist of a group of representatives responsible for making consensus-based strategic and management decisions for the project. It will oversee the project implementation; review compliance with the Royal Thai Government, UNDP and GCF requirements; and ensure implementation of the management plan for the risks identified. The Board will be comprised of:

- An Executive (role represented by National Implementing Partner) that holds the project ownership and chairs the Board. The Executive will be RID (Chair).
- A Senior Supplier representative providing guidance regarding the technical feasibility of the project, compliance with donor requirements, and rules pertaining to use of project resources. This role will be fulfilled by UNDP in its capacity as GCF Accredited Entity;
- Senior Beneficiary representatives who ensures the realisation of project benefits from the perspective of project beneficiaries

111. The National Project Director is a senior representative from RID who is responsible for the overall direction, strategic guidance, and timely delivery of project outputs.

112. The Project Manager will run the project on a day-to-day basis within the parameters laid down by the Project Board. The Project Manager will end when the final project terminal evaluation report, and other documentation required by the GCF and UNDP, has been completed and submitted to UNDP. Project Manager is responsible for day-to-day management and decision-making for the project. The National Project Director's prime responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost.

113. UNDP will transfer funds to the Royal Irrigation Department (RID), under the Ministry of Agriculture and Cooperative as "implementing partner". Once approved, UNDP Thailand will transfer funds to RID for implementation. The internal governmental implementation arrangements shall be formalized through a project board established by the Director General of RID, under the MoAC ("**Project Board**"), agreed minutes of the Project Board decision, memorandum of understanding and work plan. Furthermore, RID has agreed to select the King Mongkut Institute of Technology North Bangkok (KMUTNB) and GIZ based in Thailand as the Responsible Parties for Implementation (Output 1, Output 2/Activity 2.2 respectively). The engagement of Responsible Parties shall follow the IP regulations and rules. Below is a diagram of the funds flow:



114. A UNDP micro assessment was conducted for RID as implementing partner/executing entity, across all major categories RID is considered low risk. In addition, the UNDP micro assessment were conducted for KMUTNB and GIZ as the Responsible Parties for Output 1 and Output 2/ Activity 2.2 respectively. Please see Annex 22 for further details.

115. UNDP provides a three-tier oversight and quality assurance role involving UNDP staff in Country Offices and at regional and headquarters levels. The quality assurance role supports the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. Project Assurance must be independent of the Project Management function; the Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. The project assurance role is covered by the accredited entity fee provided by the GCF. As an Accredited Entity to the GCF, UNDP is required to deliver GCF-specific oversight and quality assurance services including: (i) Day-to-day oversight supervision, (ii) Oversight of project completion, (iii) Oversight of project reporting.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

116. The planned interventions target the Yom and Nan river basins, in Uttaradit, Phitsanulok, and Sukhothai provinces, which are key to cope with increasing climate-induced flood and drought risks to enhance the resilience of the population living in the wider Chao Phraya River Basin, including the Bangkok Metropolitan Region with around 15m inhabitants. As described in Section C.2, the river basins currently provide a critical function of managing the river flow and regulating irrigation, whereas the last water infrastructure improvement in these areas was in 1994-1995. However, in light of climate change the current precipitation patterns change and more frequent consecutive dry periods and severe flooding in wet seasons test the limits of the existing infrastructure. This enhances the flood risks as well as poses a serious threat to the already marginalised poor local farmers and agriculture day labourers during dry periods due to higher risks of losses and damages of yields and feasibility to pursue certain livelihood options. As existing water management infrastructure is no longer able to effectively respond to climate change induced changes in rainfall patterns, enlarging the storage capacity of irrigation systems, natural canals, or monkey cheeks for keeping excess water for later use (i.e. the second dry season rice) is needed.

117. Alternative financing options to address transformative change for the most vulnerable farmers as well as to enhance the wider water management, particularly introducing innovative EbA approaches, are limited, as the planned interventions do not yield aggregate, large-scale financial reflows. A Financial Appraisal was conducted for the project, please see Annex 3. It found that the proposed activities under Output 1 – technical training to staff of RID and TMD, strengthening the climate forecasting and water management tool, dissemination and distribution of climate risk information to at-risk households through technology – aim to strengthen climate risk-informed planning, decision making and inter-ministerial cooperation, generating benefits which are of larger public good in nature such as making available easily consumable climate risk-information for households and significant improvement in capacities of government agencies. Similarly, public good nature of benefits will be generated from Output 3, which will train farmers on climate risk-informed decision making in their agricultural practices, provide technical training on building on-farm

water management interventions for farmers and help farmers develop business skills for developing market access. While the proposed activities under Output 2 build grey infrastructure in the form of upgraded floodgates and canal embankments, the resulting benefits are larger good in nature without directly identifiable and measurable financial flows to either the project beneficiaries or the project sponsor. The increased storage capacity improves access to water for agriculture and public consumption during dry seasons and the increased drainage capacity mitigates / minimizes the impact on / losses to agricultural lands. It must also be noted here that the direct beneficiaries from the investment in grey infrastructure are the most-vulnerable groups in the region with little or no ability to afford increased cash outflows. Therefore, the proposed project will not generate any “directly identifiable, attributable and quantifiable” incremental financial revenues or produce any “directly identifiable, attributable and quantifiable” tangible financial cost recovery/savings of resources to either the project owner (Royal Thai Government - RTG) or to the project beneficiaries (direct beneficiaries in the targeted project area) during and after implementation of the proposed project. Due to this larger public good nature of the benefits derived from this funding proposal, repayment of any kind of loan from the project benefits / beneficiaries is not feasible.

118. GCF involvement is critical for this proposal since there is an overwhelming evidence of that amidst other potential factors such as increasing population, degradation of ecosystem and poor management of the grey infrastructure, climate change being can be attributed as the only major factor behind the increasing frequency and severity of extreme climate-related flooding and drought events in the project area of The Greater Chao Phraya river basin. In addition, alternative financing options to address transformative change for the most vulnerable farmers are limited as the planned interventions do not yield any aggregate, large scale financial reflows.

119. An integrated project such as this one, however, can usher in the paradigm shift needed to adapt to the climate impacts on the water and agriculture sectors in Thailand. The project will inform water planning and investment not only in the Yom-Nan river basin, but also the broader Chao Phraya river basin. Upscaling and replication will support Thailand's most vulnerable/poor communities, which have a largely rural profile. Further, the project can prove catalytic in mobilizing additional resources, as is already evident in the co-financing from Krungsri Bank to strengthen financial literacy among farmers, and the parallel financing from BAAC's in the form of a USD 16M credit for farmers practicing climate informed and sustainable agriculture. In addition, UNDP Thailand is laying the groundwork for greater engagement of the private sector. UNDP Thailand has partnered with the Securities and Exchange Commission (approximately 750 listed companies) and Global Compact Network Thailand (50 members) to raise awareness among businesses on shifting from ESG to SDGs and integrating SDGs Impact Measurement and Management Framework into their business and investment strategies. The first seminar was held on 24 February with over 400 participants attending the events, including representatives from listed companies, investors, financial regulators, government officials, and academia. In 2020, UNDP Thailand is planning to focus on introducing the SDG impact assessment framework called the UNDP Business Call to Action's Impact Lab to the market. UNDP Thailand will also select some institutional investors to test the UNDP's SDGs Assurance Standard on Private Equity and Bond. To promote the localization of SDGs and awareness raising among SMEs, UNDP Thailand is partnering with Thai Chamber of Commerce, who has the local presence in all 76 provinces across Thailand. UNDP Thailand will provide sustainability related workshops to local SMEs in their network as well as work with these businesses on how to engage SDGs lens into their business activities. And in the pipeline, UNDP Thailand is discussion on project development on plastic waste management with Unilever, accelerating Social Enterprises with True Digital Park, community investment funds with Singha Estate, as well as farming technology demonstration with the Siam KUBOTA. Agriculture is a critical sector in Thailand, and each of these initiative creates opportunities to engage private and finance sector entities in related work to support climate resilience. This information has been incorporated into the FP as well. Such Private sector participation in the geography and the sustainable agriculture and water management sector is limited to Corporate Social Responsibility (CSR) activities and spending budgets, which the project is already leveraging. Another potential option of Donor aid was explored but it typically does not include / understand evolving climate change threats and is usually focused on humanitarian activities.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

120. The proposed project has been designed in close consultation with and involvement of relevant government agencies and technical line departments, development organisations (e.g. GIZ, UNDP) and existing Water User Groups in the target river basin. These consultations and discussions are complemented by best practice approaches in water management, climate-resilient agriculture, ecosystem based adaptation (EbA), and climate information and modelling as detailed in the Feasibility Report (Annex 2). Building on this foundation, the project ensures that the investments as well as the results of the interventions are sustained beyond the project period and in the longer-term through the following elements of project design and implementation.

121. **Building on existing and established networks:** The project will strengthen existing networks and coordination mechanisms, including Water User groups and farmer field schools (existent in project area), as well as housing climate information data centres in combination with existing monitoring and information centres – providing location and staff after the project duration timeframe is complete. The farmer field trainings to enhance their capacity to cope with climate change impacts and learn adaptation strategies, will partly be developed through farmer field schools and other existing local networks. This can ensure the sustainability as these knowledge hubs and networks will allow the aggregation of traditional knowledge and practices with climate-resilient technologies and innovative practices.

122. **Supporting ecological systems:** GCF resources will be used to promote the application of ecological principles to water management to trigger a paradigm shift within the RID and RTG's water management approaches towards more integrative solutions (including EbA) following a landscape-ecosystem approach based on the river system. Building on the experience of GIZ in the field EbA, the project will work on technical capacity development of key decision makers and technical officials to mainstream the, for Thailand, new approach to water management into their water and irrigation planning processes. The technical capacity building will be undertaken at both national and subnational (project area) levels to reach a variety of key public officials to sustain the efforts and reach long-term project goals.

123. The project invests in institutional capacity building, so for longevity as well as for integrating cross-sectoral and inter-agency approaches to shared project goals. It, further, promotes institutional planning and coordination across government officials and communities. Project outputs will also contribute to enhancing organisational capacity of farmers to plan for and implement climate-risk informed local water management solutions and integrate climate information and advisories for water management ensuring their financial and human resource viability post-project. The approach is to strongly engage communities at several levels of planning and execution. By doing so, the project will not only ensure that the investments respond to beneficiary needs but also ensure that community organisations, particularly including women, will have capacity to keep improving approaches, even as climate variability increases and seasons become more unpredictable.

124. **Co-investments by government institutions, financial sector and communities:** The project leverages domestic co-financing with a total co-financing contribution of the Royal Thai Government through the Royal Irrigation Department (RID) of USD 16.26 million during the 5-year implementation period, and an additional USD 17.26 million for operation and maintenance beyond the GCF-funded project period. RTG resources will support investments into water management infrastructures, technical capacity development of public officials, IRDM upgrade, skills transfer to farmers, market linkage creation, and the promotion of climate change resilient livelihoods promote. They will also support implementation of resilient drinking water solutions, especially shouldering the cost of project and post-project O&M for the implemented infrastructure, as well as aim to install the remaining 11 proposed water management infrastructures in the Yom and Nan river basins in the coming decades through their normal budget allocations (in case they can't be constituted through EbA approaches). In addition, the project leverages financial sector resources with USD 113,883 from the Krungsri Bank (part of GCF accredited entity MUFG) to conduct financial literacy training for local farmers in the project area. To support further investments by farmers, BAAC has committed USD 16 million in credit to support farmers applying climate informed practices. Investments may include, for instance, climate smart farming technology and supplies, online marketing and sale of their organic rice, vegetable, fruit, startup business capital, social enterprise set up and cash flow. The BAAC financial product will be designed through a collaborative process, supported by the GCF project.

125. **Ex-post plan for Operations and Maintenance of observing equipment, water management investment and IoT:** An O&M plan (project and for post-project O&M), including the budgeting for the human and financial resources required for O&M for the project investments, is presented in Annex 21. The plan reflects local ownership and commitment for the long-term sustainability of the project activities and outcomes. The costs of developing a long-term strategy for O&M, is provided for in the first two years, and the strategy will be reassessed towards project completion (final two years). GCF resources will finance and leverage financing to support the human and technical resources required for O&M initially, with a decreasing contribution towards the end of the project lifetime, after which

domestic financing (from the budgets of designated authorities as reflected in the co-financing letters) will continue to support O&M. RID will maintain facilities under Output 1 and 2, and other MoAC agencies will run Output 3 under future government regular budget. Please see Annex 21 for the O&M plan.

126. As the IoT equipment (including drones) will be owned by district government, local authorities will be ultimately responsible for O&M. Some of the O&M cost is built into the sale price and warranty from the dealers in the domestic market (Thailand). As IoT is promoted in Thailand, there are various avenues available to secure finance for any additional O&M as well as for further replication and upscaling. These include: a) the regular public budget, distributed to government departments under MoAC, b) the regular public budget for capacity building to farmers and innovation research and c) the Communication Authority of Thailand (CAT). District authorities will provide commitments for O&M prior to any roll-out of IoT.

127. **Ownership of equipment to be provided by the project:** Most of the equipment will be owned by the farmers themselves (eg. water pumps, water level gauges), while others will be shared and therefore owned by farmer groups/cooperatives (eg cultivator, rotary weeder). For shared equipment, procedures will follow existing borrowing system, managed by the cooperative. Following this borrowing system, the day-to-day operations (fuel, manpower) would be the responsibility of the borrower. Maintenance would be responsibility of the cooperative, with costs covered by the group's bank account. Ownership will follow the same set up as during the project. The project will support farmers groups in the budgeting process if needed, to ensure funds are planned for regular maintenance of equipment. Project results will be tracked and shared with farmers and cooperatives to inform decision making for further replication/scale up. Should additional funds be needed for further replication/scale up, the BAAC credit will be available for investments in adaptation.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount			Currency		
		17.534			million USD (\$)		
GCF financial instrument		Amount	Tenor	Grace period	Pricing		
(i)	Senior loans	N/A	N/A	N/A	N/A		
(ii)	Subordinated loans	N/A	N/A	N/A	N/A		
(iii)	Equity	N/A	N/A		N/A		
(iv)	Guarantees	N/A					
(v)	Reimbursable grants	N/A					
(vi)	Grants	17.534					
(vii)	Result-based payments	N/A					
(b) Co-financing information		Total amount			Currency		
		16.378			million USD (\$)		
Name of institution		Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Royal Irrigation Dept		In kind	16.264	million USD (\$)	N/A	N/A	N/A
Krungsri Bank		In kind	0.114	million USD (\$)	N/A	N/A	N/A
(c) Total financing (c) = (a)+(b)		Amount			Currency		
		33.911			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)		N/A					

C.2. Financing by component

128. The project seeks USD 17.534 million in GCF support with a total co-financing of USD 16.378 million which includes co-financing of USD 16.264 million from the Royal Thai Government (RTG) through the Royal Irrigation Department (RID) during the implementation period for operations and maintenance (O&M) of the GCF investment, as well as for (i) repair and improvement of the existing irrigation schemes, (ii) upgrading of the Khlong Muang Chang canal regulator and (iii) improvement of the Khong Ta Key culvert; and co-financing from Krungsri Bank (a member of MUFG-global financial group) that will contribute USD 113,883 during the project implementation period to provide financial literacy training to farmers in the project area – complementing GCF-funded activities under Output 3. In addition, the RTG has also committed USD 17.254 million for O&M for 15 years beyond the GCF-funded project period.

129. In combination, GCF funds and co-financing will be used to realize 3 project outputs, namely to (i) enhance climate and risk informed planning in the water and agricultural sectors, (ii) improve water through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts and (iii) reduce volatility of agriculture livelihoods in drought and flood prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets. To further support farmers access finance for adaptation measures, the Bank for Agriculture and Agricultural Cooperatives (BAAC) has pledged a line of credit of USD 16 million through financial products with incentives for farmers to practice climate resilience. A letter of commitment is included in Annex 13. This has been counted as parallel financing for this project.

130. The Co-financing from RTG, through RID, will be a budget allocation which RID will disburse for the implementation of the Activities referred to above. With respect to the Co-financing to be provided by Krungsri Bank, the arrangement shall be formalized through a memorandum of understanding and work plan. The commitment and

application of co-financing will be monitored and reported by the Accredited entity in line with the representations and covenants set out in the AMA and FAA.

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Enhancing climate resilience in Thailand through effective water management and sustainable agriculture	1. Enhance climate and risk informed planning in the water and agricultural sectors through improved climate information and cross sectoral coordination	3.215	1.683	Grants	1.532	In-kind	RID
	2. Improve water management through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts	23.125	9.640	Grants	13.485	In-kind	RID
	3. Reduce volatility of agriculture livelihoods in drought and flood prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets	5.978	5.617	Grants	0.361	In-kind	RID, Krungsri Bank
	Project Management	1.593	0.593	Grants	1.000	In-kind	RID

Indicative total cost (USD)	33.911	17.534	16.378
C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)			
C.3.1 Does GCF funding finance capacity building activities?		Yes ☒ No ☐	
C.3.2. Does GCF funding finance technology development/transfer?		Yes ☒ No ☐	
131. Under Output 1, the project will support capacity and technological needs for tailored climate information to inform climate resilient water management and agriculture planning, while under Output 2 the project will support capacity for climate sensitive integrated water resources management. Output 3 includes training to agriculture extension officers to support farmers with climate resilient agriculture, as well as equipment related to precision agriculture and development of an app to sell agriculture products. The total requested for these activities is approximately USD 4 million.			

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

132. The project will contribute to following GCF Fund-Level Impacts for adaptation:
- Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions through the promotion of climate-resilient, sustainable and diversified livelihoods for 62,000 beneficiaries in the Yom and Nan river basins (Fund-Level Impact A1.0)
 - Increased resilience of health and well-being, and food and water security for through enhanced access to irrigation and water management practices (Fund-Level Impact A2.0)
 - Increased resilience of infrastructure and the built environment to climate change threats, through the upgrades of the critical water regulating facilities of Khlong Muang Chang and Khlong Ta Kay Reg and EbA measures to reduce flood risks for 25 million people in the Chao Phraya river basin as well as build environment, including Metropolitan Bangkok Area with 15 million inhabitants (Fund-Level impact A3.0)
 - And Increased resilience of ecosystems and ecosystem services, through the implementation and promotion of EbA measures (Fund-Level Impact A4.0)
133. The project will reduce vulnerability by enhancing adaptive capacity and resilience of rural populations. By applying a gender-sensitive approach, agriculture based communities will receive climate information and support to apply that information towards climate resilient agriculture practices, thereby reducing volatility brought about by climate change induced floods and droughts. The project will also support institutional capacity for climate-responsive planning and development in water and agriculture management through the expected increase in use of climate information in decision making. And support to upgrading critical infrastructure will enhance ability of water retention and management in the Yom and Nan river basins to cope with an intensification of extreme rain events and enhance the capacity to store water for irrigation purposes during increasingly drier dry periods; and strengthened institutional capacities, knowledge, and learning for climate-resilient water management and livelihoods security.
134. The Yom and Nan river basins are of critical importance, as water management in this area would not only ease the severity of flooding for farmers in the vicinity, but also further downstream in the broader Great Chao Phraya river basin. In addition, adequate water management will reduce drought related risks to crops and agriculture-based livelihoods in the project region. The enhancements detailed would minimize loss and damage related to flooding and drought - benefiting vulnerable agriculture households and relieving pressure on public resources for related relief and programmes.
135. The project incorporates lessons learned and best practices from several successful international efforts in order to enable a transformative impact through the climate informed improvement of water resource management and climate change resilient livelihoods planning. These incorporated best practices include the utilisation of Ecosystem-based Adaptation (EbA) solutions for climate resilient water management and livelihood development. This integration of EbA approaches has the potential to be transformational to Thailand's traditional planning concepts for water management that focus widely on hard structure interventions and could also be replicated (scaled-up) in the other river basins in Thailand (total 25).
136. In addition, the policy support activities under the proposed GCF project and capacity development, e.g. in the field of water management modelling and forecasting under the updated IRDM tool and integration of EbA methods into climate-resilient planning and solution prioritisation processes, has the potential to be replicated in and set new standards for government entities involved in water management planning and livelihoods promotion in Thailand.

2. Paradigm shift potential (max. 500 words, approximately 1 page)

137. The proposed project integrates sustainability strategies and pathways for replication and scale to catalyze impacts beyond the project duration. RTG aims to build on this project to showcase innovative solutions in the field of climate resilient water management and livelihoods promotion. Successful implementation of this project can transform traditional approaches of hard structure focused interventions for water management, provide access of farmers to tailored climate information, alternative livelihood options (including a focus on organic production), create vertical and horizontal linkages to buyers and suppliers, and provide RID with a new planning tool for localized irrigation and yield forecasting. All these climate change responsive solutions can be replicated and scaled up through the integration into planning and implementation in other river basins in Thailand.

138. In particular, the following activities advance the potential for transformative impact beyond the project and form a paradigm shift potential of the project:

- The upgrade and enhancement of a rainfall forecasting and operating tool. The project will enable RID, and its key academic institutions, to build on the IRDM tool to allow RID staff to forecast local rainfall patterns and runoff simulation. These forecasting and operating models is a key decision-making tool to inform water management to prevent floods and provide sufficient water for irrigation during dry periods. The tailored and localised information will, furthermore, be provided to subnational public entities and farmers (end-users) through mobile phone applications. This can inform local and farm-level decision making in light of increasing climate change induced flood and drought risks, as well as enhance transparency and water distribution decision-making in order to mitigate potential water distribution conflicts between water infrastructure operators and local stakeholders.
- Ensure costly investments in grey infrastructure are climate informed. By integrating climate change into water management infrastructure planning, the project supports cost-efficient investments – extending the life of the investment, and reducing costs related to loss and damage related to climate impacts and subsequent upgrades.
- Develop RID's capacities to integrate ecosystem-based adaptation (EbA) in water management. While EbA measures have been tested in Thailand, RID's interventions to enhance water management in the Chao Phraya river basin, and other basins in Thailand are traditionally focused on engineering solutions that utilize grey infrastructure. This project will develop the evidence base to support complementing grey infrastructure with ecosystem-based adaptation (EbA) measures. Given the high upfront costs of grey infrastructure, the successful implementation of this project could showcase a effective and sustainable solutions, which are also lower cost. This can be a scalable solution to increase the resilience of Thailand's water infrastructure and, consequently, of the farmers dependent on the irrigation system in light of increasing climate change impacts, while rehabilitating ecosystems in Thailand's river basins.
- The project sets out the process of identifying local level agricultural community vulnerabilities to climate change impacts and livelihood response for farmers, as well as setting forth best practice measures and strategies in the planning and management. In light of increasing climate change impacts it is inevitable for the farmers to incorporate these increasing risks into their farming activities and livelihood planning. The project will enable the farmers, through capacity building, financial training (delivered through a private sector entity), and the introduction of adaptation technologies, alternative climate-resilient crops, and farming techniques to cope with increasing risks.
- Integration of finance/private sector actors (e.g. Krungsri Bank and BAAC) into implementation of EbA technologies, support for climate resilient and sustainable agriculture investments and capacity building of farmers through financial trainings. In addition, the project will create linkages between farmers and buyers of climate resilient agriculture products.

D.3. Sustainable development (max. 500 words, approximately 1 page)

139. The project generates a number of co-benefits, not only technological advancement in climate change informed analysis, but also contributions towards social and economic development:

- Reduced losses and damages due to enhanced planning for investment and preparation for slow onset climate change impacts as well as extreme events:
- More efficient use of water resources, in line with sufficiency economy goals of the country
- Promotion of less water-intensive agriculture practices to relieve pressure on water resources
- Greater predictability and security for agriculture planning at household level

Environmental co-benefits

140. EbA measures and sustainable agriculture will promote positive environmental externalities such as soil/water quality, conservation and biodiversity. Investments in the water retention capacity of the area will also benefit ecosystem services. A study undertaken catchment wide on the upper parts of the Chao Phraya River Basin by Prattomchai (et al., 2014) looked at projected groundwater recharge under changing climate scenarios during the period of 2026 – 2040, based on GCMs (MIROC, ESM, CHEM, HadGEM2, ES HadGEM and GFDL ESM2M) under scenarios RCP2.6, RCP4.5, RCP6.0, and RCP8.5. The study found that annual groundwater recharge (storage) is likely to reduce by 12.9% as a result of the significant decrease in rainfall volumes in the April to June annual periods.⁵¹ It is expected that the retaining of water through retention options would have a positive influence on groundwater recharge, by allowing waters to be retained long enough to permeate to groundwater levels.

Economic and social co-benefits

141. By reducing volatility through support to sustainable agriculture, and expanding access to markets and finance, the project supports viability of agriculture livelihoods. As vulnerable groups are prioritized for support, the project will support poverty alleviation, particularly for women as more than half of beneficiary households are female-headed. Greater viability in the sector can not only increase agriculture productivity and related income-generating capacity, but it can also help address challenges of youth employment and rural-urban migration. By partnering with Krungsri Bank and BAAC, the project will support financial literacy and help provide access to finance to support climate resilient agriculture practices, thereby increasing competitiveness and economic growth potential.

142. The project will contribute to several SDGs, namely: 1 (no poverty) and 2 (zero hunger) by focusing on vulnerable communities with higher incidence of poverty, 6 (clean water and sanitation) by improving water access during drought periods, 8 (decent work and economic growth) by reducing volatility of agriculture livelihoods, 15 (life on land) given the environmental benefits of EbA and 17 Partnerships for the Goals through a collaborative effort with government, UNDP, GCF and finance institutions (i.e. Krungsri and BAAC). As the number of female headed households in agriculture is growing, the project can also contribute to SDG 5 (gender equality).

D.4. Needs of recipient (max. 500 words, approximately 1 page)

143. Over the last four decades, Thailand has made remarkable progress in social and economic development. However, the high exposure to climate change risks and impacts pose a serious threat to the country's economy and could undermine some of the development gains achieved through investment by the RTG, partners, and communities. Water management in the context of changing climatic conditions is an adaptation priority. The country has relied largely on grey infrastructure, with related costs of necessary upgrades being formidable. Further, agriculture is a critical sector for Thailand, which due to changes in rainfall patterns is being impacted. IWRM has been stressed as a key to sustainable development going forward, and collaboration and joint planning across sectors will be necessary, which to date has been limited.

144. In terms of vulnerability, rural areas, such as the proposed project area, will face significant climate-induced risks due to the relatively higher exposure to climate change impacts, as well as their socio-economic profiles. Poverty in Thailand predominately has a rural-profile, which fluctuates according to vulnerabilities in the agricultural sector – such as faltering economic growth, falling agricultural prices, and ongoing droughts.⁵² In 2013, for example, 80 percent of the country's poor 7.3 million live in rural areas, and a further 6.7 million were living within 20 percent above the national poverty line.⁵³ 48 percent of Thailand's provinces have a poverty head count of 10 percent or less, and a further 25 percent of provinces have a poverty headcount rate of between 11 and 20 percent.⁵⁴

⁵¹ Masud, M., Soni, P., Shrestha, S. & Tripathi, N. 2016, 'Changes in Climate Extremes over North Thailand, 1960 – 2009', *Journal of Climatology*, Research Article, 206:4289454, Pp 1-18

⁵² Overview, Thailand: <http://www.worldbank.org/en/country/thailand/overview>

⁵³ Overview, Thailand: <http://www.worldbank.org/en/country/thailand/overview>

⁵⁴ National Office of Statistics, 2015

145. Most of Thailand's rural society is engaged in agriculture related activities which generate around \$40 billion in revenue, over 10 percent of national gross domestic product (GDP)^{55, 56} and account for 74 percent of water consumption in Thailand.⁵⁷ The sectors role in job creation in the rural areas is higher than the national average of around 50 percent of society being dependent on the agricultural sector for their livelihoods. The total number of households engaged in agriculture in Thailand is almost 6 million (5,910,791 households) comprising a total 23,686,102 (almost 24 million)⁵⁸ of the country's 67 million national population. The agricultural sector in Thailand additionally, directly and indirectly, employs roughly around half of the country's population.⁵⁹ Overall, Thailand's agriculture sector is dominated by smallholder farmers - most of who own their land outright, although tenancy is widespread in parts of the Central Plain. On average nationally, farmers own land sizes of around 20 Rai (equivalent to 3.2 hectares). However, in the project area, land ownership sizes tend to be larger, ranging from 25 to 29 Rai (4.00 - 4.64 hectares) per household with the area also employing a significant portion of the area's population in seasonal laboring.

146. Almost half of the population of Thailand is engaged in agricultural sector, where incomes are relatively low. The project target area was selected based on vulnerability and urgency, given proximity of communities to the river basin and reliance on water for agriculture as a livelihood. Proportionally, Sukhothai, Phitsanulok and Uttaradit provinces, included as part of the Northern Central region, have relatively high poverty levels. Impacts from climate change are more severely felt by those with higher levels of poverty, predominately due to the high level of dependence on natural resources, and in general, a limited capacity in terms of access to resources for responding to climate vulnerability and extremes.⁶⁰ Higher levels of poverty for the Northern-Central region in turn, means that this area also has an increased level of vulnerability to climate change impacts, with lower levels of access to resources for adaptation strategies and in which to build climate resilience.

147. The beneficial areas of this project are in the areas of 3 provinces of the Northern region of Thailand: Phitsanulok, Sukhothai and Uttaradit, where gross provincial product (GPP) from 2012-2014 of these 3 provinces was approximately 15.47% of the Northern region GPP. GPP of these 3 provinces from agriculture sector in 2014 were 33.7%, 42.0% and 32.4% where the per capita GPP in 2014 were 98,126, 72,548 and 84,003 THB per person, respectively. This compares with Bangkok and the surrounding area at 376,463 THB, Nong Bua Lamphu with 46,804 as the lowest, and the national average of Thailand at 193,395 THB per capita.

Targeted population at village level (2015, figures provided by RID)

Province	District	Households	Population	Male	Female
Utaradit	Pichai	3,342	10,361	4,979	5,382
Sukhothai	Srinakorn	128	396	190	206
Sukhothai	Khongkairas	5,420	16,802	8,068	8,734
Sukhothai	Sawankha Lok	431	1,336	645	691
Phitsanulok	Promphiram	6,739	20,891	10,100	10,791
Phitsanulok	Bangrakham	3,314	10,273	4,981	5,292
Phitsanulok	Muang	626	1,941	932	1,009
Total		20,000	62,000	29,896	32,104

148. The feasibility study (Annex 2) and stakeholder engagement summary (Annex 7(a)) includes field surveys and interviews with agriculture households in the targeted districts of the proposed GCF project. Field surveys and interviews with local administrators and community leaders of the inundated lands show that floods are not a new phenomenon to the households due to the geographical location of their premises at low-lying lands. However, great concern was expressed about increased incidence of drought and water scarcity, as well as potentially increasing periods and severity of flooding.

149. The project will combine GCF and RTG resources to address the barriers to adaptation investments and catalyse community, public, and private sector financing to support the needs of the targeted agriculture-based communities and flood and drought prone areas. Capacities will be strengthened across key decision makers and the communities to plan for and manage investment in livelihoods and water security in the face of a changing climate.

⁵⁵ Office of the National Economic and Social Development Board, 2012

⁵⁶ World Bank. 2016, <http://data.worldbank.org/indicator/NV.AGR.TOTL.ZS>)

⁵⁷ National Water Management Strategy, Thailand. 2015

⁵⁸ 'Agricultural Statistics of Thailand for the year 2011- 2012', Thailand Office of Agricultural Economic, 2016: www.oae.go.th

⁵⁹ National Statistical Office Thailand, 2012

⁶⁰ Abeygunawardena, P. Vyas, Y., Knill, P., Foy, P., Harold, M., Steele, P., Tanner, T., Hirsch, D., Oosterman, M., Rooimas, J., Debois, M., Lamin, M., Liptow, H., Mausolf, E., Verheyen, R., Agrawala, S., Caspary, G., Paris, R., Kashap, A., Sharma, R., Mathur, A., Sharma, M., Sperling, F. 2003, *Poverty and climate change: Reducing the vulnerability of the poor through adaptation*, <http://www.oecd.org/env/cc/2502872.pdf>

Project will enhance the institutional framework for innovation, long-term planning for risk-informed water management and learning to promote an enabling environment for sustained resilience of the Yom and Nan river basins and its residents.

D.5. Country ownership (max. 500 words, approximately 1 page)

150. The proposed GCF project is fully aligned with national priorities and builds on existing government programmes, including Thailand's 20-year development plan, as well as the Water Management Strategy. GCF support will address the identified additional activities/costs related to integrating climate change risks into government programmes, thereby extending the value of these investments.

151. Thailand's Technology Needs Assessment (TNA) report formulated in 2012 identified three highly impacted sectors in urgent need of adaptation technologies, namely:

- Agriculture - in need of forecasting and early warning system technologies, crop improvement technologies, and precision farming technologies;
- Water Resource Management - in need of networking (via pipes and canals) and management of infrastructures (including zoning), seasonal climate prediction, and sensor web using observation and/or modeling data; and
- Modeling - in need of an integrated national data center, national data transfer/management process and the advanced research, weather research and forecasting (WRF - ARW) model, and an integrated model to address the need of agricultural sector and water resource management sector.

152. Acknowledging the need to improve sustainable water resources management, the 12th National Economic and Social Development Plan (2017-2021) aims to promote economic and social growth that is environmentally-friendly, specifically by:

- Reforming water management systems to ensure balance between users' demand and supply as well as provision of access to water;
- Enforcing environment acts and promote collaboration among sectors for integrated work plans and budgets;
- Developing water resources information centers; and
- And establishing water management organization at the local level, such as river basin committee and water users' association.

153. Efficiency is a priority in Thailand's adaptation strategy, which is guided by the philosophy of Sufficiency Economy – moderation, reasonableness, and the need of self-immunity for sufficient protection from impact arising from internal and external changes.⁶¹ Building on from the Sufficiency Economy approach – is the 'New Theory Farming', which compounds integrated farming systems, and the Buddhist approach of self-restraint combined with sustainable agriculture⁶². The 12th National Economic and Social Development Plan (2017 – 2021) aims to promote both ecologically sound economic and social growth. This is centred on water management system reformation to ensure user demand and supply balance with the provision of access to water.

154. Thailand's National Climate Change Policy was approved in Cabinet as well as the Strategic Plan on Climate Change (January 2008). It incorporates capacity building, research and development, awareness and public participation, and international cooperation in mitigation and adaptation. The Climate Change Master Plan (2015 – 2050) sets out a long-term framework for the management of climate change in Thailand. The Plan focuses on adaptation across the water management, flood and drought sector; agriculture and food security; human health; tourism; natural resources sector, and; human settlement and security. This project targets two of these sectors, namely: (i) water management, flood and drought; and (ii) agriculture and food security.

155. Development of Thailand's Climate Change National Adaptation Plan (NAP) commenced in 2015 and currently ongoing towards completion of the NAP. The draft version of the NAP currently contains a focus on agricultural projects, of which the Enhancing climate resilience in Thailand through effective water management and sustainable agriculture project is complementary to and will strengthen and build upon the capacity building on climate information, and the upgraded IRDM tool (under Output 1) elements.

156. The Nationally Determined Contribution (NDC) for Thailand includes a shortlisting of adaptation key efforts. These include the prioritization of efforts which:

⁶¹ Thailand's Nationally Determined Contribution (NDC), 2015

⁶² Amekawa, 2010:390

- Promote and strengthen Integrated Water Resources Management (IWRM) practices into achieving water security, and effective water resource management to mitigate flood and drought
- Safeguard food security through the guidance of the Sufficiency Economy Philosophy (e.g. agriculture and land management that promotes appropriate resource allocation and economic diversification at a household level; the sustainable management of community forests in order to promote community level food security)
- Promote both sustainable agriculture and Good Agricultural Practice (GAP)
- Strengthen disaster risk reduction and reduce the population's vulnerability to climate risk and extreme weather events through enhanced awareness, coordination and local community adaptive capacity, especially within disaster prone areas, and
- Strengthen climate-modeling capacity while promoting collaboration amongst relevant agency

157. The project will also help realize Thailand's Commitment to Climate Resilient Agriculture and Digital Transformation. The project is aligned with the country's 20 Year Agricultural Development Plan (2017-2036) whose objective is to lay a foundation for long-term development leading to systematic growth and solutions to problems of the agriculture sector. The specific goals are as follows: (1) Farmers specialize in their professions, so-called "Smart Farmer;" (2) Farmer institutions have an efficiency in management, so-called "Smart Agricultural Group;" (3) Quality of agricultural products meet the customers need, in other words, being "Smart Agricultural Products;" and (4) Agricultural area and sector have potential, known as "Smart Area / Agriculture." Strategy 3: Increasing Competitiveness in the Agricultural Sector through Technology and Innovations, having 3 development guidelines which are: Develop technology and innovations to drive "Agriculture 4.0" under the "Thailand 4.0" policy; Manage the agricultural information technology for ready access and utilization by farmers and Develop agricultural research works and information toward their commercialization.

158. Furthermore, the project entails the roll-out of innovative technologies for climate solutions and advances use of IoT platforms as part of the country's "MOAC's Digital Agriculture Strategic Plan (2017-2021)" (Ministry of Agriculture and Cooperatives' Committee on ICTs, 2017). The plan maps the long-term steps to digitalize agriculture starting from preparing the ICT system for being digital agriculture. For the first five years, the strategy envisions to work on the development of paradigms and framework for digital agriculture. This strategic plan comprises of three missions, which are to develop agricultural information system for proactive management, to apply appropriate digital technology with context to agriculture sector, and to support sustainable agriculture.

159. Finally, the project was designed with extensive stakeholder consultations. Under the guidance of ONEP (office of the NDA) and RID, the project design was further informed by consultations and assessments conducted during the project development phase, which included government ministries, development partners, academia, social enterprise and communities (see Annex 7(a)).

160. The proposed project builds on past community consultations, comprising of several rounds of stakeholder discussions on adaptation priorities, including flood and drought risk management through enhanced green and grey water infrastructure, and climate-resilient agriculture-based livelihoods. Stakeholder discussions took place over the period of 2016-2019, initiated and led by the RID and in coordination with the ONEP (NDA).

161. The project, through field surveys and consultations with sub-national level decision makers, communities, and community leaders has integrated local level perspectives into the project design (see Annex 7(a)). In addition, consultations with technical officials within MoAC with expertise on water management and related modeling, economics, agriculture, and hydromet services, as well as academic institutions in order to incorporate best international and national practices, and seek wide support for the project interventions. Given their expertise and experience ecosystems-based water management in Thailand, GIZ was consulted to inform the design the EbA elements of Output 2.

162. The private sector and social business have also been consulted to ensure a pragmatic lens to the design of interventions supporting farmers. These include, Siam Organic Co Ltd, which focuses on producing healthy organic products, while applying practical business approaches informed by market research, to help small-scale producers in Thailand grow their way out of poverty. Similarly, consultations with Krungsri Bank⁶³ regarding supporting farmers with business skills resulted in a partnership for the project through co-financing for Output 3.

163. Following a multi-stakeholder approach, the project once under implementation will continue to engage with numerous stakeholders, both at a national and local level. These include the implementing partner, and civil society

⁶³ A member of MUFG, a global financial group

organizations (CSOs), NGOs, private sector organizations, academia, Office of Agriculture Economics (OAE), and other public entities. The roles and responsibilities of the various stakeholders are detailed in the Stakeholder Engagement Plan (see Annex 7(b)).

164. The proposal is aligned with UNDP's comparative advantage in the areas of capacity building, providing technical and policy support, reducing barriers and creating enabling conditions for adaptation planning and investments. Specifically, the proposed project will build upon UNDP's comparative advantage stemming from experience in working with governments and communities in Thailand and globally on: i) establishing and strengthening institutional, policy and legislative mechanisms; ii) building capacity; iii) undertaking risk assessments; iv) mainstreaming climate change adaptation, disaster risk reduction and early warning systems into development planning; and v) harnessing best practices and community-based approaches across different thematic areas for climate change adaptation and disaster risk reduction. This includes experience with initiatives focused on transferring knowledge and technology via South-South cooperation. The UNDP Country Office (CO) in Thailand is well placed to oversee the implementation of the proposed project. UNDP has worked closely with RID on a number of projects, including the ongoing Climate Change Finance project, aimed at strengthening capacities of RID to formulate climate-informed projects and secure finance for implementation. In addition, UNDP is working with the Office of Agriculture Economics (AOE) of MoAC through "Integrating Agriculture in National Adaptation Plans (NAP-Ag)" program.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

165. An economic analysis of the proposed project was carried out (see Annex 3a, b) in accordance with the *Guidelines for the Economic Analysis of Projects of United Nations Development Program*.⁶⁴ The economic efficiency of the investment was determined by computing the economic net present value (NPV) with an assumed 10% discount rate, and the economic internal rate of return (IRR). For consistency purposes, all proposals developed with the support of UNDP have opted to use a 10% discount rate, in line with the existing practice of multilateral development banks.

166. Economic values (costs and benefits) are all measured in real terms of 2019. Economic costs of the project are net of taxes, duties, and price contingencies. Furthermore, the analysis assumes a shadow wage rate of 1.00 for unskilled and semi-skilled labour in Thailand. With an official unemployment rate estimated at less than 1.5%, this assumption is appropriate. Provided that the economic cost of labour in Thailand may be expected to be lower than the market wage rate (financial cost), we expect this assumption leads to significantly over-estimating the economic cost of the project, and under-estimating the true net economic value of the project.⁶⁵

167. For purpose of the economic analysis, it is assumed that the capital assets have an average longevity of 20 years. Since, capital investment is expected to have been completed in 2025, it is assumed that this 20-year time horizon covers the period 2026 to 2045. Hence, the entire period of analysis covers the period 2021 to 2045.

168. As is common when undertaking the economic analysis of investment projects, numerous assumptions were used to delineate the "with project scenario" from the "without project scenario". These assumptions are presented and discussed in detail below. Assumptions were always made so as to under-estimate the true net economic value of the proposed investment project.

169. The project realizes two important benefits of a different nature. First, climate change (projected changes in temperature and precipitation) are expected to have an adverse impact on agricultural productivity in the project area. Secondly, extreme weather events (extreme precipitation with associated floods, as well as drought) are expected to occur at a greater frequency and greater intensity in forthcoming decades. The proposed investment project is expected to alleviate both of these impacts.

170. The net present value (NPV) and internal rate of return (IRR) of the proposed investment project are presented in the table below. The table shows the NPV to be positive under the assumed range of impacts of climate change on agricultural productivity, and an IRR in excess of 20%.

Estimated Net Present Value and Internal Rate of Return

⁶⁴ UNDP. 2015. *Guidance on the conduct and reporting of the Economic and Financial Analysis of Climate Change Adaptation and Mitigation Projects and Programmes*. UNDP.

⁶⁵ It is generally pointed out that the presence of a large informal sector explains this low level of official unemployment. Furthermore, with approximately 40% of Thailand's population engaged in agriculture, there is a high degree of underemployment and off-season unemployment in Thailand. These could equally justify the use of a shadow wage rate lower than the financial cost of labor.

Scenario ¹	Net present value (\$)	Internal rate of return (%)
Minus 5% by 2042	39,230,364	29.3
Minus 10% by 2042	43,534,130	30.8
Minus 15% by 2042	47,841,453	32.2
Minus 20% by 2042	52,148,777	33.6

¹ For purpose of clarity, the scenarios presented are the assumed impacts of climate change by 2044 on agricultural productivity in the absence of the project.

171. A sensitivity analysis indicates that the NVP of the project is robust to an increase in cost of 20%, a decrease in benefits of 20%, and a simultaneous increase in cost of 20% and decrease in benefits of 20%.

172. In addition, a financial analysis has been carried out (see Annex 3c) in accordance with the *Guidelines for the Financial Analysis of Projects of United Nations Development Program*. These guidelines clearly mandate that a financial analysis of project cash flows be computed and FIRR calculated only for those proposed project activities or outputs that can reliably result in direct and quantifiable financial revenue generation (incremental earnings from baseline) or a direct and quantifiable financial savings potential to the project owners or to the project beneficiaries.

173. In all there are 3 outputs and 8 activities that constitute this proposed project. However, none of the activities or outputs clearly result in direct and quantifiable earnings or direct and quantifiable savings to the project owners or the project beneficiaries. It is only pertinent to mention here that all the activities and outputs of this proposed project result only in non-attributable savings that are of public good in nature to the larger community of the at-risk population in the proposed project area. Hence, a financial modeling-based analysis has not been conducted and an FIRR has not been calculated for any of these activities.

174. However, considering the GCF's minimum concession policy, this project's proposed activities have been analyzed from a macro-economic and government perspective to assess the need for GCF grant as the only feasible financial instrument to fund the project activities. Hence, taking into consideration the factors such as the public good nature of benefits arising out of the project, climate change being the key driver, disadvantaged socio-economic background of the target population, and the catalytic nature of GCF grants, the following is recommended:

- With 69% of budget coming from co-financing, GCF grant requested through the funding proposal is tailored to the incremental cost to make the total project investment viable and hence the right level of concessionality has been sought.
- Royal Thai Government (RTG) has no incremental ability to stretch its contributions either from internal budgets (due to the innovative, first-time nature of EbA-based project) or from external private capital participation (due to lack of tangible financial reflows)
- The nature of the benefits and the socio-economic background of the vulnerable population does not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
- The nature of the beneficiaries and the level of essentiality of the water management service to their livelihood do not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
- Hence, it is recommended that in order to reduce / close the existing financing and knowledge gaps and barriers to improve resilience of the at-risk population in the proposed project area to climate change-induced hazards, catalytic capital in the form of GCF grants are essential.

175. To ensure sustainability of the investment, the RTG has committed USD 17.254 million for O&M for 15 years beyond the GCF-funded project period.

Best practices

176. A feasibility study was conducted as part of development activities for the proposed GCF project. Conclusions and recommendations of the feasibility study have been incorporated into the design of the project, these are summarized below.

Climate information for water management and agriculture planning

177. Key lessons and recommendations for climate data management and information have been developed according to the TNA. The IRDM is an existing tool developed utilised by RID that can calculate the amount of rainfall and inflow of water into reservoirs on the rainy season and design water release schedules for the next 12 months with a relatively high level of accuracy. Its mathematical models can be used to forecast inflows into different water sources, increase farming areas along the Chao Phraya River basin, work out water management plans for individual irrigation zones and estimate agricultural yield and revenue. Under this project the tool will be upgraded to provide forecasts for shorter time frames (e.g. monthly or weekly) and tailor and distribute risk information to the needs of subnational

decision makers and farmers/ end-users. This can, ultimately, enable farmers to enhance their resilience against climate change induced droughts and floods through better water management planning.

Strengthened water management infrastructure and EbA measures for greater resilience to projected climate change
178. The proposed measures in the project seek to support more effective water management and an increase in the number of irrigated farmlands. Overall, in terms of recommended proposed improvement measures, the interventions primarily consist of measure for increased water storage within the irrigation network, and the use of water retention ponds for irrigation purposes within the dry season; enhancements to manage flood events by responding to higher return periods in agricultural land flood prevention; and increased protection of farmland against flooding and enabling two harvests per year. Specifically – activities to achieve this fall under the: dredging of canals, maintenance and improvement of existing weirs and diversion structures, heightening embankments and dams, and constructing new retention ponds within the project area. Per the recently-conducted feasibility study undertaken by the RID, in order to adequately regulate water during wet and dry season, four water gates need to be upgraded and three existing canals need to be enlarged to ensure greater climate resilience.

179. As part of the feasibility study for this proposal, RID requested additional assistance from Gesellschaft fuer Internationale Zusammenarbeit (GIZ) to propose EbA approaches for responding to the risks and impacts of the Yom-Nan area, and for integration into the proposed intervention measures included for this proposal. By nature, Ecosystem-based Adaptation measures are responsive and flexible in nature. Given the level of uncertainty in predicting future climate outcomes this is also imperative. Such measures are linked across sectors, operating in a landscape – e.g. agriculture, urban land use, water management, ecosystem settings – and in order to have a future-proof approach should be integrated in all future planning stages, consisting of integrated adaptation measures encompassing all of the relevant sectors. In order to establish such a holistic approach in integrating all the relevant aspects into future-responsive measures, the EbA measures assessed:

- Available modeling for the area: including hydrologic modeling for the hydrological regime and the probabilities of occurrence; hydraulic modeling for water levels, flow velocities, inundation maps and risk zones.
- Ecosystem-based Adaptation as a basis for sustainable climate-responsive measures, using agricultural, biodiversity, urban and water related Ecosystem based Adaptation approaches, on principles of balancing provisional, regulatory and cultural ecosystem services, and a systemic approach to utilize the interplay of all components and their effects/impacts.
- An overall integration of data management and dissemination, the infrastructure development and enhanced operation and capacity building – in developing a set of measures for intervention.

Agricultural extension and community-based trainings on climate smart agricultural practices and climate information
180. Knowledge sharing to extension officers and agricultural households on climate change impacts on agriculture in the Yom and Nan River basins needs to be undertaken that strengthens and works with existing practices and mechanisms. Interventions will support awareness raising, informed adaptation measures and training to communities, such as agriculture practices which are less water-intensive. In addition to allow for effective climate-responsive water management planning to take place – improving climate risk informed planning information will enable farmers to respond and integrate appropriately climate-informed climate smart agricultural practices appropriate to their specific location and assist to build greater climate resilience of the agricultural sector and livelihoods of households within the Yom and Nan River Basin area.

181. While further development of climate adaptive agricultural interventions is to be established through a consultative process during the project phase with farmers in the Yom and Nan River Basin, a number of key activities were identified in the feasibility phase:

- Climate-smart-ecological adaptation processes
- Integrated agriculture practices and training
- Conservation agriculture practices and training
- Support for additional on-farm water storage to assist with drought impacts
- Diversification of crop options appropriate to the locality
- Support in appropriate market development and linkages for farmers of the diversified crops.
- Promotion of interspersing of trees on farms to assist on farm water retention in soil and assisting temperature stabilisation and protection against temperature increases.
- Habitat enhancement for fish: re-stocking of irrigation canals with habitat for fish and non-fish species (increasing of food security options).
- Diversified options with 'back up' of less water-intensive crop choices.

Water Management in terms of agricultural practices

182. The efficient management of water will increase water supplies for the project area in Tung Bang Rakam Floodplain and also be able to help in retaining water flow to the lower Yom River and the Chao Phraya River Basins at a certain level. However, the storage of runoff within the area would definitely increase the water supply but also would have adverse impacts on farmlands and farming patterns. Therefore, it is necessary to set up the appropriate level of water storage in accordance with the cultivating timeline of farmers.

183. During the middle of rainy season with heavy rains and runoff draining from the upper area, all water regulator facilities should be opened in order to drain water out of the area. This is to be prepared for a large volume of runoff in the mid-season till the end of the rainy season. During the end of the rainy season when water level in the Yom River and the flooding water over the lowland subside and get close to the planned storage level, the existing water regulator facilities should be closed to store water then the farming activities can be started (the period around mid-October to late-October). The storage water volume would cover the large area of farmlands in Tung Bang Rakam floodplain at the level of less than 1.20 meters from the ground level. It would be difficult to drain the entire area before the cultivation of the second rice crops and it would cause the loss of storage water. A recommendation for the farmers is that the farming should begin in the area inundated for less than 0.20 meter. The preparations for rice cultivation with transplanting rice seedlings technique are possible to be done in the area inundated with low level of water. Farmers should prepare the farmlands and nurse the seedlings at the same time. The preparation for the seedling nursery should be in small pieces of land equipped with water pumps to drain those submerged pieces of land. Otherwise farmers should pool up to utilize the land on the high ground as their nursery. Two weeks after the preparation activities on the least flooded farmlands (of less than 0.20 meter of inundation), when water from the surroundings would be drawn for the farming activities, flood level is expected to be subsided. Then the preparation works can be done in the next less flooded farmlands and such this step can be continued until covering the entire project area. The more farmland is cultivated, the less pieces of land submerged. Farms located at the lowest elevation level of this planned flood retention area could be used for farming eight weeks after the first plots of farmlands were cultivated.

184. According to a graph of relationship between the volume of water demand and the decreased flood volume, the water demand in the area was found to be higher than the amount of flood water retained by about 58.63 million cubic meters. This shortage amount of water should be delivered from outside sources that would be from the Yom River (possibly flow from the Ping River Basin) and Khlong Bang Kaeo Canal (possibly flow from the Nan River Basin through Khlong Wang Khon, Khlong Mem Canals under operation of the Plai Chum Pol Project). The water flow should be started at the ninth week of the farming season around the end of December till early January. The in-season rice cultivation would start from the mid of April. The rainfed cultivation techniques would be applied. The harvesting should be done before the mid of August when a large amount of runoff would flow into the area.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's [Performance Measurement Frameworks](#) under the [Results Management Framework](#) to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

- ☐ Shift to low-emission sustainable development pathways
- ☒ Increased climate resilient sustainable development

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	N/A t CO ₂ eq
	Lifetime	N/A t CO ₂ eq
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	33,911,323 USD
	(b) Requested GCF amount	17,533,500 USD
	(c) Expected lifetime emission reductions	_____ t CO ₂ eq
	(d) Estimated cost per t CO ₂ eq (d = a / c)	_____ Choose an item. / t CO ₂ eq
	(e) Estimated GCF cost per t CO ₂ eq removed (e = b / c)	_____ Choose an item. / t CO ₂ eq
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	16,377,823 USD
	(g) Public source co-financed	16,263,940 USD
	(h) Private source finance leveraged	113,883 USD
	(i) Total Leverage ratio (i = f / b)	0.934
	(j) Public source co-financing ratio (j = g / b)	0.928
	(k) Private source leverage ratio (k = h / b)	0.006
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	62,000 32,104 (52%) female 29,896 (48%) male
	Indirect	471,561 (population of target districts) 235,781 (50%) female 235,781 (50%) male 25,000,000 (greater Chao Phraya River basin) 12,500,000 (50%) female 12,500,000 (50%) male
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	0.1% of Thailand population 62,000 55% female
	Indirect	1% of Thailand population (indirect beneficiaries in the project area) 235,781 (50%) female 235,781 (50%) male 39% of Thailand population (indirect beneficiaries in the greater Chao Phraya River Basin)

	12,500,000 (50%) female 12,500,000 (50%) male

E.3. Fund-level impacts

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	A 1.1 Change in expected losses of lives and economic assets (US\$) due to the impact of extreme climate-related disasters in the geographic area of the GCF intervention	Gender-disaggregated household surveys undertaken by the project for all households in target area Annual agriculture report (e.g. OAE's Agricultural Statistics of Thailand)	USD 4 million in losses	USD 0.03 million reduction in losses	USD 2 million reduction in losses	Estimated flood damage cost reduction at each return period in case without and with project Annex 3b_Econ Analysis
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	A1.2 Number of males and females benefiting from the adoption of diversified, climate resilient livelihood options (including fisheries, agriculture, tourism, etc.)	Gender-disaggregated household surveys undertaken by the project for all households in target area Annual agriculture report (e.g. OAE's Agricultural Statistics of Thailand)	0	Male = 7,750 Female = 7,750	Male = 29,896 Female = 32,104	Training and capacity building measures will result in change responses for agricultural activities and practices. Provision of climate information tailored to stakeholders' needs will result in climate information being integrated into planning and decision-making of farmers and government stakeholders, and which is climate-risk responsive.
A2.0 Increased resilience of health and well-being, and food and water security	A2.3 Number of males and females with year-round access to reliable and safe water supply despite	Gender-disaggregated household surveys undertaken by the project for all households in target area RID reports - water flows,	0	Male = 7,750 Female = 7,750	Male = 29,896 Female = 32,104	Water infrastructure completion and ongoing maintenance retained to effectively continue supply of water.

	<i>climate shocks and stresses⁶⁶</i>	water retention data at monitoring stations both in targeted area and in downstream areas for changes in flood impacts Annual agriculture report (e.g. OAE's Agricultural Statistics of Thailand)				Water access and climate smart agricultural measures will improve food security outcomes.
<i>A3.0 Increased resilience of infrastructure and the built environment to climate change</i>	<i>A3.1 Number of physical assests made more resilient to climate variability and change, considering human benefits</i>	Site visits RID reports - water flows, water retention data at monitoring stations both in targeted area and in downstream areas for changes in flood impacts	0 Existing infrastructure (canal head regulars along Khlong Muagn Chang and Khlong Ta Kay) has not been upgraded to respond to climate impacts	0 USD 0	2 USD 5 million	Extreme weather does not damage infrastructure works.
<i>A4.0 Improved resilience of ecosystems and ecosystem services.</i>	<i>A4.1 Coverage/scale of ecosystems protected and strengthened in response to climate</i>	Site visits, reports, River Basin Master Plan (ONWR); River Basin Committee meeting report (RBC Secretariat)	0	120 km ²	200 km ²	Assessment identified suitable locations for EbA measures to complement water infrastructure

⁶⁶ Beneficiaries are expected to benefit from both A1.0 and A2.0. Water supply in this case refers to water for agriculture purposes. Precision agriculture supported by the project will define minimum amounts to support productivity.

	<i>variability and change.</i> ⁶⁷					
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E.4. Fund-level outcomes

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term)	Final	
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	<i>A7.1 Use by vulnerable households, communities, businesses and public-sector services of Fund-supported tools instruments, strategies and activities to respond to climate change and variability</i> ⁶⁸	Gender-disaggregated household surveys undertaken by the project for all households in target area.	Household Level = 1 0 credit available to farmers practicing climate resilient farming (co-designed by project)	Household Level = 2 0	Household Level = 4 USD16 million	Climate-information and guidance on on-farm measures and/or crop alternatives are adopted by beneficiary households BAAC commitment to parallel finance through financial product

E.5. Project/programme performance indicators

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
1. Enhance climate and risk informed planning in the water and agricultural sectors through improved climate information and cross sectoral coordination	1.1. Level ⁶⁹ of integration of climate related information into water and agriculture planning 1.2. Number of people trained on integrating climate information into planning	Reports, integration into policy, planning and programming documents	Level = 1 0 staff from TMD, RID and other stakeholders	Level = 2 15 (30% women)	Level = 4 30 (30% women)	Government commitment to climate-informed planning
2. Improve water management through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts	2.1. Climate risk informed upgrades and enhancements to the Yom and Nan river basin area, updating the canal and regulator network to effectively manage water for both flood and drought periods for efficient use.	Total number of assets constructed and/or modified, according to type or upgrade/strengthening	0 Existing infrastructure (canal head regulators/flood gates along Khlong Muang Chang and	3 flood gates covering water distribution area of 120km ²	5 upgrades/modifications, complemented by EbA, covering water distribution area of 200km ²	Co-financing by RID to support infrastructure investment (3 flood gates) and other works Extreme weather does not damage infrastructure works.

⁶⁷ Midterm and Final Targets can be further defined during year 1 once measures are identified/selected

⁶⁸ Level of adoption is based on a scale where 1 = limited to no practice of climate-informed agriculture; Level 2 = households aware of climate impacts on agriculture and know how/where to get related technical support; Level 3 = households engaged with support activities and applying climate information and practices partially; Level 4 = households knowledgeable about climate-resilient practices and fully applying practices

⁶⁹ Level of integration is based on a scale where 1 = limited to no integration; Level 2 = climate information included in planning processes for water or agriculture planning; Level 3 = climate information included in planning processes for both sectors; Level 4 = evidence of collaborative and climate informed planning between water and agriculture sectors

			Khlong Ta Kay Reg) have not been upgraded and designs are not informed by climate information; infrastructure not complemented by EbA measures			Climate information integration into upgrade designs will mitigate potential damage to infrastructure works from flood. Assessment identified suitable locations for EbA measures
3. Reduce volatility of agriculture livelihoods in drought and flood prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets	3.1. Level of action and application of climate information towards climate resilient practices by beneficiary households	Gender-disaggregated household surveys (project randomized control trials); site visits; reports; project monitoring;	0 farm ponds 0 community nurseries in subdistricts 0 Online market platform to increase visibility and marketability of farmers' products	0 0 0	100 22 1	Climate-information is tailored and disseminated for use by farmers Guidance on on-farm measures and/or crop alternatives is provided to farmer through extension services, and is adopted by beneficiary households
	3.2. Number of households with access to finance for climate resilient farming investment	Gender-disaggregated household surveys (project randomized control trials); Annual BAAC reports	0 agriculture households practicing climate resilient agriculture (e.g. on farm EbA measures, water saving technologies/loT) with access to finance (measured as household or head of	10,000	20,000	BAAC commitment to parallel finance through financial product

			household , may also include small enterprise)			
	3.3. Number of listed companies contributing to climate resilient agriculture through contracts to purchase sustainably produced products or investment to support climate resilient agriculture (in-kind/cash contributions)	Project reports, annual co-financing reporting; company reports and PR materials	0	70	210	Continued UNDP support to companies for integration of SDG impact measurement and management into business operations and strategy. UNDP has network of 700 listed companies of the Thailand stock market

E.6. Activities

Activity	Description	Sub-activities	Deliverables
Activity 1.1. Strengthening capacity to generate tailored climate information to inform water management and agriculture planning	Activity 1.1 focuses on development of climate and risk information that is tailored for water and agriculture planning, with related capacity building.	Sub-activities: 1.1.1. Development of Terms of Reference (ToR) for the climate information processing unit mandate 1.1.2. Procurement of hardware and software for enhanced generation of climate and environmental information, and related training 1.1.3. Technical training to 30 staff from TMD, RID and relevant agencies on integration of climate change into water management planning, including consideration of agriculture needs 1.1.4. Development of tailored climate information products to inform planning, and one-to-two week, monthly, seasonal, annual and long-term climate forecast models	Deliverables: - Monitoring equipment - server, flow radar, meteorological station, CCTV, drone, lake water level measurement equipment, groundwater measurement equipment, water level measurement in a cropping plot, soil moisture meter, spare parts for small telemetry system - Capacity building/Training workshops - Technical expertise - Professional services - Tailored climate information products
Activity 1.2. Facilitating inter-ministerial coordination for climate-informed and integrated planning	Activity 1.2 supports inter-agency cooperation for climate-informed, integrated planning.	Sub-activities: 1.2.1. Development of SOPs for sharing of climate information across relevant departments 1.2.2. Workshops and meetings for sensitization and collaboration in sharing of information and identifying opportunity for collaboration and complementarity	Deliverables: - SOPs - Inter-ministerial meetings - Technical expertise - Workshops

Activity 1.3. Expanding access to climate information for application at the household level	Activity 1.3. focuses on making information and bulletins available for climate-informed household decision making. Sub-activities:	Sub-activities: 1.3.1. Development of communication strategy for dissemination of information, including timing and media 1.3.2. Development of information briefs and reports released for circulation in local and social media (dissemination informed by communications strategy) 1.3.3. Development of smartphone and Line app with climate information that can be applied at the agriculture household level 1.3.4. Development of complementary website for interactive information in line with information apps or households	Deliverables: • Technical expertise • Communications products
Activity 2.1. Climate-informed engineering designs for the 13 schemes of the Yom-Nan river basin and upgrade of 2 water infrastructure	Strengthening of water management infrastructure and existing irrigation networks and canals.	Sub-activities: 2.1.1. Technical support to review and enhance existing engineering designs for the 13 schemes of the Yom-Nan river basin area, to ensure adequate integration of climate change 2.1.2. Construction activities to upgrade of 2 canal head regulators along Khlong Muang Chang and Khlong Ta Kay Reg in the Yom-Nan river basin area	Deliverables: • Climate-informed engineering designs for 13 schemes of the Yom-Nan river basin • 2 canal head regulators along Khlong Muang Chang and Khlong Ta Kay Reg in the Yom-Nan river basin area • Technical support • Professional services
Activity 2.2. Complementing of grey infrastructure with EbA measures and integration of EbA approaches into water management policy and planning	Activity 2.2 complements upgrades to grey infrastructure with EbA measures. EbA measures will be informed by a diagnostic analysis which includes vulnerability and risk assessment, cause-impact chains and economic evaluation. The activity will also support climate-sensitive IWRM through capacity building.	Sub-activities: 2.2.1 Conduct of diagnostic analysis incl. vulnerability and risk assessment, modelling, cause-impact chains and economic evaluation 2.2.2. Based on the diagnostic analysis conduct joint prioritization of EbA solutions, finalization of design for three selected EbA measures, including related training and guidance materials 2.2.3. Implementation of identified ecosystem-based adaptation measures 2.2.4. Integration of ecosystem-based adaptation measures into water management rehabilitation plans: soil conservation measures, floodplain development and conservation, river restoration, fish migration passes, and	Deliverables: • Diagnostic analysis incl. vulnerability and risk assessment, modelling, cause-impact chains and economic evaluation • EbA measures applied at 3 sites • Water management rehabilitation plans, which include EbA measures • Lessons learned documents

		longitudinal connection of the Yom River 2.2.5. Documentation of lessons learned in terms of linking local (river basin) level planning and implementation experience with national level policy developments [contribution to the implementation of the NAP/NDC]	
Activity 3.1. Application of climate information in household agriculture planning and strengthening of related support through extension services	Activity 3.1 will ensure that farmers receive technical support through extension services to apply climate information and implement climate resilient measure in agriculture activities.	Sub-activities: 3.1.1. Assist in the development of district level adaptation action plans through technical assistance 3.1.2. Training of 20 provincial agricultural extension and land development officers in the target provinces to undertake climate-risk informed agricultural extension services 3.1.3. Training of agricultural extension and land development officers for the project area in climate-smart agricultural practices and management; including appropriate crops and inclusion of climate risk responsive farm planning 3.1.4. Precision agriculture demonstration and related dissemination of data 3.1.5. Technical assistance to support farmers to apply vulnerability assessment and climate information to agriculture decisions at the household level (e.g. crop choice and crop calendar) 3.1.6. Training through farmer fields schools in each sub-district to complement and support the activities of agricultural services (extension)	Deliverables: • Technical expertise/consultancies for district level adaptation action plans and support to farmers • Capacity building/workshops • Software and hardware related to Climate Resilience and Smart Farming/Precision Agriculture demonstration
Activity 3.2. Implementation of on-farm climate resilient measures to improve drought and flood resilience and improved access to finance for sustainable agriculture	Activity 3.2 will support vulnerable farmers to employ on-farm measures that would improve water access during dry periods. Community nurseries will also be established to support EbA measures of Output 2.2.	Sub-activities: 3.2.1. Technical expertise for development of on-farm EbA guidelines for application by farmers 3.2.2. Technical assistance and investment to support to farmers related to on-farm ponds and enhancing on-farm measures to protect irrigation channels (e.g. vegetating on farm canal buffer zones, planting shade cover areas to minimize evaporation of water along canal ways, and vegetating irrigation	Deliverables: • Technical expertise to support EbA approaches at farmer household level • On-farm measures (incl 100 on farms ponds) and community nurseries in 22 sub-districts, and small scale equipment for farmers • Financial instrument available through BAAC to support farmer investments in climate resilient agriculture

		infrastructure for flood protection) 3.2.3. Procurement of small scale equipment for farmers to support water conservation and soil management 3.2.4. Establishment of community nurseries to provide alternative livelihood income generation during drought-flood/non-farming periods, and to cultivate and supply climate smart crops, vegetables, and vegetation species for EbA measures implemented in Output 2 3.2.5. Co-design of financial instrument with BAAC to support farmers with adaptation investments towards climate resilient agriculture	
Activity 3.3. Capacity building for farmers to support market access for climate resilient agriculture products	Activity 3.3 focuses on the critical knowledge sharing and knowledge transfer to support market access for climate resilient agriculture products. With co-financing support from Krungsri Bank, farmers will also improve financial literacy so that they can better avail themselves of available markets and financial mechanisms for their products.	Sub-activities: 3.3.1. Farmer to farmer knowledge sharing workshops, including reciprocal learning processes (e.g. site visits by farmer field schools to project schemes) 3.3.2. Establishment of online market platform to increase visibility and marketability of farmers' products 3.3.3. Training to farmers to support market access for farmers, including capacity building related to financial literacy, linkages to buyers and markets, requirements for contract farming (incl. hospitals, schools, restaurants), linkages through rice cooperatives, linkages to local gatherer and processors, linkages to private sector and community support agriculture (CSA) for larger markets at Phitsanulok province, and comprising of larger market buyers, processor, cooperative	Deliverables: • Workshops/capacity building • Online agriculture product market platform

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

185. Project-level monitoring and evaluation will be undertaken in compliance with the [UNDP POPP](#) and the [UNDP Evaluation Policy](#).

Oversight and monitoring responsibilities:

186. The primary responsibility for day-to-day project monitoring and implementation rests with the Project Manager. The Project Manager will develop annual work plans to ensure the efficient implementation of the project. The Project Manager will inform the Project Board and the UNDP Country Office of any delays or difficulties during implementation, including the implementation of the Monitoring & Evaluation (M&E) plan, so that the appropriate support and corrective

measures can be adopted. The Project Manager will also ensure that all project staff maintain a high level of transparency, responsibility and accountability in monitoring and reporting project results.

187. The UNDP Country Office will support the Project Manager as needed, including through annual supervision missions. The UNDP Country Office is responsible for complying with UNDP project-level M&E requirements as outlined in the [UNDP POPP](#). Additional M&E, implementation quality assurance, and troubleshooting support will be provided by the UNDP Regional Technical Advisor as needed. The project target groups and stakeholders including the NDA Focal Point will be involved as much as possible in project-level M&E.

188. A project inception workshop will be held after the UNDP project document is signed by all relevant parties to: a) re-orient project stakeholders to the project strategy and discuss any changes in the overall context that influence project implementation; b) discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms; c) review the results framework, re-assess baselines as needed, and discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E plan; d) review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; e) plan and schedule Project Board meetings and finalize the first year annual work plan. The Project Manager will prepare the inception report no later than one month after the inception workshop. The final inception report will be cleared by the UNDP Country Office and the UNDP Regional Technical Advisor, and will be approved by the Project Board.

189. The Project Manager, the UNDP Country Office, and the UNDP Regional Technical Advisor will provide objective input to the Annual Project Report (APR) for each year of project implementation. The Project Manager will ensure that the indicators included in the project results framework are monitored annually well in advance of the APR submission deadline and will objectively report progress in the APR. The annual APR will be shared with the Project Board and other stakeholders. The UNDP Country Office will coordinate the input of the NDA Focal Point and other stakeholders to the APR. The quality rating of the previous year's APR will be used to inform the preparation of the next APR. The final project APR, along with the terminal evaluation report and corresponding management response, will serve as the final project report package.

190. An independent mid-term review process will be undertaken, and the findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the project's duration. The terms of reference, the review process and the final MTR report will follow the standard templates and guidance available on the [UNDP Evaluation Resource Center](#). The final MTR report will be cleared by the UNDP Country Office and the UNDP Regional Technical Advisor, and will be approved by the Project Board. The final MTR report will be available in English.

GCF evaluation requirements:

191. An independent terminal evaluation (TE) will take place no later than three months prior to operational closure of the project. The terms of reference, the review process and the final TE report will follow the standard templates and guidance available on the [UNDP Evaluation Resource Center](#). The final TE report will be cleared by the UNDP Country Office and the UNDP Regional Technical Advisor, and will be approved by the Project Board. The TE report will be available in English.

192. The UNDP Country Office will include the planned project terminal evaluation in the UNDP Country Office evaluation plan, and will upload the final terminal evaluation report in English and the management response to the public UNDP Evaluation Resource Centre (ERC) (erc.undp.org). The MTR and TE will be carried out by an independent evaluator. The evaluation report prepared by the independent evaluator is then quality assessed and rated by the UNDP Independent Evaluation Office.

193. The UNDP Country Office will retain all M&E records for this project for up to seven years after project financial closure in order to support ex-post evaluations. A detailed M&E budget, monitoring plan and evaluation plan will be included in the UNDP project document.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

A number of risks of varied probability have been identified for the project, as set out in detail at G2. These risks relate to project sustainability in terms of infrastructure maintenance in the long term, environmental risks to project investments, limited institutional capacity, project beneficiary target resistance/non-adoption, project timeline disruptions due extreme events (natural disasters, political events).

Selected Risk Factor 1

Category	Probability	Impact
Technical and operational	Low	High
Description		
Infrastructure investments (i.e. flood gates) are not maintained and do not serve intended purpose		
Mitigation Measure(s)		
Maintenance of the investments fall under the mandate of the executing entity, RID. Under its mandate, RID is responsible for maintaining water infrastructure. Co-financing from RID includes O&M during project implementation, and RID has committed resources for post-project O&M. The risk is therefore considered low probability.		

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Environmental risks associated with water infrastructure investments		
Mitigation Measure(s)		
Measures will be taken to minimize any sediment entering waterways as a result of earthworks (e.g. bioengineering around the canals, upgrading of flood gates). Risks are expected to be 'moderate', a safeguards plan has been developed and related costs have been integrated into the project budget. (Please see Annex 6(b))		

Selected Risk Factor 3

Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Project interventions do not have intended impacts, because behavior that is not conducive to the project objective continues (e.g. crop choice/calendar or related water use not informed by climate change, extraction of groundwater continues, etc.).		
Mitigation Measure(s)		
Project interventions are expected to reduce volatility of agriculture in the project area, through improved water management and the provision of climate information, related support to apply that information in planning, and support to beneficiaries through capacity building and sensitization to support transformational change. Climate data will be integrated into water planning and management, and related training provided to government staff - ensuring informed adaptation to changing climatic conditions and increasing climate risks in Thailand.		
Communities will be engaged throughout project implementation. Technical expertise will be provided through the project to ensure communities and district government staff have access to technical advice. Through regular monitoring, success of interventions will be measured and communicated to communities to provide assurance.		

Selected Risk Factor 4

Category	Probability	Impact
Technical and operational	Low	Low
Description		

Staff turnover or lack of technical capacity within executing entity		
Mitigation Measure(s)		
A capacity needs assessment has been undertaken, and RID was assessed as low risk across major categories. RID has capacity to implement the project. The project follows UNDP's National Implementation Modality (NIM), with support provided by UNDP for specific activities agreed between UNDP and RID.		
Selected Risk Factor 5		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Extreme event disrupts implementation or damages investments, resulting in delays and/or additional costs. COVID-19 and COVID-19 related restrictions delay implementation.		
Mitigation Measure(s)		
Related to extreme events, timing of activities during implementation will be scheduled to minimize risk particularly around the rainy season. The project team, with support from UNDP Thailand and national partners, will also ensure implementation follows national instructions related to containing the spread of COVID-19. Where possible the project will employ virtual methods to conduct consultations and training. Where necessary, timing of activities particularly if in a vulnerable area, may have to be reviewed and the timing adjusted to ensure compliance to government restrictions, and the safety of beneficiaries, partners and the project team.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

194. The project has the potential to cause moderate environmental and social impacts, and has been assessed as a Category B project. These include impacts to water quality through sediment movement during the replacement of existing infrastructure, channel and retention basin works and ecosystem-based adaptation (EbA) approaches. Noise and air quality may also be impacted during these works. Appropriate and relevant avoidance and mitigation options have been proposed in the ESMF, which if put in place, will significantly reduce the potential impacts of the project to an acceptable level. The mitigation measures include the development of site-specific erosion, drainage and sediment control plans that will be developed and implemented as a mitigation measure. The plans will include disposal plans for soil and silt removed during earthworks, ensuring that the surrounding communities and environment are not adversely impacted. Where possible, the project will undertake beneficial reuse of any excess sediment. Minor impacts include increased waste. Measures to avoid this impact include the prefabrication of materials, including for example, the water control infrastructure to reduce this risk of waste being generated.

195. The project does not require any land acquisition and/or resettlement. None of the interventions will require the displacement of people, involve economic displacement or will be conducted in protected areas or sensitive locations.

196. Thailand currently has an established regime for grievances. Notwithstanding, an independent two-tier Grievance Redress Mechanism structure was developed to address all complaints and/or grievances in the project. This Grievance Redress Mechanism complies with Thailand and UNDP Safeguard procedures.

197. Budgeting for environmental interventions and the application of risk mitigation measures to enhance positive impacts within the main river basins of Thailand is an investment in the future as it will reduce the environmental and social liability at local, regional and national levels.

198. Please see Annex 6 for further details.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

199. While the impacts of climate change affect the agricultural sectors and the country, climate change and its impacts are not gender neutral. Inequitable access to and control over natural resources, particularly land, water, and forests, as well as exclusion from the policy- and decision-making processes that affect resource distribution and their livelihood. These disproportionately affect the rural poor, who rely on natural resources for the livelihood.

200. While there have been significant advances in general for women in Thailand, the potentially growing number of women-headed households in agriculture require particular attention to their needs and challenges in the context of climate change.

201. The project is aligned with national gender equality and women's empowerment related policies and programmes as well as international commitments under the Paris Agreement, CEDAW, SDGs, etc. The project will aim to support transformational change towards gender equality and women's empowerment and will:

- Seek to understand in greater depth the potentially gender-differentiated knowledge, experience, and skills of women and men at sub-district, district and provincial levels as well as their needs, challenges, and vulnerabilities (e.g. around access to resources, decision-making, etc.) in order to strengthen project interventions including information sharing between implementers and beneficiaries, decision-making around adaptation at community and household levels, production, post-harvest, and market development decisions, etc. This can be done through introducing/piloting approaches such as gender-responsive participatory Integrated Climate Services for Agriculture (PICSA) and piloting and scaling up household methodologies (HHM).^{70 71} Several organizations have championed these approaches which have been applied in several in different regions to strengthen agricultural livelihoods, production, and value chain/market development, etc.

⁷⁰ For more about PICSA approaches, see, for example <https://ccafs.cgiar.org/participatory-integrated-climate-services-agriculture-picsa#.XOIq5Y9IDIU>

⁷¹ For more about Household Methodologies (HHM), see, for example

<https://www.ifad.org/documents/38714170/40198517/Household+methodologie+-+harnessing+the+family%27s+potential+for+change.pdf/cb0ab278-bfb4-4b4c-a237-e7841bc9e9aa>

For more on Household Methodologies, particularly the Gender Action Learning Systems (GALS) approach, See Mayoux, L. 2017. Gender Action Learning for Sustainability at Scale: Pilot in Kyrgyzstan, May – Nov 2016 https://gamechangenetwork.org/wp-content/uploads/2016/05/GALS-Process-for-RWEE-Naryn-Pilot-ConsolidatedReport_Feb2017_ed.pdf

- Ensure capacity building includes the active participation of women and men and incorporates, where relevant, issues around gender and climate information services, market development, farmer engagement, irrigation, use of innovative technology, etc.
- Mitigate any negative impacts on rural women and girls (e.g. in terms of benefit sharing, labor division of labor, access to resources, access to technology and skills development, etc.) Where possible, enhance role of new generation, both female and male youth (aged 18-25) to build a new generation of farmers equipped with knowledge and skills to build climate resilience in their community.
- Include a Gender Specialist on the project staff to support data collection and mainstreaming of gender issues (and where relevant, strategic approaches targeted specifically at women/girls across all areas of the project;
- Link with institutions who can provide training to agricultural extension workers on uptake of household methodologies (e.g. Gender Action Learning Systems or GALS) in their work with women and men in vulnerable rural households and communities to complement, strengthen, and leverage farmer to farmer sharing of climate information, Value Chain (VC) and market access opportunities
- Build/strengthen linkages with institutions/networks that can strengthen gender approaches throughout the project (e.g. academic institute, agriculture and other relevant research centres, etc.). The project should also consider linking to existing women's groups, housewife groups, credit groups.
- Build data collection and management in the process of capacity development at all levels. Technical support with integrated gender and social inclusion for each project/program.
- Ensure that good practices and lessons learned from the project around integrating participatory, gender transformative approaches in climate adaptation are documented and shared among stakeholders including line ministries, policymakers, and government/non-government decision-makers/planners/implementers at national and sub-national level and beyond (e.g. regional/international). This may also contribute to reporting on international commitments such as NDCs, etc.

202. Please see gender assessment and action plan (Annex 8) for further details.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

203. The financial management and procurement of this project will follow UNDP financial rules and regulations available [here](#)⁷².

204. Further guidance is outlined in the financial resources management section of the UNDP Programme and Operations Policies and Procedures available [here](#).

205. UNDP has comprehensive procurement policies in place as outlined in the 'Contracts and Procurement' section of UNDP's Programme and Operations Policies and Procedures (POPP). The policies outline formal procurement standards and guidelines across each phase of the procurement process, and they apply to all procurements in UNDP. See [here](#).

206. The project will be implemented following the National Implementation Modality (NIM) following NIM guidelines available [here](#)⁷³. UNDP will ascertain the national capacities of the Implementing Partner/Executing Entity by undertaking an evaluation of capacity following the Framework for Cash Transfers to Implementing Partners/Executing Entities (part of the Harmonized Approach to Cash Transfers - [HACT](#)).

For more on adapting GALS in agricultural extension, Value Chain (VC) development and Village Savings and Loans Associations (VSLA), see PELUM. 2016. Models of Integrating GALS in Functional Adult Literacy, Agricultural Extension, Advocacy, Village Savings and Lending Associations https://uganda.oxfam.org/sites/uganda.oxfam.org/files/file_attachments/Adapting-GALS-in-development-programmes-FINAL.pdf

⁷² In line with UNDP National Implementation (NIM) Guidelines, the Government is responsible for the management and procurement of the Project to achieve project outputs. Government regulations, rules and procedures therefore apply to project implementation to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. UNDP Financial Regulations section 16.05 state:

- "The administration by executing entities or, under the harmonized operational modalities, implementing partners, of resources obtained from or through UNDP shall be carried out under their respective financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP.

- Where the financial governance of an executing entity or, under the harmonized operational modalities, implementing partner, does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition that of UNDP shall apply."

⁷³ In line with UNDP NIM Guidelines, the Project will be implemented following Government regulations, rules and procedures. The Government will ensure that their respective procedures do not contravene the principles of UNDP Financial Rules and Regulations.

207. All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies⁷⁴.

208. The NIM Guidelines are a formal part of UNDP's policies and procedures, as set out in the UNDP Programme and Operations Policies and Procedures (POPP). The NIM Guidelines were corporately developed and adopted by UNDP, and are fully compliant with UNDP's procurement and financial management rules and regulations.

209. The national executing entity is MOAC - also referred to as the national 'Implementing Partner' in UNDP terminology - is required to implement the project in compliance with UNDP rules and regulations, policies and procedures (including the NIM Guidelines). In legal terms, this is ensured through the national Government's signature of the Special Fund to Thailand, together with a UNDP project document which will be signed by the Implementing Partner to govern the use of the funds. Both of these documents require compliance. Prior to signature of the project document, all national Implementing Partners need to have undergone a Harmonized Approach to Cash Transfer (HACT) assessment by UNDP to assess capacities to implement the project. During implementation, UNDP will provide oversight and quality assurance in accordance with its policies and procedures, and any specific requirements in the Accreditation Master Agreement (AMA) and project confirmation to be agreed with the GCF. This may include, but is not limited to, monitoring missions, spot checks, facilitation and participation in project board meetings, quarterly progress and annual implementation reviews, and audits at project level or at Implementing partner level on the resources received from UNDP.

210. The Harmonized Approach to Cash Transfer (HACT) framework consists of four processes: (1) macro assessments; (2) micro assessments; (3) cash transfers and disbursements; and (4) assurance activities. Assurance activities include planning, periodic on-site reviews (spot checks), programmatic monitoring, scheduled audits and special audits. During micro-assessment, there can weaknesses identified for which actions are required to addresses the gaps. When a spot check finds that the gaps are not addressed it will mean that the level of assurance activities will have to remain higher and modalities of engaging with that implementing partner will have to be reviewed if necessary. All details are available [here](#).

211. The project will be audited in accordance with UNDP policies and procedures on audits, informed by and together with any specific requirements agreed in the AMA. According to the current audit policies, UNDP will be appointing the auditors. In UNDP scheduled audits are performed during the programme cycle as per UNDP assurance/audit plans, on the basis of the implementing partner's risk rating and UNDP's guidelines. A scheduled audit is used to determine whether the funds transferred to the implementing partner were used for the appropriate purpose and in accordance with the work plan. A scheduled audit can consist of a financial audit or an internal control audit⁷⁵.

212. GCF resources will be provided to the implementing partner, less any agreed cost recovery amount. Under UNDP's national implementation modality, UNDP advances cash funds on a quarterly basis to the implementing partner (executing entity) for the implementation of agreed and approved programme activities, in accordance with UNDP standard policies and the NIM Guidelines. The implementing partner reports back expenditure via a financial report on quarterly basis to UNDP. Any additional requirements will be as in accordance with the AMA as and when it is agreed.

G.4. Disclosure of funding proposal

☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

⁷⁴ In line with UNDP NIM Guidelines, and in accordance with UNDP Policies and Procedures on audits, the Project shall be audited in accordance with HACT audit framework whereby the project can assign the national Supreme Audit Institution (SAI) to conduct financial audits.

⁷⁵ In line with UNDP NIM Guidelines, and in accordance with UNDP Policies and Procedures on audits, the Project shall be audited in accordance with HACT audit framework whereby the project can assign the national Supreme Audit Institution (SAI) to conduct financial audits.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- | | | |
|--------------------------|-------------|---|
| ☐ | Annex 1 | NDA no-objection letter |
| ☐ | Annex 2 | Feasibility study |
| ☐ | Annex 3 (a) | Economic analysis |
| ☐ | Annex 3 (b) | Economic analysis (excel) |
| ☐ | Annex 3 (c) | Financial analysis |
| ☐ | Annex 4 | Detailed budget plan |
| ☐ | Annex 5 | Implementation timetable including key project/programme milestones |
| ☐ | Annex 6 (a) | Social and environmental screening procedure (SESP) |
| ☐ | Annex 6 (b) | Environmental and social management framework (ESMF) |
| ☐ | Annex 6 (c) | Environmental and Social Safeguards (ESS) report |
| ☐ | Annex 7 (a) | Stakeholder consultation report |
| ☐ | Annex 7 (b) | Stakeholder engagement plan |
| ☐ | Annex 8 | Gender assessment and action plan |
| ☐ | Annex 9 | Legal due diligence (regulation, taxation and insurance) N/A |
| ☐ | Annex 10 | Procurement and HR plans |
| ☐ | Annex 11 | Monitoring and evaluation plan |
| ☐ | Annex 12 | AE fee request |
| ☐ | Annex 13 | Co-financing commitment letters |
| ☐ | Annex 14 | Term sheet |

H.2. Other annexes as applicable

- | | | |
|--------------------------|--------------|--|
| ☐ | Annex 15 | Evidence of internal approval |
| ☐ | Annex 16 | Maps indicating the location of proposed interventions |
| ☐ | Annex 17 | Multi-country project/programme information N/A |
| ☐ | Annex 18 (a) | Appraisal report |
| ☐ | Annex 18 (b) | Evaluation of the baseline project |
| ☐ | Annex 19 | Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity N/A |
| ☐ | Annex 20a | Partner Capacity Assessment Tool (PCAT) |
| ☐ | Annex 20b | UNDP micro-assessment reports |
| ☐ | Annex 21 | Operations and maintenance plan |
| ☐ | Annex 22 | Yom-Nan operation and maintenance project |
| ☐ | Annex 23 | Survey of climate change impact on community livelihoods, the Yom-Nan irrigation project |
| ☐ | Annex 24 (a) | Responses to GCF comments on Concept Note |
| ☐ | Annex 24 (b) | Responses to GCF comments on Proposal |
| ☐ | Annex 25 | (Draft) LOA on UNDP support services |

- | | | |
|--------------------------|----------|---|
| ☐ | Annex 26 | Climate Change in Chao Phraya Basin |
| ☐ | | OAG Letter to RID on LOA |
| ☐ | | Full package of UNDP responses to iTAP questions and comments |
| ☐ | | UNDP responses to iTAP Assessments |

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)

No. 1006.5/2295



Ministry of Natural Resources and Environment
92 Soi Phahol Yothin 7, Phahol Yothin Road
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Thailand
Tel. +662 265 6500 Ext. 6849
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12 October B.E. 2560 (2017)

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by United Nations Development Programme (UNDP) regarding Enhancing Climate Resilience in Thailand through Effective Water Management and Sustainable Agriculture

We refer to the project "Enhancing Climate Resilience in Thailand through Effective Water Management and Sustainable Agriculture." in Thailand as included in the funding proposal submitted by United Nations Development Programme (UNDP) to us on May 31, 2017.

The undersigned is the duly authorized representative of the National Designated Authority of the Kingdom of Thailand, Dr. Wijarn Simachaya, Permanent Secretary, Ministry of Natural Resources and Environment.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Kingdom of Thailand has no-objection to the project, as included in the funding proposal;

(b)...

- (b) The project, as included in the funding proposal, is in conformity with national priorities under Thailand Country Programme (TCP), climate change policies, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project, as included in the funding proposal, is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project, as included in the funding proposal, has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours sincerely,

W. Simeach

Permanent Secretary

Ministry of Natural Resources and Environment

Green Climate Fund Board Secretariat
Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea
Email: info@gcfund.org
Tel: +82.32.458.6059 (KST)

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Enhancing Climate Resilience in Thailand through Effective Water Management and Sustainable Agriculture
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	United Nations Development Programme (UNDP)
Environmental and social safeguards (ESS) category	Category B
Location - specific location(s) of project or target country or location(s) of programme	Phitsanulok, Sukhothai, and Uttaradit provinces of Thailand
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, September 2, 2021
Language(s) of disclosure	English and Thai
Explanation on language	Thai is the official language of Thailand and understandable to the stakeholders.
Link to disclosure	English and Thai The Environmental and Social Management Framework (ESMF) UNDP in Thailand
Other link(s)	Royal Irrigation Department Website (English and Thai): CC-GCF (rid.go.th)
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)" (Annex 6a of the FP).
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, September 2, 2021
Language(s) of disclosure	English and Thai
Explanation on language	Thai is the official language of Thailand and understandable to the stakeholders.
Link to disclosure	English and Thai The Environmental and Social Management Framework (ESMF) UNDP in Thailand
Other link(s)	Royal Irrigation Department Website (English and Thai): CC-GCF (rid.go.th)
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)" (Annex 6a of the FP).
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A

Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	Social and Environmental Safeguards Procedure (SESP)/ Thursday, September 2, 2021
Language(s) of disclosure	English and Thai
Explanation on language	Thai is the official language of Thailand
Link to disclosure	English and Thai Social and Environmental Screening Procedure (SESP) UNDP in Thailand
Other link(s)	Royal Irrigation Department Website (English and Thai): CC-GCF (rid.go.th)
Remarks	The SESP is a UNDP document that is used for identifying social and environmental risks and mitigation measures. It does not replace or conflict with GCF ESS requirements and policies.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Thursday, September 2, 2021
Place	Yom Nan Operation and Maintenance Project Regional Irrigation Office 3 8 moo 8 Thathong Muang, Phitsanulok 65000
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 4, 2021

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP170

Proposal name:	Enhancing climate resilience in Thailand through effective water management and sustainable agriculture
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Thailand
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project contributes to partial protection of downstream areas, including vulnerable farmers and downstream urban areas in Bangkok, from increasingly frequent and intense flooding via the integration of 'green with grey' infrastructure leads to enhanced flood prevention and more resilient ecosystems and livelihoods.	The Ecosystem Based Adaptation approaches for flood prevention require extensive participatory planning to be undertaken with communities, regional government, and experts. This has been integrated into the workplan and identified as important implementation stage.
Thailand is a country regarded as in having amongst the most severe risks and vulnerabilities from climate change ¹ . The project addresses issues pertinent to the country and regarded as high priority within the National Action Plan. The project is the first project to the Green Climate Fund for Thailand and has high level support from the Thai government. This will facilitate potential upscale of the models, and development of grey-green infrastructure models, innovative technology and supporting communities in adaptive livelihood development (through parallel finance credit line).	There should be a plan for further upscale across Thailand's watersheds and downstream areas developed during the project utilising this strong national support.
The project includes significant financial contribution from the Royal Thai Government and Krungsri Bank and offers potential of a credit line from parallel finance for farmers from the Bank of Agriculture and Agricultural Cooperatives (BAAC) which has potential for transformative upscale of climate smart agriculture.	The co-finance from financial institution needs to support smallholder farmers and small enterprise to upscale the adoption of climate smart agriculture. An implementation plan to upscale activities should be developed during implementation.

¹ Eckstein et al (2021). Global Climate Risk Index. Germanwatch.

Strengths	Points of caution
The project demonstrates an innovative pilot to implement best practice in water use efficiency (e.g., Alternate Wet Dry / Systems Rice Intensification) and innovative technology in precision agriculture linked to sensors reporting on soil moisture levels to provide information to supporting effective water level change through efficient management of the irrigation system as has been demonstrated by International Rice Research Institute. This cultivation technique can reduce GHG emissions by up to 50% and water use by 30%; digital technology assists in its effective application at site level. This is highly innovative and is in line with Thailand's strategic policies to support agriculture, digital transformation, and climate change; in line with Thailand's Digital Agriculture Strategy and 4.0 New Economic Development Plan which aims to harness digital technologies to make Thailand more competitive, manage resources efficient, improve livelihoods and adaptative to climate change. Thailand is well-placed to support development of digital innovation into the agricultural sector.	There is need to ensure strong community involvement throughout project implementation, so that farmer behavior that is not climate resilient (e.g., unsuitable crop choice or farming calendar) is discontinued. The improved rice cultivation methodologies require efficient management of irrigation system to utilize water effectively and requires cooperation between farmers utilizing the same irrigation source. Developing, testing and scaling-up innovative technologies should be a priority for the Green Climate Fund. They should not be evaluated under the same criteria for 'cost-effectiveness' as well-tested business-as-usual applications since the future values of the technology is not yet incorporated into the model. GCF can help support and facilitate the innovation transition from innovation to adoption.
The project demonstrates best practice in water use efficiency (e.g. Systems Rice Intensification) and innovative technology in precision agriculture linked to sensors reporting on soil moisture levels to provide information to supporting effective water level change through efficient management of the irrigation system. This has potential to be upscaled and contribute a paradigm shift for reducing water use, greenhouse gas emissions.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVIII, titled "List of proposed conditions and recommendations," respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Agriculture provides one of the most important sources of income and employment for Thailand; Thailand is the 6th largest producer of rice and 2nd for global exports. Climate change-induced shifts in precipitation and temperature patterns have enhanced flood and drought risk, increasing the intensity and frequency of both. Increased flood risk translates into accelerated soil erosion and impeded drainage, affecting agricultural productivity. This negatively impacts smallholder farmers who depend on agriculture for their livelihoods. The climate analysis clearly shows the following:

- (a) **Increased drought risk** – The frequency of extreme SPEI has increased, supporting an increase in drought.
 - (b) **Increased risk of extreme rainfall** – Historical station data show spatially consistent positive trends in annual rainfall and measures of daily rainfall intensity. Ensemble projections show clearly future projected increases of about 10-15% in annual rainfall.
 - (c) **Increased discharge** – Significant increases (more than 50%) in discharge from an ensemble of RCMs and hydrological model are consistent with the +10% in annual precipitation since the catchments amplify the rainfall signal.
4. The Chao Phraya River Basin Area is one of two principal river systems in Thailand. It consists of eight sub river basins: Yom, Nan, Ping, Wang, Chao Phraya, Sakae Krang, Pa Sak and Tha Chin. Working towards river basin-based integrated water management, the project integrates ecosystem-based adaptation (EbA) 'green' measures with 'traditional' grey infrastructure for flood control and irrigation, supported by enhanced technology and climate-informed planning and capacity development. The objective of the project is to improve water management, food security, and agricultural livelihoods of inhabitants in the Yom and Nam river basins to respond to the increased occurrence of climate hazards in the form of floods and droughts. Activities at the household scale are intended to directly benefit 62,000 people in the Phitsanulok, Sukhothai, and Uttaradit provinces (project sites) found in the Yom and Nan river basins, and at the basin scale to indirectly benefit 25,000,000 inhabitants of the Greater Chao Phraya River Basin, who live within or downstream of the project area. The country's national adaptation plan was developed by its country's leading scientists and climate change experts. The Thailand National Adaptation Plan recognises, and prioritizes, the importance of climate change adaptation interventions within the Chao Phraya River Basin for downstream communities.
5. The project seeks a GCF grant of USD 17.534 million, or 51.7 per cent of the total project cost (USD 33.911 million). The project is classified as environmental and social safeguards (ESS) category B. The Royal Irrigation Department of Thailand (RID) will co-finance the project at USD 16.264 million during the years of implementation, and Bank of Ayudhya (Krungsri) will co-finance the project at USD 0.114 million for supportive activities such as financial literacy training for farmers. RID has also allocated USD 17.26 million for post-project operations and maintenance.
6. **Significant parallel financing** is provided by the Bank for Agriculture and Agricultural Cooperatives (BAAC) via their financial products aimed at climate resilient agricultural investments. This is pledged as a credit line of USD 16 million to farmers, cooperatives, social enterprises, and other small to medium-sized businesses nationwide.

2.2 Component-by-component analysis

Output 1: Enhance climate and risk informed planning in the water and agricultural sectors through improved climate information and cross sectoral coordination (total cost: USD 3.215 million; GCF cost: USD 1.683 million)

7. GCF resources will support the improvement of climate risk-responsive decision-making in Thailand by enhancing quality and availability of tailored climate information for both water management and agricultural planning. This will be achieved through improving the accuracy and precision of climate projections and the dissemination of tailored climate information products to relevant stakeholders, to support the integration into policies and planning across sectors, as well as comparison of water availability and demand, in order to make the needed adjustments.

8. This output will also enhance inter-ministerial coordination in water management, enabling stronger collaboration between key decision-making bodies in Thailand's water and agriculture sectors to incorporate climate risk management and adaptation strategies into their sectoral planning processes. Importantly, tailored climate information will be generated and disseminated to agriculture households for application in household-level planning.

Output 2: Improve water management through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts (total cost: USD 23.125 million; GCF cost: USD 9.640 million)

9. This output focuses on complementing RID investments to improve water management in the Yom and Nan river basins area with ecosystem-based adaptation measures. GCF's contribution will fund the review of designs for water infrastructure upgrades at the thirteen irrigation schemes in the Yom-Nan basin, to ensure that they are adequate given future climate projections. Given the high upfront costs of the basin-wide investment required to adapt to climate change, this will support long term sustainability of government investment. GCF will also complement government co-financing and planned long-term investment by RID in the entire area of the Yom and Nan river basins by supporting the upgrade of infrastructure at two of the schemes.

10. Water management in Thailand is largely dependent on grey infrastructure, although there are examples where use of ecosystem-based measures has been established. GCF resources will support RID in identifying and implementing ecosystem-based approaches to complement grey infrastructure. While ecosystem-based measures alone cannot address all climate risk challenges, complementing grey infrastructure with ecosystem-based solutions can be a cost-efficient solution while also generating environmental co-benefits. This integration of EbA approaches has the potential to be transformational to Thailand's traditional planning concepts for water management that focus widely on hard structure interventions, through upscaling across the thirteen schemes in the Yom-Nan River Basin area and the greater Chao Phraya River Basin. The project has significant potential for upscale within the basin and to other basins in Thailand, the elevated level of support from the Thai government will facilitate this upscale.

Output 3: Reduce volatility of agriculture livelihoods in drought and flood-prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets (total cost: USD 5.978 million; GCF cost: USD 5.617 million)

11. This output will involve providing training to facilitate climate-informed decision-making related to agriculture practices, particularly implementation of on-farm interventions for farmers in the immediate project area. Training materials developed through this project will thereafter be applied throughout the agriculture sector in Thailand through farmer field training (FFT). The project area is dominated by agriculture activities, which are particularly vulnerable to climatic conditions. It is critical therefore that interventions in the project area engage farmers effectively.

12. This output will be implemented in partnership and coordination with King Mongkut's University of Technology North Bangkok (KMUTNB) and Krungsri Bank. KMUTNB will help to deliver training to farmers on applying water forecast based on climate information to household level agriculture practices, and Krungsri Bank, through co-financing activities, will support farmers in developing their business skills for greater market access. Trainings under this output will be coordinated for cost-saving purposes, but importantly also to ensure that extension officers and farmers benefit from the collaborative approach of the project – bringing together expertise from the water and agriculture departments, academia, and the financial sector, as well as the farmers themselves.

Project management costs, total: USD 1.593 million; GCF: USD 0.593 million.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

13. Thailand is identified as one of the most vulnerable countries to climate change risks and impacts due to geographical and topographical characteristics. Climate induced extreme weather events leading to increased flood and drought risk are reported as becoming more frequent and intense. Climate analysis shows significant increase in temperature, significant increase in precipitation and significant increase in water discharge.

14. This project directly targets existing water infrastructure (output 2) and the related governing institutions which need improvement, and communities which need further support (output 1 and 3). Stemming from a careful consideration of other subriver basins, the selection of the Yom and Nan river basins for infrastructure update is of particularly strategic importance as they retain the excess water flowing through Thailand's central plain, Bangkok, and Gulf of Thailand.

15. Given local social and economic conditions, it is appropriate that farming households are the focus of the project. The climate resilience of 62,000 project participants (direct beneficiaries) in the Yom Nan River Basin and 25,000,000 (indirect beneficiaries) in the Greater Chao Phraya River system are expected to be improved by the interventions, as described in Section 2. The project selects agricultural communities in Phitsanulok, Sukhothai, and Uttaradit provinces (project sites) that depend on rice production as their main source of income. One of the main interventions of the project is the focus on capacity development. Increased community adaptive capacity is needed to implement more climate resilient practices for sustainable livelihoods (outputs 1 and 3).

16. In summary, complementing ongoing work initiated by the Royal Thai Government, the project for climate-resilient water infrastructure is expected to result in enhanced livelihoods for Thailand's most vulnerable people. Enhanced climate and risk decision-making and support services are not only provided to households, but also include ministries involved in water management and agricultural planning, and organizations across the relevant value chains. The focus on agricultural livelihoods and water management is justified given Thailand's development needs and goals.

3.2 Paradigm shift potential

Scale: N/A

17. The project aims to contribute to the shift from 'traditional' water management systems to an integrated system with the eventual careful inclusion of EbA. The inclusion of EbA with grey infrastructure has been shown to be less costly and promotes resilience of ecosystems and livelihoods by conserving and protecting existing ecosystem services (e.g., flood regulation, soil retention). The project is part of an overall strategy of the Royal Thai Government to support canal and embankment improvements of the Greater Chao Phraya River system coupled with improved extension services (i.e., soil rehabilitation, climate vulnerability assessments), focusing on low-lying areas.

18. The major constraint facing farming communities lies in 1) inefficient water management; and 2) barriers to market access, both of which are tackled in this proposal. Improved information and capacity development will lead to better decision-making, especially as the project also seeks to improve management practices and market linkages. There is potential for a more profitable and climate resilient agricultural sector for project participants. This is underpinned by a series of proposed interventions seeking to build a more robust and conducive policy and institutional environment via knowledge building and exchange with responsible ministries. Innovation is also present, with the demonstration of a precision

agriculture model, which optimizes irrigation efficiency. This is coupled with the enhanced intelligence-based real-time decision-making (IRDM) tool, which provides more downscaled rainfall and run-off information for better localised advice, will improve decision-making, and improve water management. Thailand has developed a positive enabling environment for the integration of technology into smart agriculture. The IoT tool for water management has already been tested in Thailand by Kasetsart University and in Philippines by the International Rice Research Institute (<https://waterice.irri.org/how-we-work>). The project is well placed to further develop this technology with staunch support from Thai agencies and supported by Thailand's national strategies which place an emphasis on integrating smart agriculture and digital technologies to transform the agricultural sector. Testing and future development of this highly innovative system to improve community managed rice lands, save water resources and reduce greenhouse gas emissions is in line with the Green Climate Fund's strategic aims to integrate and develop innovative approaches to combat climate change. The GCF Secretariat view that as this is testing an innovative system, it should not be evaluated primarily based on cost-effectiveness criteria; since many emerging technologies and applications will encounter hurdles to move from the innovation to adoption phases.

19. With respect to post-project sustainability, the project appears well-designed and phased to provide support to the eventual planned replication to other subriver basins and farming communities, though the incremental plan for eventual upscale across all river-basins by the Thai government is not yet clear, though it is planned in Thai national policies. Building on established in-country networks and refining government responsibilities, economic incentives for communities for more climate resilient agricultural practices are expected to be created, supported by government actors. For example, outputs of adaptation plans (with legal standing) and climate responsive cropping patterns at the household level are expected with on-hand implementation of on-farm measures for improved water storage and soil conservation, coupled with bankable agro-investments.

3.3 Sustainable development potential

Scale: N/A

20. The proposal contributes to Sustainable Development Goals: (1) No Poverty, (2) Zero Hunger, (5) Gender Equality, (6) Clean Water and Sanitation, (13) Climate Action, (15) Life On Land, (17) Partnerships for the Goals. Positive environmental externalities are expected in improved water management leading to 1) efficient water use; and 2) controlled soil conservation mainly in the river and floodplain areas. There is evidence that the project would enhance groundwater recharge, leading to further water availability.

21. With Outputs 1 and 3, the linkages between government actors, local farming communities in the project sites, academic institutions (King Mongkut's University of Technology North Bangkok), and local financial institutions (co-financiers, cooperatives) are expected to increase mainly across the rice value chain. Capacity development and facilitated planning and coordination underpins these improved relationships. Coupled with the more efficient water resource use and enhanced market access, the enhanced linkages are intended to translate into financial benefits especially for smallholder farming communities, where female-led households are prioritised. Overall, significant effort is made to contribute to the project sites' overall self-sufficiency in health and food security. While the focus of this proposal is on water management for agriculture, the project may be strengthened with further discussion on the impacts on surrounding natural ecosystems.

3.4 Needs of the recipient

Scale: N/A

22. As previously noted, Thailand is facing abundant climate risks: extreme rain episodes lead to increased flood risk and drier dry seasons lead to water unavailability. Overall, the

selected project sites, Sukhothai, Phitsanulok and Uttaradit Provinces, are composed of impoverished rural communities that are geographically at risk, as they live on flood plains, and are financially underserved. With their livelihoods tied to the climate sensitive industry of agriculture, the targeted beneficiaries and their productive assets are assessed to be greatly affected by climate induced extreme weather events and will be disproportionately impacted by their effects.

23. The Provinces are part of the Yom and Nan subriver basins which form part of the Greater Chao Phraya River Basin area. This area is of critical importance for food production and flood regulation. The Yom and Nan subriver basins are in the middle section. Water is stored here to prevent flooding to Thailand's capital and economic centre, Bangkok.

24. National and local resources are leveraged by GCF funding. Green Climate Fund and RID funding is focused on enhancing infrastructure (Output 2), while GCF also funds activities supporting an enabling environment, namely capacity developing, planning, knowledge management, and coordination (Outputs 1 and 3). Other co-financiers (Krungsri Bank and BAAC) are focused on supporting the project gains during and after implementation, through supporting market access and providing further opportunities for investment.

25. Institutions benefiting from the interventions were chosen based on their sectoral contribution and decision-making power to implement the project's integrated water resource management approach. This approach targets specific institutional needs, such as the refinement of intelligence-based real-time decision-making modelling (IRDM), which involves RID, the Agricultural Research Development Agency, the National Research Council of Thailand, KMUTNB, and the Thailand Research Organization Network. Institutional coordination is also targeted, including the Thai Meteorological Department, the Hydro and Agro Informatics Institute, the Geo-Informatics and Space Technology Development Agency, the Department of Water Resources, and the Hydrographic Department of the Royal Thai Navy.

3.5 Country ownership

Scale: N/A

26. The proposal has strategic alignment with the Royal Thai Government's highest-level development and climate policy and master plan, namely the 20-year national strategic plan (2018–2037), National Economic and Social Development Plan (2017-2021), and the Climate Change Master Plan. It addresses the top two priorities detailed in the GCF country programme: water resources management and food security. Related activities have been adequately coordinated with ongoing and planned similar projects. The AE has consulted numerous stakeholders including the project participants, financial institutions, and national and local government agencies. Reported results include the need for a strong flood control system in the area with adjustments to be made in controlled water levels, intake and outlet systems, drainage system, road system, and monitoring system.

27. UNDP has a strong track record of development projects in Thailand especially focusing on agriculture and water management. It has a country office and will continue to work with key agencies and organizations. The executing entity (EE) is the Kingdom of Thailand, represented by the Ministry of Agriculture and Cooperatives (MOAC). The project is implemented by RID who will held accountable by UNDP for the use of GCF proceeds. RID and other partners such as GIZ and KMUTNB have undergone the required assessments under the Harmonised Approach to Cash Transfers Framework. This Framework represents a common operational framework for UN agencies' transfer of cash to responsible parties.

28. RID is mandated to maintain water infrastructure and is uniquely posed to cater to both hard and soft aspects of the interventions with other partners. RID is also a co-financier and contributes to both the project implementation phase and post-project operations and maintenance. This co-investment is a clear indication of government commitment.

3.6 Efficiency and effectiveness

Scale: N/A

29. The level of concessionality is assessed to be justified by the expected benefits from the proposed interventions. This is coupled with the limited financial return of adaptation activities for the benefit of vulnerable rural farming communities in northern Thailand.

30. Output 3 is expected to spearhead the creation of further economic opportunities for the project participants in selected agriculture value chains as enabled by Outputs 1 and 2, which work in the enabling environment and physical protection, respectively, all of which can be considered as public goods with low financial returns but with high economic value. Private sector parallel financing is present and amount to a significant percentage of co-financing at 93 per cent of the total project cost. This will stimulate potential revenue-generating activities and will thus be financed by non-GCF proceeds.

31. In terms of enhanced resilience, 62,000 direct beneficiaries are estimated. The project utilizes GCF investment of USD 282.80 per direct beneficiary, and USD 56.20 per total direct and indirect beneficiaries.

32. Thailand has developed a positive enabling environment for the integration of technology into smart agriculture. The IoT tool for water management has already been tested in Thailand by Kasetsart University and in Philippines by the International Rice Research Institute (<https://waterice.irri.org/how-we-work>). The project is well placed to further develop this technology with staunch support from Thai agencies and supported by Thailand's national strategies which place an emphasis on integrating smart agriculture and digital technologies to transform the agricultural sector. The GCF Secretariat view that as this is testing an innovative system, it should not be evaluated primarily based on cost-effectiveness criteria. It should be noted that emerging technologies and applications will encounter hurdles to move from the innovation to adoption phases and cannot yet be evaluated on traditional cost-effectiveness since the values, and future impacts, of the technologies are at innovation stage. This is in line with theoretical concepts of innovation development as exemplified by historical examples, from mobile phones to wireless sensors and other emerging technologies. Digital technologies, smart agriculture and building capacity for 'smart farmers will transform Thailand's approach to tackle climate change and create resilient livelihoods in the agriculture sector in the next decade.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

33. **Project background.** The project aims to provide interventions to adapt water management and agricultural livelihoods in the Yom and Nan river basins in Thailand to extreme weather events such as droughts and floods, by enhancing climate and risk planning in the water and agricultural sectors, improve water management through strengthened infrastructure by integrating EbA measures, and reducing the volatility of livelihoods in these areas through various technical and financial support. Among the expected environmental co-benefits of the interventions are improved soil and water quality including improved groundwater recharge, biodiversity conservation and enhanced ecosystem services. The social co-benefits include improved access to water supply during droughts, particularly for vulnerable groups such as female-headed households.

34. **Environmental and social risk category and safeguard instruments.** The project is classified as category B since the project will have potential limited adverse environmental and social risks and/or impacts which are site-specific, largely reversible, and can be mitigated with appropriate environmental and social management measures. The project has been screened

against the AE's Social and Environmental Standards Procedure (SESP) and an Environmental and Social Management Framework (ESMF) has been prepared including an Environmental and Social Management Plan (ESMP). The Secretariat confirms that the risk category is aligned with GCF's environmental and social risk categorization and is within the environmental and social risk accreditation level of the AE.

35. **Compliance with GCF's environmental and social safeguards (ESS) standards.**

Subparagraphs (a)–(h) below summarize the programme's compliance with GCF ESS standards and requirements:

- (a) **ESS 1: Assessment and management of environmental and social risks and impacts.** The project has submitted an ESMF with sections containing the environmental and social risk assessment and management measures as provided in the ESMP. The impact risk assessment was undertaken using the AE's SESP to assess the probability (expected, highly likely, moderately likely, or not likely) and the severity of impacts (critical, severe, moderate, minor, or negligible). The results of environmental and social risk assessment are presented and an ESMP is proposed that identifies, among other things, potential impact areas, required mitigation measures, and responsibilities of stakeholders. The potential negative impacts are on water quality through sediment movement during the replacement of existing infrastructure, channel and retention basin works; and EbA activities for wetland restoration. Noise and air quality may also be impacted during these works. Minor impacts also include increased waste from the removal of existing structures;
- (b) **ESS 2: Labour and working conditions.** The project underscores that activities will respond to the core labour standards per the International Labour Organization and the labour laws and occupational health and safety regulations of Thailand. This includes a number of performance standards in relation to labour and working conditions such as on freedom of association, elimination of discrimination in employment and occupation, elimination of forced or compulsory labour, and ensuring no forms of child labour. To limit risks and keep working conditions safe, generic contract clauses are provided as part of tender procedures that contractors need to comply with. These aspects will be monitored, and all workers will be informed and trained on the existence of a grievance mechanism in a continuous and systematic manner;
- (c) **ESS 3: Resource efficiency and pollution prevention.** The AE has identified and assessed activities that generate emissions to air, noise and vibration, discharges to water, contamination of land and other project-associated wastes. The AE incorporated measures in the ESMP to avoid and/or reduce pollution and promote sustainable use of resources. The proposed project is expected to generate sediments during construction works that might lead to water and land pollution. However, the ESMP presents measures to manage, limit and avoid pollution including an approach to monitor and report environmental risks on a daily and weekly basis. In addition, an erosion, drainage and sediment control management plan and contaminated soil disposal management guidance are provided;
- (d) **ESS 4: Community health, safety and security.** The project recognizes the fact that issues related to site security might emerge, thus security of sites is part of the requirements in the works contract and should remain in force throughout the contract period. This also includes the management of on-site construction materials, construction and sanitary waste, additional strengthening of erosion control and soil stabilization systems and other conditions resulting from contractor activities which may increase the potential for damage. As regards noise, reasonable efforts to schedule activities during normal working hours will be made to minimize nuisance to the surrounding communities. This would also mean developing a public notification and noise management plan, as necessary. The project, through its contractors, will also

ensure that a traffic management plan for anticipated service disruptions will be prepared should transport and mobility of the communities be hampered by the project's activities;

- (e) **ESS 5: Land acquisition and involuntary resettlement.** The implementation of the project will not require any resettlement. None of the interventions will require the displacement of people or involve economic displacement. There may be instances where there may be a need to utilize areas of land adjacent to where the structural interventions will be undertaken such as for water infrastructure works. However, where access is required, the land will be returned in the same condition as it was prior to any use. The ESMP further highlights that access to this type of land will only be obtained through voluntary agreements with landholders. Where a voluntary agreement cannot be concluded, the land will not be used. The project will likewise ensure full compliance with the AE's social and environmental standards guidance note on displacement and resettlement;
- (f) **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The project area and its surroundings are rich in flora and fauna, including aquatic and marine species. Two protected areas are located in proximity to the project area, but no activities will be undertaken in these areas. The ESMP has a number of performance criteria that will enable project activities to limit impacts on terrestrial, aquatic and marine flora and fauna. The management measures that will be implemented during construction activities include, among others, compliance with no clearance of vegetation outside of the designated clearing boundaries and no introduction of new weed species during construction works or implementing EbA activities. A flora and fauna monitoring programme will also be implemented;
- (g) **GCF Indigenous Peoples Policy and ESS 7: Indigenous peoples.** The ESMF notes that ethnic minorities make up 15 per cent of the total population of Thailand and the AE has undertaken screening to identify the presence of ethnic minorities in the general project area. It notes that there are no indigenous peoples and/or ethnic minorities among the smallholder farming communities within the project area but that there are Thai Khrang villages in the highland area away from the project sites. The AE notes the intention to include them in the project activities. Efforts to engage these groups will include extending knowledge on EbA and smart farming and involvement in project design and implementation, particularly with respect to EbA activities; and
- (h) **ESS 8: Cultural heritage.** The ESMF identifies three sites within the overall project footprint including a popular touristic cultural heritage site visited by tourists. Despite the presence of these sites, the AE expects that activities will be implemented with limited or no risks, following the performance criteria indicated in the ESMP. Nonetheless, should there be any unexpected cultural heritage items discovered during project works, the AE or respective EE will immediately cease work within the area and will consult with the relevant authorities to arrive at an appropriate decision. Nonetheless, should there be any unexpected cultural heritage items discovered during project works, a chance finds procedure will be followed to ensure that the project works are immediately ceased within the area and consultations are held with the relevant authorities to arrive at an appropriate decision.

36. **Implementation arrangements.** The implementing agencies will be responsible for the implementation and compliance with the ESMP, in collaboration with the various partners and contractors. The ESMP will also be part of the tender documents. The ESMP has clearly defined the roles and responsibilities of the stakeholders that will be involved during implementation. The monitoring and evaluation oversight and responsibilities are also defined. The ESMP has recognized the need to carry out capacity-building such as for project personnel

to attend induction training that covers health, safety, environmental and cultural requirements, and for workers engaged in any activity with the potential to cause serious environmental harm.

37. **Stakeholder engagement and grievance redress mechanism (GRM).** The project has conducted field surveys and interviews with beneficiary households in the targeted districts. It also included interviews with local administrators and community leaders in the areas. These consultations, including with decision-makers, communities and community leaders have integrated their concerns into the project design. The project has prepared a stakeholder engagement plan that provides an indicative list of stakeholders for the outputs and activities for the lifetime of this project. The project will continue to engage with stakeholders, including those from civil society organizations, non-governmental organizations, private sector organizations, academia and public organizations. The stakeholder engagement plan is a living document and will be revised and updated if needs arise during project inception and implementation.

38. The project has provided a GRM for complaints on environmental and social issues. A safeguards officer will be stationed at the project management unit and will be designated as the officer in charge of the GRM at the project level. A complaints register will also be maintained to keep track of grievances and monitor how they are addressed. Notwithstanding this, an independent two-tier GRM structure has been developed to address all complaints and/or grievances in the project both at the activity level and at the AE level.

4.2 Gender policy

39. The accredited entity (AE) has provided a gender assessment and gender action plan and therefore complies with the requirements of the GCF Gender Policy.

40. The AE has provided a gender assessment based on a desk review and stakeholder consultation with communities including women. The assessment indicates that Thailand has an enabling environment based on the constitution and civil and commercial code that accords women and men the same rights to own, use, and make decisions regarding land and non-land assets. It also has laws allowing for joint land entitlement and co-ownership of property for married and unmarried couples. Thailand has also ratified the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW). It has a Department of Women's Affairs and Family Development within the Ministry of Social Development and Human Security to deal with gender and women's empowerment and social inclusion issues. While favourable policies and laws exist, customary laws still influence decision-making in rural areas and have impacts that may disadvantage women's access to land, property, and inheritance. There is also a lack of clear policy directives on the integration of gender equality and limited participation of women in the formulation and implementation of climate policies and plans. This is further compounded by the lack of sex disaggregated data, which represents another key challenge in gender and social inclusion in climate change action.

41. The gender assessment attests that women play a significant role in agricultural activities in the same way that men do despite having less access to mechanized technology, agricultural inputs, finance and credit, seeds, information, and knowledge. At the same time, 40 per cent of the participants in the focus group discussions held during the assessment, indicated that, in practice, communities have access to, and decision-making power over land but are unsure of their formal ownership status. Further, 30 per cent of the Focus Group Discussion participants mentioned that new property (land as well as house) is commonly registered under the wife's name because men tend to avoid paperwork and prefer to work in the field. Ownership of land by women seems to have increased – albeit slightly – in the northern region of Thailand, where women made up 28.7 per cent of the landholders in 2013, compared to 20.7 per cent in 2003. However, a similar statistic for women landholders is not available for the project provinces.

42. Apart from their role in agriculture, women are also tasked almost exclusively with the additional roles of cleaning, cooking, childcare, care (average 5.4 hours a day on domestic tasks and childcare, compared to 2.9 hours for men) and home gardens for household consumption. Water for home consumption is also the responsibility of women, adding to their burden in periods of water stress. Young women and children are also engaged in the latter providing support to their mothers by fetching water and facing the same challenges. While the responsibility for household consumption lies with women, the management of water usage groups is usually led by men with women having the responsibilities of vice president, financing, and accounting as well as administrative positions. Social norms, inequitable power dynamics, and the gendered division of labour affect women's and men's access to resources in northern Thailand, including water. Men typically have greater rights of access to, and decision-making power over water. Women in rural areas, particularly poor and ethnic minority women, may face disproportionate water insecurity compared to rural men. Climate change increases water scarcity and water-related insecurities, even in managing water user groups, particularly for women who rely on natural resources for their livelihood.

43. The assessment indicates that five core areas of focus could address the challenges to vulnerable women and benefit communities through the project. These are: access to information about climate change and related impacts on water and agriculture; defining the localized and differentiated information needed and to be disseminated; engagement in decision-making related to water use/management and agriculture; access to technology and related training to advance climate resilient agriculture; and access to finance to invest in climate resilient agricultural practices.

44. The gender action plan provided by the AE complies with the requirement of the GCF Gender Policy. It has provided specific activities, targets (50 per cent women participation and further disaggregated by age), timelines, financial and human resources by way of gender expertise for the project to address the challenges raised. It will employ a pilot approach such as gender-responsive Participatory Integrated Climate Services for Agriculture and piloting and scaling up household methodologies expected to support the work on addressing and strengthening gender equitable intra-household work and decision-making. The action plan will also formulate a working group on gender with various actors to ensure that issues raised are addressed and solutions implemented. The project will also focus on building data collection and management while at the same time documenting gender disaggregated data in a planned vulnerability and risk assessment. It will also target women and female entrepreneurs in trainings that will build their skills and provide them with financial incentives and support to access climate resilient agricultural practices and climate change information. Overall, to address some of the underlying causes of inequality, the project will facilitate the creation of an enabling environment by working with men to alter harmful stereotypes while also designing strategies to monitor and mitigate the potential issue of gender-based violence.

4.3 Risks

4.3.1 Overall project assessment (medium risk)

45. GCF is requested to provide a grant of USD 17.5 million to adapt water management and agriculture livelihoods in the Yom and Nan river basins to climate change-induced extreme weather events in Thailand. The Royal Thai Government through its Irrigation Department and Krungsri Bank are providing in-kind contributions worth USD 16.2 million and USD 0.1 million, respectively. There is no co-financing from the AE.

46. Thailand has investment grade credit rating, and its fiscal situation may support availing concessional debt from GCF. The AE has supported grant financing from GCF based on public good nature of the project and with 48 per cent of the budget coming from co-financing and

opined that the GCF grant is needed to cover the incremental cost to make the total project viable.

4.3.2. **Accredited entity/executing entity capability to execute the current project (low risk)**

47. UNDP is the AE for this project. The AE has been operating in Thailand since 1966 and has an ongoing portfolio in the country with an annual budget range of USD 7–10 million over the past five years. The AE has its regional office representation in Thailand.

48. The Kingdom of Thailand, represented by the Ministry of Agriculture and Cooperatives, (MOAC) is the EE for this project. The AE carried out the capacity assessment of MOAC, which resulted in low risk in all major categories.

4.3.3. **Project-specific execution risks (medium risk)**

49. Construction risks: the project implementation involves canal dredging, and construction related to the floodgate. A total of 54 per cent of GCF financing (USD 9.6 million) and 82 per cent of government co-financing (USD 13.4 million) is budgeted for Component 2. Under this component, GCF resources will cover the construction cost of upgrading two floodgates for flood mitigation and improvement of water storage for agriculture. The government co-financing will support canal dredging and strengthening of embankments to improve waterway connections. The AE has identified that the construction and on-groundwork may be affected by extreme weather conditions. Therefore, the implementation plan needs to be adaptable to such conditions.

50. Operation and maintenance activities: GCF resources will complement the planned grey infrastructure with ecosystem-based solutions. Therefore, identifying the most appropriate² EbA measures for the specific project target area and responsibility for operation and maintenance are crucial for the infrastructure (e.g., floodgates) to serve the intended purpose. The design of EBA methods requires extensive participatory discussions with local stakeholders, including farmers, and has been integrated into the implementation plan. The project team plans to mobilize resources to cover the cost from different departments in the Government. As multiple departments are involved in the financing, the coordination and mobilization efforts will be important to sustainably finance the operation and maintenance costs. The fact that the Government will allocate USD 17.26 million for post-project operations and maintenance is reassuring. The irrigation department will maintain facilities under Outputs 1 and 2, and other MOAC agencies will run Output 3 under regular government budget in the future.

51. Economic analysis and project viability: an economic analysis has been carried out and results in economic internal rate of return (EIRR) ranges between 29 and 33 per cent. The analysis considered benefits mainly from the avoided cost of agriculture yield affected by climate change and damages from extreme events. The sensitivity analysis is based on the scenarios: 1) an increase in cost of 20 per cent; 2) a decrease in benefits of 20 per cent; and 3) a simultaneous increase in cost of 20 per cent and decrease in benefits of 20 per cent. The net present values under all scenarios remain positive.

52. Concessionality of GCF financing: the AE has proposed 100 per cent grant financing thus asking for maximum concessionality. While the project does not lead to any direct revenue

² Standard & Poor's credit rating for Thailand stands at BBB+ with stable outlook. Moody's credit rating for Thailand was last set at Baa1 with stable outlook.

(Accessed 1st June 2021: <https://tradingeconomics.com/thailand/rating>)

generation, the Government of Thailand and AE may consider partial concessional debt financing for the intervention based on the investment grade rating of the country. Moody's has rated the country as Baa1 (2020) and stated in its credit opinion that the credit profile of Thailand reflects strong public and external finances that give the country significant room to counter shocks, including the economic hit from the coronavirus outbreak. Moody's has also recognized lower government debt as compared to its peers and low share of foreign currency debt in overall debt of the Government.

4.3.4. Compliance risk (medium risk)

53. UNDP (the AE) has advised that they apply a zero-tolerance policy in relation to fraud and corruption. The guiding principles of UNDP's commitment to prevent, identify and address all acts of fraud and corruption have been laid out in the *UNDP Policy against Fraud and Corrupt Practices* (the *FCP Policy*), which applies to all activities and operations of UNDP, including projects and programmes.

54. The fundamental principles of UNDP's *FCP Policy* serve as the basis for and are integrated in UNDP's policy frameworks in relation to procurement, financial management, internal control and accountability and staff rules and regulations. Under the terms of the *FCP Policy* and related policies, UNDP has a commitment, when developing a new programme or project, to ensure that risks related to fraud and corrupt practices are fully identified and considered in the programme/project design and processes and that adequate and effective measures to mitigate such risks are put in place.

55. UNDP has advised that the processes and requirements for UNDP's partnership capacity assessment require UNDP to carry out a thorough due diligence of the risks in relation to the programme/project it intends to engage in, as well as in relation to the potential partners who will be involved in the implementation of such programmes/projects. Such capacity assessments include, among other things:

- (a) A screening of all entities being considered to be contracted by or partner with UNDP against the Consolidated UN Security Council Sanctions List (to which UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden).
- (b) An assessment of any history of fraud, corruption, or other fraudulent practices; and/or
- (c) Potential conflicts of interest.

56. Based on this capacity assessment, UNDP then builds an internal control output into the design of the project and ensures that the financial management of the programme/project contains adequate safeguards to prevent, monitor and address any risks and acts of fraud, corruption or anti-money laundering/counter-financing of terrorism that may be identified.

57. UNDP has also informed that they apply a Harmonized Approach to Cash Transfers (HACT), which represents a common operational (harmonized) framework for transferring cash to government and non-governmental partners (both implementing partners and responsible parties). The common objective of the HACT framework is to support a closer alignment of development aid with national priorities and to strengthen national capacities for management and accountability. It serves as a set of procedures on requesting, disbursing, providing assurance, and reporting on funds to effectively manage risks, reduce transaction costs and promote sustainable development in a coordinated manner.

58. Cash transfers (above the threshold of USD 300,000 in a programme cycle, or otherwise in all circumstances where UNDP considers the operating environment to be high risk or has no prior experience with the partner) are presumed to present a material risk to UNDP and will require a HACT micro-assessment and risk rating by qualified third-party service providers.

Spot checks, programmatic Output verification and auditing requirements are then determined and carried out at a frequency determined in line with the risk rating.

59. As per the accreditation master agreement between UNDP and GCF, UNDP will apply its own policies. These include the *FCP Policy*,³ which sets out the processes for reporting fraud and ensures protection against retaliation for any person that has reported allegations of wrongdoing or cooperated with a duly authorized audit or investigation.

60. UNDP refers to the *UNDP Policy for Protection against Retaliation* as its key "whistle-blower" protection policy.⁴ The Office of Audit and Investigations conducts audits, investigations, social and environmental compliance reviews, and disclosure of internal audit reports.⁵ UNDP also has internal control protocols consisting of an *Internal Control Framework* and *Anti-Fraud Policy*.⁶

61. UNDP has advised that it conducts partner capacity assessments in relation to all its programming and project activities. Reference is made to the *Partner Capacity Assessment Tool (PCAT)*⁷ and the *HACT Framework*.⁸

62. UNDP has confirmed that there are no payments, vouchers, or tradeable instruments of value used in activities under the project.

63. **Recommended risk rating:** The Office of Risk Management and Compliance (ORMC)/Compliance Team has conducted a review of the project in accordance with relevant GCF Board approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the ORMC/Compliance Team have determined a risk rating of 'medium' and has no objection to this request proceeding to the next steps for processing.

64. ORMC/Compliance would like to remind UNDP, as the AE, of its continuing obligations and responsibilities with regard to monitoring and reporting any risks for money laundering, terrorist financing, or prohibited practices among the intended counterparties, executing entities, beneficiaries, persons involved, or any of the proposed activities.

4.3.5. GCF portfolio concentration risk (low risk)

65. In case of approval, the impact of this proposal on the GCF portfolio risk remains non-material and within the risk appetite in terms of concentration level, results area, or single proposal.

4.3.6. Recommendation

66. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall project	Medium	

³ https://www.undp.org/content/dam/undp/documents/about/transparencydocs/UNDP_Anti-fraud_Policy_English_FINAL.pdf

⁴ https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/Ethics_Protection%20against%20Retaliation.docx

⁵ <https://www.undp.org/content/undp/en/home/account%20ability/audit.html>

⁶ <https://popp.undp.org/SitePages/POPPSubject.aspx?SBJID=384&Menu=BusinessUnit>

⁷ https://popp.undp.org/UNDP_POPP_DOCUMENT_LIBRARY/Public/Consolidated%20Risk-Based%20Partner%20Capacity%20Assess%20Tool%20.xlsm

⁸ https://popp.undp.org/UNDP_POPP_DOCUMENT_LIBRARY/Public/FRM_2014%20UNDG%20HACT%20Framework.pdf

Accredited entity (AE)/executing entity (EE) capability	Low	The implementation plans, especially for construction work, need to be adaptable for extreme weather conditions. Operation and maintenance cost and responsibility will be critical for the sustainability of the project.
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

67. The project will be implemented following UNDP's National Implementation Modality, according to the Special Fund to Thailand. As an accredited entity to GCF, UNDP will deliver GCF-specific oversight and quality assurance services including: (i) day-to-day oversight supervision; (ii) oversight of project completion; and (iii) oversight of project reporting.

68. The EE for the project, Ministry of Agriculture and Cooperatives (MOAC), will assign the Royal Irrigation Department (RID), to implement the project outputs and will be accountable to UNDP for managing the project, including monitoring and evaluation of project interventions, achieving project outcomes, and for the effective use of UNDP resources. A UNDP micro assessment was conducted for RID, across all major categories and RID is considered low risk.

69. UNDP will transfer funds to RID, under MOAC for implementation. Furthermore, RID has agreed to select the KMUTNB and GIZ based in Thailand as the responsible parties for implementation (Output 1, Output 2/Activity 2.2 respectively). The engagement of responsible parties shall follow the Implementing Partner (in GCF parlance, Executing Entity) regulations and rules. UNDP micro assessments were also conducted for KMUTNB and GIZ as the responsible parties.

70. A project manager will run the project on a day-to-day basis within the parameters laid down by the Project Board. The project manager's contract will end when the final project terminal evaluation report, and other documentation required by GCF and UNDP, has been completed and submitted to UNDP. The project manager is responsible for day-to-day management and decision-making for the project.

71. The financial management and procurement of this project will follow UNDP financial rules and regulations. The project will be audited in accordance with UNDP policies and procedures on audits, informed by and together with any specific requirements agreed in the accreditation master agreement. According to the current audit policies, UNDP will appoint the auditors.

4.5 Results monitoring and reporting

72. As an adaptation project, the intervention aims at enhancing climate resilience in the Yom and Nam river basins of Thailand through efficient water management and resilient agricultural practices. It is expected to benefit 62,000 direct and 471,561 indirect beneficiaries in the basin.

73. Overall, the funding proposal and log frame have been found to adequately apply the results management framework and performance measurement framework indicators. The outcomes, impacts, and the related indicators are aligned with GCF requirements. Pertinent information on Means of Verification, baseline and targets have been provided. Similarly, the log-frame contains clear presentation of project-level results, their indicators and other required information in a concise manner. Their diligent application during implementation would facilitate results monitoring, progress reporting, performance assessment and finally their delivery on the ground.

74. The implementation timetable has been found to follow the required structure and format. The milestones and deliverables are well articulated and reporting timelines are provided as required.

4.6 Legal assessment

75. The accreditation master agreement was signed with the accredited entity on 5 August 2016 and became effective on 23 November 2016.

76. The AE has provided a certificate confirming that it has obtained all internal approvals and has the capacity and authority to implement the project.

77. The proposed project will be implemented in the Kingdom of Thailand, a country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The Secretariat submitted a revised draft of the privileges and immunities agreement to the Government of Thailand on 5 October 2018. However, no response has been received yet.

78. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made only after GCF has obtained satisfactory protection against litigation and expropriation in the country or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

79. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the Secretariat within 180 days from the date of Board approval; and
- (b) Completion of legal due diligence to the satisfaction of the Secretariat.

Independent Technical Advisory Panel's assessment of FP170

Proposal name:	Enhancing climate resilience in Thailand through effective water management and sustainable agriculture
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Thailand
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel¹

1.1 Impact potential *Scale: N/A*

1.1.1. Adaptation impact

- Thailand's vulnerability to climate change resides in the geographical and socioeconomic context: an extensive coastline, a high dependence of rural communities on agriculture, and extensive urban areas located on flood-prone plains. The proposal argues that the most significant impacts would be the increased risk of floods and droughts, and that the risk has already risen since 1955, as a consequence of a higher incidence of extreme rainfall events and increased rainfall variability. The proposal states that, according to projections, these climate extremes will increase both in frequency and intensity.^{2, 3}
- The proposed project seeks to enhance the water sector's climate change adaptation capacity and agriculture livelihoods in the Yom and Nan river basins, by committing to establish an integrated water resources management (IWRM) regime that focuses on year-round water management, taking into consideration wet-season-specific floods and dry-season-specific droughts in an integrated approach. In doing so, both the major concerns of floods and droughts are likely to be addressed, where flood control through hydraulic structures will be complemented by early warning and by establishing ecosystem-based adaptation (EbA), and drought will be managed by using water that is likely to be retained in reservoirs as well as by enhancing infiltration of water in groundwater aquifers by means of water storage and induced recharge of aquifers. Once the availability of water for use during the dry season is ensured for compensating the growing moisture deficit, irrigation facilities are expected to ameliorate drought and facilitate crop production even under climate change.
- Adaptation in the Yom and Nan basins is likely to be achieved through three main components: (1) informed planning in the water and agricultural sectors through improved climate information and cross-sectorial coordination; (2) improved water management through strengthened infrastructure complemented by EbA measures; and (3) improved climate resilience of farmers through support for incorporating climate forecasts and knowledge of climate change risks into planning, and for developing the market for agricultural products.

¹ This assessment is based on the funding proposal submitted to the independent Technical Advisory Panel on 12 July 2021.

² Funding proposal, pages 2 and 5.

³ The project's climate rationale is discussed in section 1.4 of this assessment.

4. Component 1 is focused on improving the availability and quality of climate information and supporting its dissemination, and on facilitating interministerial coordination. The Royal Irrigation Department's (RID) Intelligence-based Real-time Decision-Making Modelling (IRDM) tool,⁴ which currently aids the RID in managing water in the Chao Phraya River basin, would be updated to increase the level of forecasting precision as well as for generating projections for shorter time frames (currently, projections are made on an annual basis). Interministerial coordination for climate-informed and integrated planning would be facilitated through technical support in developing standard operating procedures (SOPs) for information-sharing across government stakeholders. Dissemination of climate information would involve the development of a communication strategy and the use of briefs, reports and smartphone apps.

5. Although activity 1.1 includes the procurement and installation of climate and water monitoring equipment for a value of approximately USD 800,000, the original submission of this proposal did not include a description of this equipment. Upon the request of the independent Technical Advisory Panel (TAP), the accredited entity (AE) submitted a breakdown of the costs of the equipment. The detailed budget of the present submission was updated to include a list of equipment to be installed. However, as indicated below, the independent TAP questions the adequacy of some of the proposed equipment to achieve the proposed objectives.

6. Through component 2, the project would strengthen water infrastructure through hard engineering works complemented with ecosystem-based adaptation (EbA) measures. Activities related to grey infrastructure include the review and enhancement of the existing engineering designs for the 13 schemes of the Yom-Nan river basin area, to ensure the designs take into consideration climate change adaptation, and construction works to upgrade two canal head regulators along Khlong Muang Chang and Khlong Ta Kay Reg in the Yom-Nan river basin area. With increased rainfall run-off under climate change, these regulators (i.e. water infrastructures) would ameliorate flood vulnerability at the lower reaches of the Yom-Nan river basin.⁵

7. EbA measures are perceived to be complementary to the hard flood-controlling measures, with specific objectives to further strengthen flood amelioration and to enhance the adaptive capacity of affected communities within the river basin. EbA measures could include support for the implementation of integrated farming practices (such as rice-fish management), revegetation of riverbanks, wetlands management techniques, erosion control, and soil conservation measures. The German Agency for International Cooperation (GIZ) is currently developing, together with the Thai water sector, an EbA Guidebook to be used as a national standard for the systematic identification, prioritization and selection of EbA measures. The Guidebook will be complemented by a Code of Practice that will cover specific design issues of

⁴ IRDM is a collaborative effort involving five organizations: Royal Irrigation Department, the Agricultural Research Development Agency, the National Research Council of Thailand, the King Mongkut's University of Technology North Bangkok, and the Thailand Research Organizations Network. IRDM is a milestone in enhancing the institutional coordination between the agricultural and irrigation departments, as well as in developing the public forecasting and modelling capacity. IRDM can calculate the amount of rainfall and inflow of water into reservoirs during the wet season and design water release schedules for 12-month periods, with a relatively high level of accuracy. Its mathematical models can be used to forecast inflows into different water sources, support water management plans for individual irrigation zones and estimate agricultural yield and revenue.

⁵ In 2012, hydrological modelling (Mike-11 ECOLab&AD) for the Yom-Nan river basin was conducted by the Department of Water Resources, under the Ministry of Natural Resources and Environment. The model represented river profiles and modelled water levels under four different scenarios: (i) no water reservoir construction; (ii) no dam at the upper Yom river; (iii) reservoir construction; and (iv) dam construction at the upper Yom river, for both the year 2010 and future years (2015-2020). A number of models were used to model different aspects of the Yom-Nan river basin, including MIKE FLOOD for flood, MIKE11 ECOLab&AD for water quality, and MIKE BASIN for water consumption. These models are updated every year, especially after flood events when river cross-sections can change. However, based on the TAP's findings from the interview with the AE, hydrological modelling including climate change projections has not been carried out. Therefore, it is not possible to ascertain the impact of climate change on flooding and on the general river basin regime.

selected EbA measures. The EbA Guidebook and Code of Practice are expected to be ready by the end of 2021 and will be published by the Thai Office of the National Water Resources (ONWR).

8. EbA measures would be designed using modelling techniques to assess the potential impact on the hydrological regime, in order to support the prioritization of EbA measures and cost-benefit analyses. The AE further explained that EbA measures would include a mixture of best practices and lessons learned in past and ongoing projects, which are briefly described in the feasibility study. Interventions would be designed through the following process: (1) geographical scaling (mapping of hot spots) to define the geographic boundaries of the pilot areas; identification of sub-regions within each project area and mapping of major hydro system features and economic activities; (2) scoping by assessing environmental and socioeconomic impacts and establishing priorities among the major concerns and issues as well as tracing the root causative factors behind the selected concerns and issues; (3) data collection; (4) modelling for verification and quantification of root-cause relationships with either physically deterministic or conceptual models (hydrologic modelling with TalsimNG software and/or hydraulic 2D modelling as required; erosion and sedimentation modelling with Modified Uniform Soil Loss Equation (MUSLE); water quality modelling with GISMO software; assessment to identify risks and uncertainties; identification and ranking of potential EbA measures).⁶

9. The funding proposal includes a more thorough description of EbA measures.⁷ The AE stated that “EbA measures will be implemented in and around the two water infrastructure interventions of the project area – Klong Mueang and Klong Takae. These areas are mostly dominated by the paddy fields, 27,936 rai and 58,964 rai are the cultivated areas, respectively. While the EbA measures are not necessarily oriented to agricultural production, they will support year-round better water management by contributing to IWRM and provide ecosystem services which will benefit cultivated areas.”⁸ The independent TAP understands that the “27,936 rai and 58,964 rai” do not correspond to the coverage of EbA interventions, but to the cultivated areas around the water infrastructure works, that would supposedly be benefited by EbA. Without an estimated spatial coverage of the actual area that would be directly modified by EbA interventions (including aquatic and adjoining wetland-forest covered areas), it is not possible to assess the per hectare cost of EbA interventions or the estimation of target beneficiaries for the proposed activity 2.2.

10. The design and implementation of EbA measures would include multiple measures in three locations and would cost USD 1.7 M (activity 2.2G in project budget). Potential EbA sites have been narrowed down to Pichai Srinakorn, Khonkairas, Sawankha Lok, Promphiram, Bangrakham and Muang Phitsahulok. As indicated in paragraph 9 above, the method for estimating beneficiaries of this activity is not described in the proposal.

11. Component 3 is focused on enhancing the climate resilience of farmers. The project would provide training and technical assistance to farmers and extension officers on the application of climate forecasts and knowledge of climate change risks in household agriculture planning, including the development of district-level adaptation action plans and implementation of climate-smart agricultural practices.

12. The project would support on-farm interventions to improve drought and flood resilience through technical assistance for the development of on-farm EbA guidelines, the procurement of small scale equipment for farmers (such as water pumps, small cultivators, on-farm water level measurement, etc.), the construction of 100 ponds as micro-scale water

⁶ Explanations given in AE’s responses, 22 September 2020.

⁷ Feasibility Study, section 4.2.1. Measures would include wetland / floodplain / oxbow lake restoration; living weirs; riparian zone rehabilitation / bank stabilization by vetiver grass, buffer zones, sedimentation traps and/or an appropriate combination of such measures.

⁸ See AE responses to independent TAP comments, dated 29th June and 21st July, 2021.

reservoirs, the establishment of community nurseries, the co-design of a financial instrument with the Bank of Agriculture and Agriculture Cooperatives (BAAC) to support farmers with adaptation investments, and capacity-building for farmers to support market access for their products. BAAC has committed USD 16 million for this instrument, as parallel financing. Upon request from the independent TAP, the AE provided a commitment letter from BAAC.⁹

13. Activity 3.1 includes the implementation, at a demonstration scale, of what is called precision agriculture or smart farming. This approach to agricultural production involves the use of modern observation and communication technology to generate and use geo-referenced information for the control of agricultural systems, integrating characteristics of soil, plants and climate. Through the use of these systems, algorithms executed by machines and based on parameters provided by machines, make an increasing number of decisions. For adequate implementation of precision agriculture to address droughts, the evapotranspiration level of the specific crops in the targeted area must be well understood. Hourly evapotranspiration data or, at least, daily data, as against data on water availability at root zones and water deficit, are needed. Although the use of precision agriculture has been applied in extreme water-stressed agriculture systems across the world with the aim of saving as much water as possible during irrigation applications, the intention to use precision agriculture in water-abundant areas such as the Yom-Nan basin appears technologically feasible but not viable in the economic sense.

14. Within activity 3.1, the project would install five Internet of Things (IoT) platforms, which would include IoT sensors, drones with multispectral cameras for surveying agricultural areas, hardware and software for data processing, storage and communication, and technical support and related training. Although it is widely recognized that precision agriculture has the potential to increase production and reduce the need for inputs with large-scale and/or high-cost production, the use of such advanced technology by poor, vulnerable farmers has numerous challenges. For example, the platforms include drone kits for spraying liquids (water or organic fertilizers), which can carry only 15 litres and would cost USD 20,000 each. Recognizing that smallholders in the Yom-Nan basin are among the most vulnerable population under climate change, there is no clear reason why agricultural activities including precision irrigation should be mechanized in small farms. The selection criteria for this activity stipulate that the landowner must have an agricultural area not larger than 1,6 hectares).¹⁰ There is no analysis in the feasibility study (FS) in relation to the potential annual increment of productivity in pilot farms as against annual depreciation of the investment plus the annualized operation and maintenance (O&M) costs. Questioned by the TAP, the AE responded that: "A separate economic cost-benefit analysis of farm-level precision agriculture platforms – that include IoT investments and drone technology – does suggest a benefit-cost ratio greater than one and a positive NPV [net present value]. This analysis includes the full costs of investment (upfront capital costs) along with annual O&M costs. The primary benefit for farmers that adopt the technology platform is increased farming income as the investment allows them to grow higher yield varieties and in marginal seasons. This is all under the overhang of a changing climate i.e., under the assumption farm incomes will erode as the climate changes." The economic analysis referred to is not included in the proposal. Apart from upfront investment requirements for precision agriculture involving modern technologies, there is a significant investment requirement for operationalization of precision agricultural practices, which often becomes a barrier to adoption of such techniques, especially at smallholder levels. In the absence of adequate analytical explanations as well as a roadmap for the extension of such techniques among smallholders, there is considerable uncertainty regarding the immediate uptake of the proposed intervention within the project implementation period.

⁹ Annex 13: Co-financing commitment letters, part 2.

¹⁰ Response to TAP comments at B.27, response 5.

15. Information regarding the time required for complete realization of the capital investment from the proceeds of increased productivity has not been provided in the FS. It appears difficult to understand whether there is any economic viability at all of precision agriculture to address potential water stress in the farms indicated, keeping in view their average potential decline in productivity due to claimed climate-change-induced increased water stress. Since this type of technology is not developed in Thailand, parts will need to be imported, which may pose a threat to the sustainability in the medium and long term.

16. The proposal does not provide an evaluation of the reach these platforms would have in terms of direct beneficiaries (people and agricultural area), or of the potential economic efficiency of this approach. Questioned about this matter, the AE responded that “Based upon survey of existing data of the target area and criterion of beneficiary’s selection, 10 farms in each district will be selected for demonstrating as pilot farms. Total pilot farms will be 70 farms from 7 districts¹¹; Pichai Srinakorn, Khonkairas, Sawankha Lok, Promphiram, Bangrakham and Muang Phitsahulok...” and that an economic evaluation by the BAAC “...confirmed that using drone and IoT can increase efficiency of farming by reducing water consumption, cost of soil testing, fertilizer mixing, seeding and fertilizing. The intervention will further establish best practices which can be replicated and up-scaled (further supported by the credit line to be made available through BAAC).” The AE indicated that this activity would serve as a pilot project to demonstrate results and accelerate the introduction of precision agriculture.

17. Although the proposal states that IoT equipment would be owned and maintained by the district governments,¹² the operation and maintenance (O&M) costs were not found in the O&M plan (annex 21) of the first submission of this FP (B.27). Questioned by the TAP on this matter, the AE responded that “... O&M cost is built into the sale price and warrantee from the dealers in the domestic market (Thailand). The dealers include an insurance and drone permit as one price. The after-sale service is generally provided by the dealers.” Even with the included insurance and warrantee, it is not clear how the dealers would cover the operation costs or the maintenance costs after purchase or for how long they would do so. Also, the dealers would have a natural incentive to promote the use of precision agriculture and IoT in farms with minimum O&M cost, which may generate unfairness in the distribution of assistance and may also make the application of this technology less efficient. The AE also added that “the project management unit will support resource mobilization for further replication and upscaling, and O&M cost (if applicable) from the regular public budget, distributed to government departments under the Ministry of Agricultural and Cooperatives (MoAC). Additional resources for IoT/O&M can be mobilized through the regular public budget for capacity-building to farmers and innovation research...” and that “there is high possibility to mobilize the O&M cost from the CAT (Communication Authority of Thailand) that has promoted the IoT for smart farming.” The economic analysis in the current submission (B.30) included the O&M costs for the IoT platforms, estimated in USD 107,000 per year after year five of project implementation. However, the O&M Plan does not include a description of the composition of this cost and the method used to estimate it. Asked by the TAP, the AE described the method used to estimate O&M costs of IoT platforms.¹³

18. Each IoT platform would cost USD 270,000, or USD 27,000 for each farm, making it unlikely for small farmers (up to 1.6 ha) to adopt it or even operate and maintain it by their own means. Following the estimation presented of the O&M costs, for groups of ten farmers sharing a platform to use this technology, each farmer would have an annual O&M cost of USD 2,000. In this case, being a pilot, the government would cover these costs, but for widespread

¹¹ The original submission included the installation of seven IoT platforms (one for each district), which was reduced to five in the present submission. Hence, 50 farms would participate in the pilot programme.

¹² Funding proposal, paragraph 98.

¹³ Response to TAP comments, question 8.

implementation by small farmers, it is uncertain how these systems would be operated and maintained. The use of this technology would not only be unaffordable for small farmers but would also require very high technical skills for O&M, thereby relying on external support and imported parts.

19. In response to concerns raised by the independent TAP, the present submission reduces the installation of IoT platforms from 7 to 5,¹⁴ reducing the budget for this activity by approximately USD 0.5 million, which was transferred to activity 3.2, which supports farmers with low tech/simpler interventions (e.g. on farm ponds, small equipment, etc.). The total budget for component 3 remains unchanged.

20. The independent TAP noted that project performance indicators do not include targets for this activity. In response to this comment, the AE suggested adding a performance indicator to assess the rate of IoT adoption by gender, age, socioeconomic profile of household, and the impact on productivity. The AE further clarified that “The benefit from IoT is captured under indicator A1.2, ‘Number of males and females benefiting from the adoption of diversified, climate resilient livelihood options (including fisheries, agriculture, tourism, etc.)’. Any further upscaling using the BAAC credit would be captured under indicator 3.2, ‘Number of households with access to finance for climate resilient farming investment’.”¹⁵

21. The Department of Royal Rainmaking and Agricultural Aviation (DRRAA) undertakes an annual average of 5,000 rain-seeding flights.¹⁶ Such activities may have an effect on precipitation patterns. The project does not include a description of reach and impact of the cloud-seeding programme being carried out by this governmental agency. Questioned by the independent TAP, the AE responded that despite the fact that cloud-seeding activities have not been the subject of long-term research, “published research on the effect of rainmaking in Thailand did not find a statistically significant relationship between cloud seeding and increases in rainfall amount or rainfall area.”¹⁷ The AE also added, “the seeding programme is still mostly active during drier pre-monsoon periods, before the southwest monsoon. The seeding programme is therefore not expected to affect rainfall during flooding events, which occur later during the year when the southwest monsoon is more intense.”

22. Direct beneficiaries of project interventions are estimated to be the 62,000 people who live in the project site area, with indirect beneficiaries reaching approximately 470,000 people that live in project districts, plus 25 million people living in the greater Chao Phraya River Basin.

1.2 Paradigm shift potential

Scale: N/A

1.2.1. Innovation

23. Although most of the project revolves around expanding and improving existing climate and water monitoring and management systems, and redesigning and constructing water infrastructure (floodgates and canals), the proposal includes some innovative activities in the form of EbA measures. Although concrete EbA measures are not yet defined, the present submission includes an accurate description of the potential alternatives, of which three would be selected for implementation based on a “diagnostic analysis including vulnerability and risk assessment, modelling, cause-impact chains and economic evaluation”.¹⁸ The implementation of

¹⁴ A previous submission of this project included seven pilot platforms.

¹⁵ AE responses to TAP assessments, 21 July 2021.

¹⁶ Royal Rainmaking Project, Wikipedia.

¹⁷ Woodley, W.L., Rosenfeld, D. and Silverman B.A. (2003). On-Top Glaciogenic Cloud Seeding in Thailand. Part I: The Demonstration Experiment. *Journal of Applied Meteorology*, vol. 42, no. 7, pp. 920–938. Parts I and II available at <https://www.jstor.org/stable/10.2307/26185817>; <<https://www.jstor.org/stable/10.2307/26185818>

¹⁸ Funding proposal, page 23.

EbA measures for climate resilient water management and livelihood development would represent an innovative approach for Thailand, for the first time integrating both grey and green approaches to water drainage, storage, diversion and irrigation. This paradigm shift to a hybrid approach to adaptation has potential to maximize effectiveness and cost-effectiveness.¹⁹ The project is also innovative in integrating climate risk management into IWRM, employing solutions which simultaneously tackle drought and flood by diverting floodwaters to replenish groundwater for use in dry periods, whilst also mitigating the floods.

24. Precision agriculture is also in its early days globally, and although BAAC launched a soft loan for purchasing drones in 2019, the disbursement rate was low.²⁰ This could be because precision agriculture is still at the introduction stage in Thailand, or because the idea of using such expensive equipment did not make economic sense to Thai small farmers. Perhaps the potential returns are too small compared to the capital cost, the incremental loss of opportunity costs through depreciation and annualized O&M costs, despite the fact that the BAAC has created access to credit. Despite institutional incentives, the current lack of enthusiasm involving the advanced precision agriculture technology may be an indication that the innovation is mostly theoretical. Overall, the funding proposal fails to provide adequate explanation justifying the use of precision agriculture. Although innovative, it is costly (both from a CapEx and an OpEx basis), thereby hindering diffusion of this advanced technology to low-income small farmers in the region.

1.2.2. Potential for knowledge and learning

25. The project involves capacity-building, technical training and enhancement for inter-agency coordination as cross-cutting activities in all components.

26. Component 1 is focused on improving the capacity for generating, communicating and using climate and hydrological information. These goals would be achieved through several actions, including:

- (a) The provision of hardware and software for enhanced generation of climate and environmental information;
- (b) Technical training to 30 staff from the Thai Meteorological Department (TMD), Royal Irrigation Department (RID) and relevant agencies on integration of climate change into water management and agricultural planning;
- (c) Development of tailored climate information products and climate forecast models;
- (d) Development of SOPs for sharing of climate information across relevant departments;
- (e) Workshops and meetings for sensitization and collaboration in sharing information and supporting collaboration;
- (f) Development of a communication strategy for dissemination of information;
- (g) Development of information briefs and reports released for circulation in local and social media;
- (h) Development of smartphone and LINE app with climate information to be applied at the agriculture household level; and
- (i) Development of a complementary website.

¹⁹ See, for example, World Bank Group and World Resources Institute (2019) *Green and Gray: Creating Next Generation Infrastructure*.

²⁰ Responses to TAP comments at B.27, response 5.

27. The risk of cross-posting or poaching of TMD and RID and other relevant agency personnel trained under this project may put at risk the effectiveness of this capacity-building. Appropriate safeguards need to be in place, such as a minimum number of years for remaining in a post.

1.2.3. Contribution to the creation of an enabling environment

28. The RID has committed, through a co-financing letter, to provide the necessary funds for O&M of activities beyond completion of the project, as estimated in the O&M Plan.

29. Project activities that would promote project sustainability beyond project lifespan include the design of a financial instrument with BAAC to support farmers with adaptation investments towards climate resilient agriculture, knowledge-sharing and training activities for farmers to support market access, and the establishment of an online market platform to increase visibility and marketability of farmers' products.

1.2.4. Contribution to the regulatory framework and policies

30. The integration of climate risk knowledge into policies and planning across the water and agriculture sectors would be sought through the provision of technical training to 30 staff from TMD, RID and relevant agencies on integration of climate change knowledge into water management and agriculture planning, and through the development of tailored climate information products to inform planning.

1.2.5. Scalability and replicability

31. Given that the entire country faces similar challenges regarding floods and droughts, and considering that capacity-building would benefit national institutions (RID and TMD), the project has good potential for scaling up project activities to other regions of Thailand without equally increasing the total costs of implementation. Parallel financing from BAAC in the form of credit for farmers practicing climate-informed and sustainable agriculture would also support scaling up. However, scaling up of practicing precision agriculture without increasing the total cost of implementation appears over-optimistic.

32. The proposal presents a clear theory of change describing the existing barriers and root causes that undermine the capacity for mitigating flood and drought risks, and how they relate to project activities.

1.3 Sustainable development potential

Scale: N/A

1.3.1. Environmental co-benefits

33. The most significant environmental co-benefits would come from the implementation of EbA measures and conservation agriculture, and include reduced soil erosion, improved surface water quality and groundwater recharge, and enhanced biodiversity.

1.3.2. Social co-benefits

34. The main social benefits include increased food and water security, which would translate into enhanced food security and better health conditions of farming families.

1.3.3. Economic co-benefits

35. The project's support to climate resilient agriculture and to expanding access to markets and financing would enhance income generation for the vulnerable farmers targeted by the project, and possibly generate jobs for young people.

36. Enhanced climate forecasting and the use of the IRDM tool, together with the development of communication tools, would provide people with more time to safeguard their assets or harvest the crops to avoid damage from an impending extreme weather event.

1.3.4. Gender-sensitive development impact

37. Existing gender inequality and provisions to ensure equal gender participation and mitigation of any negative impact on women and girls are clearly explained in the Gender Assessment and Action Plan.

1.4 Needs of the recipient

Scale: N/A

1.4.1. Vulnerability of the country and climate rationale

38. The proposal argues that wet seasons are becoming more intense, with the target basin being increasingly unable to manage slow run-off to the central plains and the greater Bangkok area. Further, the proposal states that dry seasons are becoming drier, with the water that the infrastructure is able to retain being insufficient to cover the irrigation needs of farmers. The increased run-off of more intense precipitation events could also be accelerating soil erosion and impacting groundwater resources.

39. The climate rationale presented in a previous submission of this proposal was questioned by the independent TAP for using gridded data sets instead of real observed data from stations for evaluating historical trends. The present submission states that it "utilized a daily rainfall and air temperature dataset from 17 stations within the Chao Phraya basin provided by Thai Meteorological Department" complemented by "freely available global gridded data products".²¹ This observed data complemented with gridded data is used for trend analysis, for calculating extreme rainfall, temperature and drought indices, and for validating climate models. Model-based outputs are only useful when a model is selected on the basis of its performance to reproduce observational data trends. The process is called "validation", and is intended for the country or the region in question. If a model output is used without proper validation, the outputs are inadequate for making projections on which to base and design a project.

40. The validation of the model ensembles used for climate projections is made against computer-assisted reanalysis data sets²² developed by the Climate Research Unit (CRU) and the Global Precipitation Climatology Centre (GPCC). The feasibility study presented in the first submission was missing a comparison between observed and gridded data. The AE later submitted charts for monthly and annual rainfall and monthly and annual mean temperature showing a comparison between data from observations, CRU, GPCC and Climate Hazards group Infrared Precipitation with Stations (CHIRPS).²³ Since the CHIRPS dataset takes into account satellite observations of rainfall data, calibrated against point-source based observational datasets, the alternative validation of rainfall data involving computer generated gridded datasets is acceptable. The proposal deals with seasonal high rainfall and run-off to cause floods as well as a dry seasonal lack of rainfall leading to high evapotranspiration and droughts.

²¹ Annex 26, section 1.3.1.

²² Annex 26, figures 14a, 14b, 15a, 15b.

²³ AE responses to TAP comments, 29 June 2021.

Therefore, the use of CHIRPS rainfall datasets for the validation of gridded data and the use of the latter for the validation of projection models is also acceptable.

41. The study concludes that while precipitation has become less frequent in most parts of Thailand, with a significant reduction in the occurrence of consecutive wet days, and total annual precipitation remaining mostly unchanged, precipitation events have become more intense.²⁴ Such claims, however, remain questionable in the absence of rainfall intensity analysis involving the peak rainfall season. The trends shown in the study are mostly based on a non-significant relationship involving rainfall-related indices, and their extent of change is not very high. For example, the number of rainy days per year and in May to October fell by only one day over a full decade, and the simple daily intensity index increased by only 0.17 mm/day/decade. Regarding rainfall historical data, annex 26 concludes that: "Trends are weak and inconsistent and should not be interpreted as systematic drivers behind historical extreme events (such as the 2011 flood), nor as expressions of anthropogenic climate change."

42. Despite such weaknesses in trend analyses, it is easily recognized that the total rainfall is increasing against a reduction in the number of rainy days, which has resulted in intensification of rainfall during the rainy season. In such a condition, rainfall run-off will increase and tend to generate flow volumes to overtop riverbanks and cause floods.

43. Flood is a phenomenon that occurs when there is more water than available drainage capacity. In a river basin, flood generally occurs as a seasonal phenomenon, when the water flow at any given point exceeds drainage capacity and water level rises above a locally perceived 'danger level'. The normal flood can become an 'extreme flood' event if the water level surpasses a perceived threshold beyond the known and defined danger level. Flood can also occur due to congestion, especially when the river is choked due to siltation or other obstacles. Urban floods occur when there is an extreme rainfall event (e.g. equal or more than a rainfall event surpassing an upper threshold in several consecutive hours; for example, the accumulation of 100 mm or more rainfall in about 24 hours) that exceeds the urban drainage capacity. In a river basin, unusual and sudden (flash) flood can occur if a large volume of water is drained (for some reason, like breaking of a dam, or outburst of a cloud, or breach of a glacier lake to trigger a glacial lake outburst flood) beyond a threshold in excess of the drainage capacity of any given point in the river basin. In a previous submission, the independent TAP could not find any supporting information in the funding proposal defining flood in the target river basin. No relationship was made in the proposal between any defined extreme rainfall event and a given water flow threshold beyond which floods occur. There was no sense of cause and effect in the discussion in the feasibility study. The present submission defines flood as when the flow of the Chao Phraya River exceeds 2,800 cubic metres per second (m^3/s), which happened during the devastating 2011 floods.

44. The feasibility study presents data, showing that the validation of the models used for climate projections also show the projections for the MRI-AGCM3.2s atmospheric model. This is used for rainfall projections, and discharge projections are obtained from these through hydrological modelling. The feasibility study presents the results from a study by Wichakul et al. (2014), which estimated that there would be increases in run-off of more than 20 per cent at the beginning of the Chao Phraya River due to increases in mid-season rainfall. Available data²⁵ show the maximum daily discharge corresponding to different return periods for present climate, near future climate and future climate for four locations. As per the data presented by the AE, at the beginning of the Chao Phraya River, a flow of 2,800 m^3/s has a return period of five years at present. In the near future a flood with a return period of five years would produce

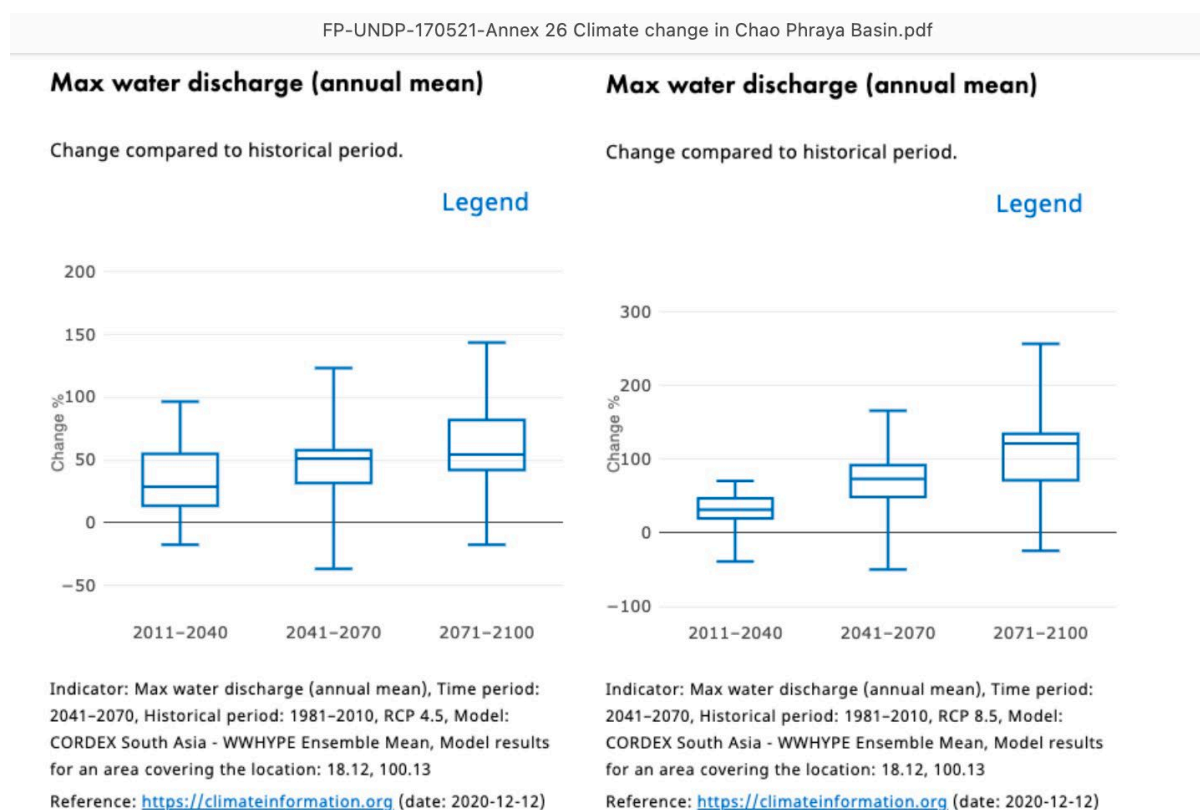
²⁴ Feasibility study, page 40.

²⁵ Annex 26, figure 29.

a discharge of 3,300 m³/s. This clearly exhibits that under climate change, flood intensity will significantly increase due to a significant increase in run-off.

45. Also in annex 26, in section 6.2, which deals with mitigation of flood risks, it is concluded that: “Given the results of the modelling above and the uncertainty associated with future projections of rainfall it is recommended that interventions to mitigate flood risks be based on current flood risks, known areas where flooding risks pose a threat, and to the extent possible allow for flexibility to adapt to changes in future rainfall distributions and flooding as they evolve.” The AE clarified that project interventions “have been selected as highest priority activities, based on in-depth modelling of current flood risks by the Thailand Government...”. Since the Yom-Nan river basin is managed by hydraulic structures, hydrological modelling must show at which levels of discharge beyond an upper threshold flooding actually occurs beyond the operation of the management structures. The AE clarified that the maximum discharge levels presented in figure 79 of the feasibility study for each section of the Chao Phraya River represent flood thresholds and take into account the current water management infrastructure.²⁶

Figure 19: Changes in maximum water discharge at Phrae (northern Chao Phraya) for different future periods under RCP4.5 (left) and RCP8.5 (right)



46. The independent TAP acknowledges the fact that, as per the above-mentioned modelling results, the possibility of greater flooding in future decades cannot be ruled out and the system will only get wetter, despite the two regulators in the Yom-Nan river basin. Under such circumstances, even under the RCP4.5 scenario, the investment in water infrastructure (i.e. flood regulators) can be considered to be a climate-driven investment, despite having a weak evidence base regarding flooding induced by climate change.

²⁶ AE responses to TAP comments, 21 July 2021.

47. Usually in monsoon-influenced regions across the globe, agricultural drought as a phenomenon occurs, if at all, only in non-monsoon months. In the broader climatological perspective, Thailand's climate is influenced by the El Niño-Southern Oscillation (ENSO). Therefore, the potential influence of the El Niño phase of ENSO in contributing to increased droughts cannot be ruled out, especially under climate change. Precipitation in non-monsoon months was found to be increasing (low confidence) at a rate of 10.8mm/decade.²⁷ Increased rainfall during the non-monsoon period would not exacerbate flooding, rather it might compensate for evapotranspirative losses from the topsoil, thereby reducing drought impacts. This increase in rainfall, as well as the increase in temperature over all the months, have been analysed to examine the past and projected drought-related indices such as the standard precipitation index (SPI), standardized precipitation and evapotranspiration index (SPEI) and potential evapotranspiration (PET). While SPI and SPEI are indices widely used for understanding landscape-level drought, PET is widely used for understanding crop-specific drought and for calculating crop water deficits – a quantitative index for analysing droughts.

48. Although annex 26 (Climate Change in Chao Phraya Basin) concludes that “Trends in extreme rainfall indices are weak, and do not show any consistent sign, nor clear spatial pattern,” the PET analyses exhibit an increasing drought intensity in the near future (2021–2050) for all months. This indicates that the perceived compensation from increased rainfall during the wet season would not be adequate to prevent the occurrence of drought, which would require irrigation water for dry season paddy cultivation in the target areas. Moreover, the 12-month SPEI trends in the four sub-basins including the Yom-Nan basin clearly show that the projected values in future years would continue to decline, leading to higher levels of persisting droughts. Accordingly, the estimated water deficits would be likely to increase under such drought conditions, as presented by model outputs.

1.4.2. Vulnerable groups

49. The project targets farming households and farming communities, which are the groups most vulnerable to flooding and drought in Thailand. The design phase of the project included a survey that revealed that 96 per cent of households in the Yom-Nan River basin were dependent on rice farming as a source of income. Selection criteria prioritized vulnerability and urgency, considering the proximity of communities to the river basin and their reliance on water for agriculture.

1.4.3. Economic and social development

50. Thailand's rural population is mostly engaged in agriculture, which generates over 10 per cent of the gross domestic product, and accounts for 74 per cent of water consumption. Almost 6 million households in Thailand depend on agriculture as their source of income. The agricultural sector in Thailand employs approximately half of the country's population and is, overall, dominated by smallholder farms which, in the project area, have an average land size between 4.00 and 4.64 hectares.

51. In 2015, Thailand became an upper-middle income economy with average per capita annual income of USD 5,775. Over the last four decades, Thailand has made remarkable progress in social and economic development. Poverty declined from 67 per cent in 1986 to 11 per cent in 2014. However, income inequality is evident between urban and rural populations. Poverty is higher in the northern-central region, where the target area is located.

1.4.4. Absence of alternative sources of financing

²⁷ Feasibility study, page 32.

52. Given the public good nature of benefits, and that project interventions would not generate direct revenues, alternative financing options to address climate change adaptation in the agricultural and water sectors are limited. The proposed line of credit of USD 16 million from BAAC through financial products with incentives for farmers to practice climate resilience may address climate finance availability from financial institutes; however, this does not ensure that it will address current issues and challenges of accessibility and affordability of climate finance for small farm holdings which neither the feasibility study nor the funding proposal have presented clearly.

1.4.5. Need for strengthening institutions and implementation capacity

53. The need for strengthening institutions, including technical and management skills and inter-agency coordination and collaboration are recognized and widely addressed in the proposal through the capacity-building and institutional strengthening activities described.

1.5 Country ownership

Scale: N/A

1.5.1. Alignment with national climate strategy and policies

54. As described in the proposal, the project is aligned with the country's priorities and policies including Thailand's 20-year development plan, as well as the Water Management Strategy, the Technology Needs Assessment report formulated in 2012, the 12th National Economic and Social Development Plan (2017–2021), the national Climate Change Master Plan (2015–2050), the Climate Change National Adaptation Plan and Nationally Determined Contribution.

1.5.2. Capacity of accredited entities or executing entities to deliver

55. The accredited entity, the UNDP, has worked with RID on a number of projects, including the ongoing Climate Change Finance project, related to RID's capacity to formulate climate-informed projects and secure finance for implementation. UNDP is also working with the Office of Agriculture Economics of the Ministry of Agriculture and Cooperatives, in the "Integrating Agriculture in National Adaptation Plans (NAP-Ag)" programme.

1.5.3. Engagement with civil society organizations and other relevant stakeholders

56. The inception phase included several rounds of stakeholder discussions over the period from 2016 to 2019, on adaptation priorities, including the implementation of EbA measures, construction of grey infrastructure and increased climate resiliency for agriculture-based livelihoods. Subnational level decision makers, the private sector, farmers and community leaders were engaged through field surveys and consultation forums. Academic institutions and technical officials within MoAC with expertise in water and agriculture were also consulted in order to inform the project design and seek support for the project interventions. The Stakeholder Consultation Report describes the parties consulted and the feedback received.

1.6 Efficiency and effectiveness

Scale: N/A

1.6.1. Cost-effectiveness and efficiency



57. Given that the benefits represent a public good, and that the project interventions would not generate direct revenues, it is unlikely that GCF financing would crowd out private investments.

58. The proposed financial structure seems reasonable to achieve the project objectives.

1.6.2. Amount of co-financing

59. Co-financing is relatively high: GCF would finance 52 per cent of total project cost and the rest would be financed by the government of Thailand (almost 48 per cent) and the Krungsri Bank (0,3 per cent).

1.6.3. Financial viability

60. The cost-benefit analysis includes four *without project* scenarios: 5 per cent, 10 per cent, 15 per cent and 20 per cent reduction in agricultural production by 2042. As economic benefits, the analysis considers the avoided damage of floods and the increased agricultural production. The resulting net present value and internal rate of return are summarized in the following table provided in the Economic Analysis (tables 17 and 20):

Table 17: Estimated net present value and internal rate of return

Scenario ¹	Net present value (\$)	Internal rate of return (%)
Minus 5% by 2042	36,765,294	26.8
Minus 10% by 2042	41,269,784	28.2
Minus 15% by 2042	45,781,255	29.5
Minus 20% by 2042	50,292,239	30.8

¹ For purpose of clarity, the scenarios presented are the assumed impacts of climate change by 2045 on agricultural productivity in the absence of the project.

61. Even with a simultaneous increase in cost of 20 per cent and a decrease in benefits of 20 per cent, the economic efficiency of the project remains significant:

Table 20: Estimated net present value and internal rate of return

Scenario ¹	Net present value (\$)	Internal rate of return (%)
Minus 5% by 2042	15,627,504	16.3
Minus 10% by 2042	19,231,096	17.5
Minus 15% by 2042	22,840,273	18.6
Minus 20% by 2042	26,449,060	19.7

62. The basis for calculating the future flood damages in the *with project* scenario, as presented in table 15 of the economic analysis, was not clear. Questioned by the TAP on this matter, the AE presented a description of the calculation process made to estimate avoided flood damage, which was done by relating recorded flood damages for different return periods with water level at gauging stations and flow.²⁸

63. As indicated in paragraphs 13 and 23 of this assessment, the benefits of replication of IoT, including their O&M, would be extremely difficult for the farmers in view of high initial costs and the absence of any information regarding potential cost recovery by them. The classic cost-benefit analysis does not take such information into account. Even if bank loans are being offered, the penetration of precision agriculture might remain questionable due to cost-related uncertainties. These can make part of the project financially untenable for a large segment of the target beneficiaries.

²⁸ Responses to TAP comments, question 9.

1.6.4. Best practices

64. EbA can be considered a best practice for improving climate resilience in a cost-efficient manner. GIZ is currently developing an EbA Guidebook to be used as a national standard for the systematic identification, prioritization and selection of EbA measures at river basin scale.

65. Potential EbA measures include Monkey Cheek, which is a flood control concept that focuses on building reservoirs to retain rainwater and alleviate discharge, vetiver grass to prevent soil erosion and landslides, the “living weir” concept to stabilize soils and riverbanks, forest restoration using temporary weirs, wastewater treatment using grass filtration, lagoons, wetlands and mangroves, and conservation agriculture practices.

66. Monkey Cheek projects should involve intense public consultations and adequate compensation to farmers who lose their lands to water reservoirs. The details regarding costs and benefits are not provided in the funding proposal and associated documents.

II. Overall remarks from the independent Technical Advisory Panel

67. The available evidence base suggests that both floods and drought might take place in the near future in the targeted basin. Although the extent of increase in intensity of such hazards remains highly uncertain, the continuing occurrence of both phenomena is likely in the Yom-Nan basin.

68. In the baseline period (based on CRU data), the occurrence of drought events (SPEI values) was increasing. Thereby, an increase in drought driven by climate change could be established in the Chao Phraya River basin.

69. Despite a weak evidence base, the data indicate that flooding will likely be a problem in the Chao Phraya basin. The incidence of higher levels of rainfall has been observed (not so much in the Chao Phraya basin itself, but in the country, especially in the southern sub-catchments). The model outputs indicate that run-off in the Chao Phraya basin will increase by 20–25 per cent by the 2040s. The return period of floods at the Chao Phraya headwaters will also become shorter, meaning that there will be more flood incidents in the medium future than in the past. Since there have been past floods, floods will likely continue to occur at a higher frequency.

70. For these reasons, the independent TAP endorses this funding proposal with one condition:

(a) The AE must present, prior to the first disbursement by the GCF under the project, (i) a feasibility assessment (based on existing case-studies and platforms in South East Asia) on the implementation of the project’s proposed utilization of an Internet of Things (IoT) irrigation system for rice cultivation; this should include a costing analysis, review of technical requirements, review of equipment suppliers, plan for development of a Thai version of an IoT irrigation software platform; and (ii) a plan for development and scaling up such IoT irrigation systems by the Thai government as part of its plans for digital transformation and sustainability.

71. The risk of cross posting or poaching of TMD and RID and other relevant agency personnel trained under this project may put at risk the effectiveness of this capacity-building. The independent TAP recommends that appropriate safeguards need to be in place, such as a minimum number of years for remaining in a post.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP170)

Proposal name:	Enhancing climate resilience in Thailand through effective water management and sustainable agriculture
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Thailand
Project/programme size:	Small

Impact potential

We note iTAP's assessment that by addressing both the major concerns of floods and droughts, the project is expected to facilitate climate resilient crop production. The project will, through integration of EbA and infrastructure establish a regime of integrated water resource management for year-round water security.

Below, we address specific comments presented in this assessment, referring also to prior responses.

Paragraph 9

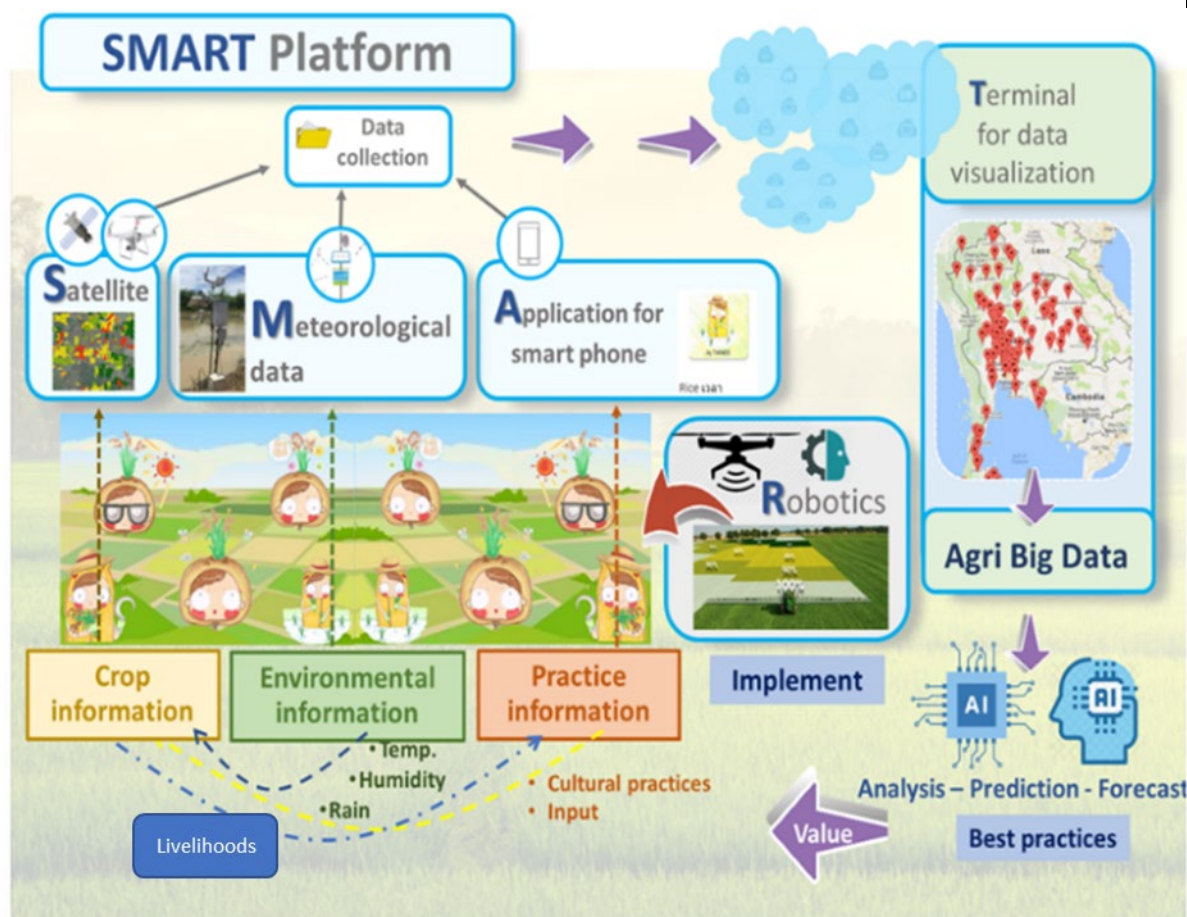
The comment on the per hectare cost of EbA interventions or the estimation of target beneficiaries is well noted. Since the EbA measures will be undertaken in and around the cultivated areas, the spatial coverage is indeed estimated based on these areas. The estimation will be vetted and refined during the participatory process with the communities whereby the final determination of the EbA measures will be made. A comprehensive Annex 18(b) - Lessons Learned from Baseline Projects has been included in the FP package detailing the process by which EbA measures will be identified/selected. In addition, the proposal package includes the safeguards measures to be put in place to protect farmers in the area. Please refer to Annex 6(b) – Environmental and Social Management Framework.

Paragraph 14

We underscore that the government is proposing the establishment of five platforms (each covering an aggregate of 10 farms) to pilot precision agriculture technology across project-targeted farms that will benefit from the investment of capital from GCF funds and O&M by the districts. Through a consultative process, the set of farms will be selected to serve as pilots whereby the farmers benefit from the higher yields and incomes and the data collected from these farms serves the government vision to improve climate resilient agriculture planning in the sector and drive future investments. The platform based on an IoT technology, has already been tested in other SE Asian countries including Thailand, and will address the issue of improved water management in the aggregated irrigation system thereby dealing with increased water demands predicted from increased evapotranspiration and increased temperature. The smallholder farmers are not expected to invest any funds to facilitate this pilot and the O&M, post-project, will continue to be the responsibility of the districts. It is also important to note that drones referred to in the ITAP comments serve multiple purposes,



particularly through remote sensing, data collection on crop health and herbicide application, and are not intended as the source of irrigation for the farms. Irrigated water will be applied through communal system of irrigation channels. The below picture is provided to clarify the use of precision agriculture involving Internet of Things (IoT) and drones:



IoT is indeed an innovative technology (approx. \$1.3M of the total GCF funding request) that is being adopted by the government across its many investments (government and donor funded) and there is a commitment for O&M and vision for upscaling that will ensure that these technologies become cost-effective over time. The precision agriculture interventions will be based on a successful research conducted by the Kasetsart University Thailand in collaboration with the Ministry of Agriculture and Cooperatives and international best practices, such as International Rice Research Institute application in SE Asia. The research indicates the IoT system is especially effectively for multiple purposes:

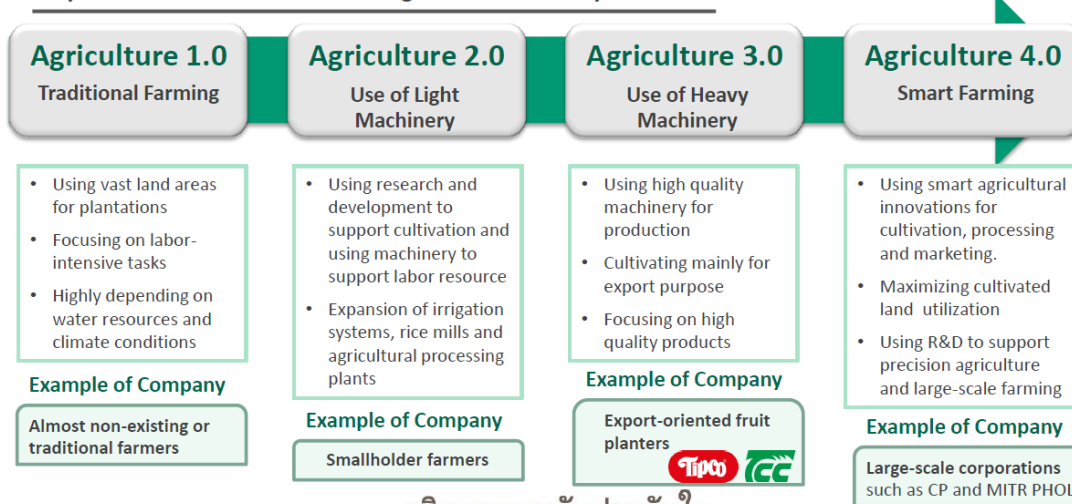
- Assist timing for planting;
- Land preparation;
- Drone usage for remote sensing (crop health data collection)_and efficient herbicide/pesticide application if this is needed;
- Enabling farmers to effectively manage water levels through better management of soil moisture levels (as monitored by IoT sensors)
- Data collection for enabling policy and planning with local government; and
- Climate data forecasting for better decision making.

On iTAP's comment related to cost-benefit analysis, as per standard practice for the projects, the economic analysis for the overall project has been undertaken. This includes the cost-benefit analysis of farm-level precision agriculture platforms – that include IoT investments and drone technology – and it does result in a benefit-cost ratio greater than one and a positive NPV [net present value] for the project in its entirety. This analysis includes the full costs of investment (upfront capital costs) along with estimated annual O&M costs of this activity. The primary benefit for farmers that adopt the technology platform is increased farming income as the investment allows them to grow higher yield varieties and in marginal seasons. This is all under the overhang of a changing climate i.e., under the assumption farm incomes will erode as the climate changes.

As acknowledged in the assessment, the intent is to demonstrate results to support Thai Government's strategy to accelerate smart agriculture. The project will support this through testing the application in a new context with the view to working with Universities, private sector, financial institutions, communities and district agencies for upscale.

With regards to replication and upscaling, the expectation is not that the project-targeted smallholders will adopt this technology en-masse. Rather, as part of the vision, the results will support early adoption and uptake by larger farms or groups of farms in Thailand, including through support from the BAAC credit thereby easing the way for a wider market for smart agriculture. This paves the way for innovative technologies to take root, leading to overall reduction of costs and wider adoption in the medium- to long-term. Please see the transition plan below from the National Strategy Thailand 4.0 to shift traditional agriculture into smart agriculture, highlighting the need for IoT and precision farming.

Key Characteristics of Thailand's Agricultural Industry



บริการทุกระดับประทับใจ

Source: Asian International Journal of Social Sciences

Analysis done in the project will help define the appropriate number of rai (farm area) where precision agriculture results in benefits. Results of the pilot will be shared widely to inform investments by other districts, small farmer groups or by larger farms.

Paragraph 15

We maintain that piloting technology does not necessitate that it is locally sourced. There are no pre-selected firms to provide this support, or preference of international vs local providers. In addition, there are Thai companies working in this space. For example:

<https://www.arv.co.th/varuna/>. During project implementation, an open, competitive procurement process will ensure cost-effectiveness and value for money, with considerations given to training, maintenance and warranties available.

Paragraph 17

We recognize that O&M costs related to IoT and drones are not fully clear as it is a new industry and the estimations are based on existing case studies. Dealers are not expected to cover O&M per se, but protections offered by warrantees would affect actual O&M costs. It is to be noted that in the project, district authorities would be responsible for O&M and related resource commitments (e.g., personnel/finance). Further, before any investments are commenced, the districts will be expected to provide a letter of support for long term O&M (beyond the project and supplier contracts), as the O&M estimates based on final site-selection and supplier commitments are finalized during implementation.

Paradigm shift potential

We note the iTAP's acknowledgement of the pioneering and innovative approach the project is advancing for water management, integrating grey and green approaches. We also note that project is promoting an innovative approach in integrating climate risk management into IWRM.

Regarding iTAP concerns on the approximately US1.3M estimated for introduction of innovative precision agriculture technology in the project, we address specific concerns below. (Also refer to the response in the previous section)

Paragraph 24

It is indeed, as iTAP notes, true that precision agriculture is still in the introductory stage in Thailand. We would like to posit that new technologies require testing; new innovations do not have the network of retailers, repair and service entities as one would find with well tested technology. The early adoption and piloting of innovative technology should not be evaluated in terms of standalone cost-effectiveness, since it is not at a stage for mass rollout (when the technology would not be described as 'innovative').

The Government of Thailand has requested UNDP to help facilitate the testing and piloting of technology to support its National Strategy-Thailand 4.0: the government's industrial economic model, which aims to shift traditional agriculture into smart agriculture. The advancement of technology in agriculture is also in line with Thailand's 20 year Agricultural Development Plan (2017-2036) and "MOAC's Digital Agriculture Strategic Plan (2017-2021)". The market value of smart farming in Thailand amounted to around 128.7 million U.S. dollars in 2018, and was forecasted to reach 269.9 million U.S. dollars in 2022. There is growing literature to suggest precision agriculture can be viable for Thailand, notably: "Small Scale Farmers under Thailand's Smart Farm System" by Ministry of Agriculture and Cooperatives; "Precision Agriculture for Rice Production in Thailand" by Kasetsart University; "A comprehensive solution for landscape-scale sustainable water management in rice" by International Rice Research Institute. Around the country, there are many examples and piloting projects with industry partners on precision agriculture and usage of drone for farming: for instance, telecommunications firm DTAC is working with Department of Agricultural Extension on new IoT-based precision farming solution. In recognition of

Thailand's interest in precision farming, an international conference is planned for January 2022, providing political and financial momentum for the country to accelerate these innovations. (<https://waset.org/advances-in-precision-farming-technologies-conference-in-january-2022-in-bangkok>)

In light of this, we stress that the project intends to focus on paving the way for the paradigm shift envisioned by the Thai Government. The iTAP review comments state that: "Although innovative, it is costly (both from a CapEx and an OpEx basis), thereby limiting diffusion of this advanced technology to low-income small farmers in the region." It is not expected that smallholder farmers in the project would upscale to full IoT individually or immediately. A few aggregate farms benefit from the introduction of the technology, with full support for capital and O&M costs (beyond the project), while serving as pilots that allow the government to consolidate successes and failures on their path to digitalization of agriculture over time.

Paragraph 27

Regarding concern of staff poaching, capacity building in the relevant departments would be sustained through the following approaches: (i) Training of Trainers approach for the department staff and (ii) Codification of the training materials and dissemination. These approaches could help address any staff turnover by institutionalizing the training and, to an extent, broadening the reach of the capacity building efforts.

Sustainable development potential

We note iTAP's acknowledgement of the environmental (reduced soil erosion, improved surface water quality and groundwater recharge, and enhanced biodiversity), social (enhanced food and water security and better health conditions), economic (enhanced income generation and safeguarding against loss of assets), and gender (provisions to address gender inequality) benefits of the project.

Needs of the recipient

Comments from iTAP are well noted on the reach of the project to benefit the most vulnerable and its contribution to socio-economic development.

We appreciate the discussions with iTAP and the opportunity provided to clarify several aspects of the climate rationale for the project. As noted above we have used a combination of multiple (all globally available) sources of both observed and projected future climate data, including Thai weather stations from the basin, CRU, GPCC, CHIRPS and ensembles of CMIP 5 & 6 and CORDEX climate models, both for RCP4.5 and RCP8.5 (see Annex 2, Feasibility Study, sections 1.7.3 & 1.7.4.). We used these multiple data sources, a calibrated hydrological model of the Chao Phraya basin and available literature to assess the evidence for increases in rainfall intensity, surface runoff and water discharge (to assess the likelihood that flooding risk will increase under climate change). All data sources and variables indicate likely increases in flooding hazard and risk (see FP paragraphs 11, 12, 16 & 17 and Annex 2, Feasibility Study sections 1.7.4 and 1.7.5). The same multiple data sources were used to evaluate indicators of agricultural drought, including potential evapotranspiration (PET), SPEI, crop and irrigation water requirements (CWR & IWR), all of which indicated increasing agricultural drought risk (see FP paragraphs 13 & 17 and Annex 2, Feasibility Study sections

1.7.4 and 1.7.5). As recommended within the recently released IPCC AR6 WGI report (chapter 10), we compared these different sources of data and information to ensure that observed trends and projected future changes (from bias corrected GCMs and RCMs) were consistent and not contradictory in terms of the expected direction (sign) of change, indicating whether climate hazards and hence climate-induced risk will increase. As iTAP acknowledges, these indicators and models consistently demonstrate that flooding risk (through increases in rainfall intensity) and agricultural drought risk (through increases in PET, CWR and IWR) are likely to continue to increase over the Chao Phraya basin in the future, thereby providing the necessary rationale for project interventions to address these expected elevated climate risks (see FP, paragraph 17 for a summary).

Paragraph 51

iTAP's comment on (d) Absence of alternative sources of financing is noted. Challenges of accessibility and affordability of climate finance for small farmers is a global development challenge. In this project, the credit line provided by BAAC is new and additional, developed specifically to complement this project's interventions, with the GCF investment allowing for crowding in of this co-financing. The project will work with BAAC to design the credit to ensure that project beneficiaries will have better/enhanced access. Credit will support investments in adaptation (e.g. farm ponds, precision agriculture, etc.) taking into account the financial capacity of the farmers. Krungsri bank will also partner with the project to provide the financial literacy training to the project beneficiaries. The capacity building and investments for climate resilience, financial literacy training from Krungsri bank and credit line provided by BAAC will together facilitate a holistic approach to build creditworthiness and enable access to finance for the beneficiary farmers. This could also serve as a model for BAAC to replicate and scale up their loan programme in other regions.

Country ownership

Comments from iTAP are well noted.

Efficiency and effectiveness

iTAP's concern on efficiency of precision agriculture is noted. Refer to previous response in Impact Potential section.

Overall remarks from the independent Technical Advisory Panel:

We thank the iTAP for the endorsement of the proposal, noting the condition and the recommendation as below:

iTAP Condition: *"The AE must present, prior to the first disbursement by the GCF under the project, (i) a feasibility assessment (based on existing case-studies and platforms in South East Asia) on the implementation of the project's proposed utilization of an Internet of Things (IoT)*

irrigation system for rice cultivation; this should include a costing analysis, review of technical requirements, review of equipment suppliers, plan for development of a Thai version of an IoT irrigation software platform; and (ii) a plan for development and scaling up such IoT irrigation systems by the Thai government as part of its plans for digital transformation and sustainability.”

We accept the condition, as per the elements outlined, noting the assessment is intended for the entire IOT platform (inclusive of software and hardware requirements). We also note the timeline stipulated by ITAP for the fulfilment of the condition prior to first disbursement.

As per the Government of Thailand’s request to facilitate the introduction of precision agriculture technology, the project supports testing and piloting of precision agriculture and IOT technologies. The project is fully aligned with the National Strategy-Thailand 4.0: the government’s industrial economic model, which aims to shift traditional agriculture into smart agriculture as well as the government’s 20 year Agricultural Development Plan (2017-2036) and “MOAC’s Digital Agriculture Strategic Plan (2017-2021)”. In line with these strategies and plans, the requested assessment will build on best practices from regional and country pilots and case studies, as well as growing research on precision agriculture in Thailand (by Ministry of Agriculture and Cooperatives (MOAC); Kasetsart University; International Rice Research Institute, among others).

iTAP Recommendation: *“The risk of cross posting or poaching of TMD and RID and other relevant agency personnel trained under this project may put at risk the effectiveness of this capacity-building. The independent TAP recommends that appropriate safeguards need to be in place, such as a minimum number of years for remaining in a post.”*

We note ITAP’s concern and have consulted the government. The project will facilitate sustained capacity building in the relevant departments through the following approaches: (i) Training of Trainers approach for the department staff and (ii) Codification of the training materials and dissemination. These approaches would help address sustainability issues by institutionalizing the training and broadening the reach of the capacity building efforts.



Gender Assessment, Action Plan and Budget

Enhancing climate resilience in Thailand through effective water management and sustainable agriculture

I. Introduction

This gender assessment aims to provide an overview of the gender situation in Thailand, with a specific focus on supporting climate resilience and transformational change in water management and sustainable agriculture, identifying gender issues that are relevant to the project, and examining potential gender mainstreaming opportunities at the policy and project level. The assessment is based on available data from studies conducted by the Royal Thai Government (RTG), UN agencies, and multilateral development agencies; and includes:

- Undertaking a desk review on gender-related studies conducted for Thailand
- Drawing on earlier stakeholder consultations [and engaging potential female beneficiaries of the project and incorporating feedback into the design of the proposed project]
- Incorporating information and lessons learned from past studies and assessments on gender in Thailand;
- Documentation on Thailand's relevant gender and climate commitments under the SDGs, CEDAW, and the Paris Agreement (including Nationally Determined Contributions)
- Integrating gender considerations in the project indicators, targets and activities.

II. Gender and Climate Change Related Risks in Thailand

Thailand is considered one of the sixteen countries in the "extreme risk" category that are most vulnerable to the impacts of climate change over the next thirty years (Thailand NDC). Thailand's agriculture sector is particularly at risk with 55% of the nation's total area under agricultural use. Changes in climatic conditions have the potential to destabilize agricultural productivity and impact farmers' incomes and lives.^{1 2}

While the impacts of climate change affect the agricultural sectors and the country, climate change and its impacts are not gender neutral. Inequitable access to and control over natural resources, particularly land, water, and forests, as well as exclusion from the policy- and decision-making processes that affect resource distribution and their livelihood. These disproportionately affect the poor, particularly poor rural women, members of indigenous and ethnic minorities, who rely on natural resources for the livelihood.

¹ Talberth, J., Reyta, K., 2014. Climate Change in the Lower Mekong Basin - An Analysis of Economic Values at Risk, USAID Mekong Adaptation and Resilience to Climate Change (USAID Mekong ARCC). World Resources Institute, Washington, D.C.

² Plan International. 2018. Climate change, young women and girls: Vulnerability, impacts, and adaptation in Northern Thailand.

Women as well as men are also both important change agents and critical actors in actions to address climate change and in adopting adaptation options to strengthen resilience of communities and households. Their potentially different knowledge and skills as well as needs and priorities are important to consider in the the design, implementation, monitoring, and evaluation of climate change and natural resources projects.

III. Existing Gender Inequality in Thailand

The **Gender Inequality Index (GII)** measures gender inequalities across three important dimensions of human development—*reproductive health*, measured by maternal mortality ratio and adolescent birth rates; *empowerment*, measured by proportion of parliamentary seats occupied by females and proportion of adult females and males aged 25 years and older with at least some secondary education; and *economic status*, expressed as labour market participation and measured by labour force participation rate of female and male populations aged 15 years and older. The GII exposes differences in the distribution of achievements between women and men and measures the human development costs of gender inequality. The higher the GII value the greater the disparities between females and males and the greater loss to human development.

The 2017 GII sheds light on the position of women in 160 countries; it yields insights in gender gaps in major areas of human development. The component indicators highlight areas in need of critical policy intervention and stimulate proactive thinking and public policy to overcome systematic gender-based disadvantages.

Thailand has a GII value of 0.393, ranking it 93 out of 160 countries in the 2017 index (see Table 1). In Thailand, 4.8 percent of parliamentary seats are held by women, and 42.4 percent of adult women have reached at least a secondary level of education compared to 47.5 percent of their male counterparts. For every 100,000 live births, 20 women die from pregnancy related causes; and the adolescent birth rate is 51.9 births per 1,000 women of ages 15-19. Female participation in the labour market is 60.5 percent compared to 77.3 for men.³ In comparison, the Philippines and Viet Nam are ranked at 97 and 67 respectively on this index. Unemployment rate for youth (2019) is men 3.4%, women 5.3%, while the share of youth not in employment, education or training (2019) is men 11.2%, women 18.6%.⁴

³ UNDP. 2018 Human Development Indices and Indicators: 2018 Statistical Update. Briefing note for countries on the 2018 Statistical Update

⁴ ILOSTAT, 2019

Table 1: Thailand's GII relative to selected countries and groups (2017) Source: UNDP. 2018 Human Development Indices and Indicators: 2018 Statistical Update. Briefing note for countries on the 2018 Statistical Update

	GII value	GII Rank	Maternal mortality ratio	Adolescent birth rate	Female seats in parliament (%)	Population with at least some secondary education (%)		Labour force participation rate (%)	
						F	M	F	M
Thailand	0.393	93	20	51.9	4.8	42.4	47.5	60.5	77.3
Philippines	0.427	97	114	60.5	29.1	76.6	72.4	49.6	75.1
Viet Nam	0.304	67	54	27.3	26.7	66.2	77.7	73.2	83.5
E Asia and the Pacific	0.312	-	62	22.4	19.8	67.8	75.5	60.1	70.3
High HDI	0.289	-	38	26.6	22.3	69.5	75.7	55.0	75.5

Maternal mortality ratio is expressed in number of deaths per 100,000 live births and adolescent birth rate is expressed in number of births per 1,000 women ages 15 – 19.

Based on the sex-disaggregated Human Development Index (HDI), the **Gender Development Index (GDI)** is defined as a ratio of female to male HDI. The GDI measures gender inequalities across three dimensions of human development; health (measured by women's and men's life expectancy at birth); education (measured by expected years of schooling for children and mean years for men and women aged 25 years and older); and command over economic resources (measured by estimated Gross National Income GNI) per capita for women and men.

The GDI for 2017 is calculated for 164 countries. The 2017 female HDI is 0.753 in contrast to 0.756 for males. The GDI value for Thailand is 0.996 compared to the Philippines (1.000) and Viet Nam (1.005) (See Table 2).

Table 2: Thailand's GDI for 2017 relative to selected countries and groups Source: UNDP. 2018 Human Development Indices and Indicators: 2018 Statistical Update. Briefing note for countries on the 2018 Statistical Update

	Life expectancy at birth		Expected years of schooling		Mean years of schooling		GNI per capita		HDI value		F-M ratio
	F	M	F	M	F	M	F	M	F	M	GDI Value
Thailand	79.3	71.8	14.8	14.5	7.4	7.8	13,793	17,327	0.753	0.756	0.996
Philippines	72.8	65.9	12.9	12.3	9.5	9.2	7,582	10,705	0.699	0.698	1.000

Viet Nam	81.0	71.8	12.9	12.5	7.9	8.5	5,345	6,383	0.696	0.692	1.005
E Asia and the Pacific	76.7	72.8	13.5	13.2	7.6	8.3	10,689	10,568	0.717	0.750	0.957
High HDI	78.2	74.0	14.3	13.9	8.0	8.6	10,945	18,948	0.740	0.773	0.957

The OECD's Social Institutions and Gender Index (2019) considered 12 key indicators over the life cycle to show how Thailand (among other countries) performs across childhood, youth, adulthood, and older age. The indicators cover education, labour market and work, political representation, time use and social protection. Three indicators are for women only and the rest are given in the form of female-to-male ratio. Table 3 below shows how Thailand performs and includes the Philippines and Viet Nam for comparison. Thailand performs better than at least two thirds of countries (i.e., it is among the top third performers), better than at least one third but worse than at least one third (i.e., it is among the medium third performers), and worse than at least two thirds of countries (i.e., it is among the bottom third performers).

Table 3: Summary of Thailand's performance on the Life-course gender gap dashboard relative to selected countries (Source: OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.)

	Childhood & youth (5 indicators)			Adulthood (6 indicators)			Older age (1 indicator)			Overall (12 indicators)			Missing indicators
	Top Third	Mid Third	Bottom Third	Top Third	Mid Third	Bottom Third	Top Third	Mid Third	Bottom Third	Top third	Mid third	Bottom third	
	Number of indicators												
TH	1	1	3	3	1	2	0	0	0	4	2	5	1
PH	2	2	1	1	3	0	0	0	1	3	5	2	2
VN	2	1	1	2	2	0	0	0	0	4	3	1	4

TH = Thailand; PH = the Philippines; VN = Vietnam

The Gender Social Norms Index (GSNI) attempts to measure the prevalence of gender biases affecting gender equality. The index includes four dimensions – political, economic, educational, and physical integrity. According to GSNI, 95.47 % show a gender bias in at least one of the dimensions. 74.5 % show biases in two or more dimensions. Physical integrity is the dimension that indicate the strongest gender bias at 84.53%. This dimension uses proxies to measure attitudes to intimate partner violence and reproductive rights. For the political dimension, which measure attitudes to women as political leaders,

67.3% indicate that they have a bias. The economic and educational dimensions show 50.86% and 29.02% respectively.⁵

Per UN Women, challenges in realizing gender equality and the empowerment of women in Thailand remain in the areas of lack of sex disaggregated data, traditional attitudes and stereotypes which underpin domestic violence and violence against women, low participation of women in politics and decision-making positions, discrimination and vulnerabilities of ethnic and rural women as well as women in the informal sector, HIV prevalence, trafficking and exploitation.⁶

In addition, data drawn from draft national assessment on social dimension of CC impact report⁷, provides the following information: National poverty rate (2017) is 7.21% overall. Urban 5.49%, Rural 8.91%. Men 7.59%, Women 6.87%. Higher rates for both genders in the rural areas, 9.23% men and 8.61% women. Lower poverty rates for women in all regions. (National Statistic Office)

5.3 million people are in absolute poverty (living below national poverty line), 5.4 million are near poor (just above national poverty line). 1.5 million households are landless.

The National Economic and Social Development Board (NESDB) in its 2017 report on poverty and inequality describes the general characteristics of Thailand's poor as follows:⁸

- *Individual characteristics:* low education, working in agricultural sector, landless, low income without job benefits
- *Household characteristics:* large-sized, skipped-generation, elderly or female-headed households most vulnerable to poverty with limited opportunity to engage in income generation
- *Concentration:* southern, north-eastern and northern regions, more rural than urban.

Agriculture Sector

Poverty tends to concentrate in farming households. According to the Office of Agricultural Economics (OAE), based on 2013/2014 data, farming households in the northeast have the highest rate of poverty with a regional average of 37.27 percent, followed by the north 26.09 percent, the central region 13.86 percent, and the south 9.57 percent.⁹ One-third of Thai labor force is in agriculture. Nearly 60 million rai of farm land is prone to drought or floods, and over 80% of which is outside of irrigated area.

The 2017 NESDB report also shows the highest incidence of poverty among children and elderly populations. Due to low income in agricultural work, young agricultural workforce migrated to cities for jobs, so by 2013, 1 in 4 Thai farmers were over the age of 55, leading to rise in "skipped-generation"

⁵ UNDP (2020). Tackling social norms: A game changer for gender inequalities.

⁶ <http://asiapacific.unwomen.org/en/countries/thailand>

⁷ Busakorn Suriyasarn & Pawin Talerngsri, (draft) *Social Dimension of Climate Change Impacts in Thailand: Analysis of Risks, Policy, Planning and Finance*, UNDP (forthcoming).

⁸ NESDB, รายงานการวิเคราะห์สถานการณ์ความยากจนและความเหลื่อมล้ำในประเทศไทย ปี 2560, September 2018, p. ข.

⁹ 2013/2014 figures. Office of Agricultural Economics, ภาวะความยากจนของครัวเรือนเกษตร ปีเพาะปลูก 2556/7, <http://www.oae.go.th/view/1/ภาวะความยากจนของครัวเรือนเกษตร/TH-TH>

households (grandparents living with grandchildren). There are some indications that there are more single elderly female-headed skipped-generation households than the male-headed ones.

In 2013, 1.4 million single-parent households (7.1% of all households); majority single-mother often in difficult situation. 400,000 skipped-generation households (2% and growing trend) (NESDB, 12th Development Plan). A growing number of Thai elders (38%, even male-female split) are living in vulnerable family situation (living alone, with elderly spouse or with grandchildren). 63% living in rural area. However, more elderly female heads in skipped-generation households (meaning more elderly women with grandchildren). About 20 percent of elders in skipped-generation households have income below the poverty line¹⁰ - more than twice the national poverty rate.

In the OAE analysis, factors contributing to poverty in farming households include geographical region, access to irrigation, age and education of head of household, number of family members, dependent burden, type of farmland, household net assets, and household debt. In the OAE's view, farming households in the northeast and the north are the most likely to be poor and should be prioritized in poverty reduction interventions.¹¹ The National Economic and Social Development Board (NESDB) also stressed the need to reduce poverty with the priority for the north and northeast in the Twelfth Plan.

Roughly 40% of expenses of Thai household debts is spent on basic consumption, and 14% (formal) & 9.9% (informal) for agricultural activities, and 7.9% formal & 32.9% informal for businesses.

While the agricultural sector produces overall only 10 percent of the GDP, it is important for Thailand's rural economy, representing 31 percent of the GDP for the north, 28 percent for the south, 23 percent for the northeast, and 20 percent for the west.¹² GDP in 2014 value. NESDB. 2014 figures. Office of Agricultural Economics.¹³ The agricultural census carried out in 2013 showed an increased proportion of female land holders. In the Northern Region, women made up 28.7% of the landholders in 2013, compared to 20.7% in 2003.¹⁴

Focus group discussions were held in the project sites during project development, including 401 including as a sampling group. The number of female and male respondents was 199 and 202 persons, respectively. According to perceptions, the farmers in the four districts reported on the incidence of climate change related impacts in the long-term shifts over the last 10-20 years and adaptive strategies in coping with the shifts. The survey showed that 93% of the farmers feel a increase temperature and that the temperature becomes hotter, and 84-91% feel the change in rainfall amount and intensity, with greater concerns about drought. In response, a number of autonomous adaptation strategies have been undertaken by farmers. The most common were changing crops and/or crop patterns and timing. Some other strategies used were swapping to crops that were less water-intensive and stopping cultivating for a while to other career,

¹⁰ NESDB, *Twelfth National Economic and Social Development Plan (2017-2021)*.

¹¹ For ethnic dimension of poverty see: John Draper & Joel Sawat Selway, "A New Dataset on Horizontal Structural Ethnic Inequalities in Thailand in order to Address Sustainable Development Goal 10," *Social Indicators Research*, Springer, 2019

¹² GDP in 2014 value. NESDB.

¹³ <http://www.nso.go.th/sites/2014/Documents/info/Infographic59/2.Labor/9.Labor.pdf>

¹⁴ National Statistical Office (2013), [2013 Agricultural Census Northern Region](#), p. 17.

while the most extensively used approach by the farmers was an increasing dependence on groundwater use as a supplementary water supply. Success of autonomous adaptation can be improved through climate information and support to ensure choices are consistent with changing temperature and rainfall patterns.

Gender Division of Labour

Focus group discussions further indicated that women generally do the same activities in agricultural cultivation as men do. This is in addition to the role of caregiver for families and household work. This is consistent with a study in Northern Thailand¹⁵ where it was found that women often hold the primary responsibility for unpaid household labour (e.g. cleaning, cooking, childcare, and caring for home gardens for household food consumption). Another study by the Food and Agriculture Organization of the United Nations (FAO) into gender roles and use of technology in Thailand found that while women were generally involved in all aspects of agricultural production, while men dominate mechanized tasks. Further, women have significantly reduced access to resources (e.g. inputs, funds/credit, seeds, breeds and plant varieties, and information and knowledge)¹⁶ and therefore decision-making power when it comes to agriculture activities.

Water insecurity creates livelihood-related challenges and potential losses of income. A shortage of water can add stress to women's lives through increased workloads (fetching water). Young women and children in Northern Thailand often work support their mothers with household labour. They also contribute to agricultural labour and subsequently household's income. Climate change increases water insecurity, increasing the burden particularly for women and children as primary household caretakers and agricultural labourers.^{17 18} Focus group discussions in the project sites indicate that for managing water usage group, the leaders are usually men. However, roles and responsibilities of women in these groups include vice president, financing and accounting as well as administrative positions.

A recent study indicated that reduced fertility, access to electricity and running water have contributed to easing rural women's lives and labour burdens. Even with these, women in rural Thailand still carry the bigger burden of domestic work than men. While it is several years old and the economic and social

¹⁵ Plan International. 2018. Climate change, young women and girls: Vulnerability, impacts, and adaptation in Northern Thailand.

¹⁶ FAO. 2003. Gender-responsive technology for poverty alleviation in Thailand. August 2003 <http://www.fao.org/3/ae538e/ae538e00.htm#Contents>

¹⁷ Andajani-Sutjahjo, S., Chirawatkul, S., Saito, E., 2015. Gender and Water in Northeast Thailand: Inequalities and Women's Realities. *Journal of International Women's Studies* 16, 200–212.

¹⁸ Plan International. 2018. Climate change, young women and girls: Vulnerability, impacts, and adaptation in Northern Thailand.

contexts have evolved, a 2001 national survey noted that in rural areas, women spent an average 5.4 hours a day on domestic tasks and childcare, compared to 2.9 hours for men.¹⁹

COVID-19 and the extent to which it could impact Thailand must be mentioned. As women are the primary caretakers, illness in the family would add to the responsibilities of women. One of the measures to contain the spread of the virus has been to close schools and care facilities, this is likely to have added to the burden of unpaid care work of women. As at 30 June 2020, there have been 3,171 diagnosed cases in Thailand, 12 of which were in the project sites. Restrictions are slowly being lifted and the related socio-economic impacts will be felt for the longer term. For a population of 65 million, the relatively low number of cases can be seen as a success towards containment. There is however uncertainty going forward as restrictions continue to be lifted and international travel re-opens.

Access to Resources

Thailand's Constitution (Articles 33 and 41) and the Civil and Commercial Code (Section 1336) state that women and men have the same rights to own, use, and make decisions regarding land and non-land assets. The law allows for joint land titling and co-ownership of property for married and unmarried couples. Women and men's equal right to inheritance is protected by Thailand's Civil and Commercial Code and traditionally the youngest daughter inherits most of the property.²⁰²¹ Customary law still influences decision-making in rural areas communities, with communal rules allocating resources, decision-making, and conflict resolution in ways that may disadvantage women's access to land, property and inheritance.^{22 23} The Civil and Commercial Code stipulates that, after divorce, commonly held property (*Sin Somros*) is to be divided equally between man and woman (Thailand, 1925, Sec. 1533). Spouses manage his or her own individual property (*Sin Suan Tua*) (Thailand, 1925, Sec. 1473). In administering common marital property, either the spouses are joint managers, or one spouse must obtain consent from the other (Thailand, 1925, Sec. 1476).²⁴

Women and men have the same rights to open bank accounts, obtain credit, and access formal financial services.²⁵ Government measures such as the National Village and Urban Community Fund extends loans to villages and urban communities and provides vocational training and try to reach women and men.²⁶

¹⁹ Fox, L., Wiggins, S., Ludi, E. and Mdee, A. 2018. The lives of rural women and girls What does an inclusive agricultural transformation that empowers women look like? Bill and Melinda Gates Foundation.

²⁰ FAO (Food and Agriculture Organization of the United Nations), 2017. Gender and Land Rights Database: Thailand. Available at: http://www.fao.org/gender-landrights-database/country-profiles/countries-list/customary-law/en/?country_iso3=THA

²¹ OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

²² FAO (Food and Agriculture Organization of the United Nations), 2017. Gender and Land Rights Database: Thailand. Available at: http://www.fao.org/gender-landrights-database/country-profiles/countries-list/customary-law/en/?country_iso3=THA

²³ OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

²⁴ OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

²⁵ World Bank, 2016. Women Business and the Law: Getting to Equal. Available at: <http://wbl.worldbank.org/~media/WBG/WBL/Documents/Reports/2016/Women-Business-and-the-Law-2016.pdf>

²⁶ OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

Social norms, inequitable power dynamics, and the gendered division of labour affect women's and men's access to resources in Northern Thailand. This includes water.²⁷ Men typically have greater rights of access to, and decision-making power over water. Women in rural areas, particularly poor women of ethnic minority, may face disproportionate water insecurity compared to rural men and people in urban areas due to overlapping sources of vulnerability and marginalization (Plan International, 2018).

A recent study demonstrates some of the challenges faced at the community level around land ownership, the pattern of which is mixed in Thailand. Forty percent of the participants in focus group discussions indicated that, in practice, they have access to, and decision-making power over land but are unsure of their formal ownership status. In 30 percent of the FGDs, participants mentioned that new property (land as well as house) is commonly registered under the wife's name because men tend to avoid paperwork and prefer to work in the field.²⁸

Education

Thailand's 1999 Education Act grants "all children, without discrimination" a right to a quality education and the Ministry of Education supports all children a twelve-year education. From a gender perspective, Thailand has been very successful at promoting the right to girls' education. Between the years 2005 to 2010, the primary school completion rate for girls in school was 89 % while the completion rate for boys was 97 percent. While there are still fewer girls than boys attending primary school, the gender gap has decreased considerably from slightly more than 8 % in 1971 to slightly less than two percent in 2009.²⁹ By 2012, there was little difference in enrolment between urban and rural areas.³⁰

In 1998, the Child-Friendly School (CFS) initiative was launched to promote education for all, and many of CFS's activities are ongoing in Thailand today. Yet, girls continue to face challenges accessing education including poverty, language, and mobility/distance to school. Marginalized populations also face challenges.³¹

From the perspective of higher education, women have more opportunities than men. However, although 53 percent of science students in higher education are female, less than 24 percent of women study engineering, manufacturing and construction studies. In recent years, Thailand's Ministry of Education

²⁷ Plan International. 2018. Climate change, young women, and girls: Vulnerability, impacts and adaptation in Northern Thailand.

²⁸ Aktera, S., Rutsaert, P., Luis, J., Htwe, NM., San, S.S., Raharjo, B., Pustaka, A. 2017. Women's empowerment and gender equity in agriculture: A different perspective from Southeast Asia. in Food Policy, Vol. 69, May 2017. pp 270-279.

²⁹ The Borgen Project. 10 Extremely Important Facts about Girls' Education in Thailand. Blog July 5, 2018. <https://borgenproject.org/facts-about-girls-education-in-thailand/>

³⁰ Fox, L., Wiggins, S., Ludi, E. and Mdee, A. 2018. The lives of rural women and girls What does an inclusive agricultural transformation that empowers women look like? Bill and Melinda Gates Foundation.

³¹ The Borgen Project. 10 Extremely Important Facts about Girls' Education in Thailand. Blog July 5, 2018. <https://borgenproject.org/facts-about-girls-education-in-thailand/> accessed 23 May 2018.

has worked to promote Science Technology Engineering and Mathematics (STEM) education among Thai girls.³²

Health

Thailand has worked to ensure wide access to health services including sexual and reproductive health (SRH) services. A primary health care system expanded to sub-districts beginning in the 1980s and access to health care became more equitable. From a gender perspective, achievements include the following: 80% of pregnant women used prenatal care in 1987 and 92% in 2000. With Universal Health Care (UHC), use reached 100% in 2012. Birth attendance increased from 90.8% in 1990 to 99.7% in 2012. Further, contraceptive prevalence rate (CPR) increased from 14.8% in 1970 to 79.3 % in 2012. The total fertility rate also fell from 5.9 children per woman in the 1970s to 1.6 in 2011.³³

Challenges continue however, including the increase in adolescent birth rate from 31 per 1000 women aged 15 – 19 year in 2000 to 53 in 2012. The adolescent birth rate has sharply increased from 31 per 1000 women aged 15–19 years in 2000 to 53 in 2012. Unmet need for contraception is 6.9% higher among women aged 15–19 years (11.6%) than among women aged 45–49 years (8.3%).³⁴

Women still face challenges accessing safe abortions even though it is permitted by law, when conducted by a physician, if the pregnancy threatens a woman's life or health, or is the result of a sexual crime. Official reports suggest that about 30,000 abortions occur each year (including induced, spontaneous and therapeutic), but other estimates go as high as 200,000–300,000 annually, suggesting that abortion is widely practiced³⁵

Indigenous peoples in rural Thailand experience health inequities as demonstrated by a recent study in Northern Thailand.³⁶ Healthcare is limited or not available in rural areas and indigenous people are less likely to access public health services, leading to negative health consequences. Many women with preventable diseases (e.g. cervical cancer and HIV/AIDS) are indigenous and face discrimination for being women and being indigenous; this discrimination also affects their access to healthcare. Further, the unequal labour distribution in their households can also lead to unequal opportunities for indigenous women to seek healthcare, invest in educational and vocational skills, and participate in paid work.

³²The Borgen Project. 10 Extremely Important Facts about Girls' Education in Thailand. Blog July 5, 2018.

<https://borgenproject.org/facts-about-girls-education-in-thailand/>

³³ Tangcharoensathien V, Chaturachinda K, Im-em W. Commentary: Thailand: sexual and reproductive health before and after universal health coverage in 2002. *Glob Public Health*. 2015;10(2):246–248. doi:10.1080/17441692.2014.986166

³⁴ Tangcharoensathien V, Chaturachinda K, Im-em W. Commentary: Thailand: sexual and reproductive health before and after universal health coverage in 2002. *Glob Public Health*. 2015;10(2):246–248. doi:10.1080/17441692.2014.986166

³⁵ Tangcharoensathien V, Chaturachinda K, Im-em W. Commentary: Thailand: sexual and reproductive health before and after universal health coverage in 2002. *Glob Public Health*. 2015;10(2):246–248. doi:10.1080/17441692.2014.986166

³⁶ Onouma Thummapol, Sylvia Barton, Tanya Park. Healthcare Access Experiences Among Indigenous Women in Northern Rural Thailand: A Focused Ethnographic Study in *Central Asian Journal of Global Health*, Vol. 7, No. 1, 2018.

The impacts of COVID-19 in Thailand and in the world are not yet fully known. The project beneficiaries are expected to be vulnerable agriculture households, many female-headed and including elderly, which are in a high risk category.

Political Participation

Article 65 of the Constitution states that men and women in Thailand have the same rights to hold public and political office in the executive, judiciary and legislative branches (Thailand, 2007). There are no legal quotas at national or local levels. Further, the government provides incentives to encourage women's political participation, including training for female political candidates, public education campaigns, and pushing for quotas within political parties. Even though women have the same rights, they face cultural barriers to political participation. For example, patriarchal norms pose challenges to women who run for office at the local level – norms that expect them to stay at home. Election rigging, corruption, and patron-client relationships also complicate women's advancement in the political arena and their advancement.³⁷

Women are still underrepresented in public and private sectors, including in the parliament, government, judiciary and administration both at national and local levels. Thailand has ranked as one of the lowest countries for gender equality in politics in the region of Asia and the Pacific.³⁸ The percentage of women in provincial and local governments in Thailand is much lower than that of the national government. There is only one female governor from 76 provinces (1.32%).³⁹ Female Chiefs of Provincial Administrative Organizations (PAO) and Sub-district Administrative Organizations (SAO) account for 8% and 6.45% respectively.⁴⁰

In reference to agriculture, a recent study on gender equity in agriculture in Southeast Asia found that women in Thailand were active members of village and agricultural organizations and had direct contact with extension officers. Women's involvement in decision making in rice farming was observed to be greater in Thailand compared to the Philippines, Indonesia, and Myanmar – other countries in the study.⁴¹

Gender-based Violence

Patriarchal norms and attitudes in Thailand contribute to violence against women in the public and private spheres. Thailand does not currently have one overarching law addressing all forms of violence against women. The Domestic Violence Victim Protection Act of 2007 criminalizes domestic violence and perpetrators are liable to imprisonment of up to six months or a fine (Thailand, 2007b, Art. 4). The law extends to current and former partners, and includes physical and psychological violence (Thailand, 2007b). While the definition does not specify sexual and economic violence, these can fall under the law, as domestic violence is broadly defined as "any act done with the intention or in the manner likely to

³⁷ OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

³⁸ <http://www.unwomen.org/en/digital-library/publications/2019/03/women-in-politics-2019-map>

³⁹ Prae Province (appointed October 2019)

⁴⁰ Data from King Prajadhipok's Institute 2017.

⁴¹ Aktera, S., Rutsaert, P., Luis, J., Htwe, NM., San, S.S., Raharjo, B., Pustika, A. 2017. Women's empowerment and gender equity in agriculture: A different perspective from Southeast Asia. in Food Policy, Vol. 69, May 2017. pp 270-279.

cause harm to the body, mind or health of a family member” and “to exert coercion or immoral influence over a family member in order to wrongfully cause him to do, not to do or yield to any act” (Thailand, 2007b, Art. 3).⁴²

Rape is criminalized by the Penal Code of 1965, and the subsequent Criminal Code Amendment Act of 2007. The Amendment Act removed the provision allowing a husband to rape his wife. (OECD, 2019, SIGI Thailand). The Labour Protection Act of 2008 and the Penal Code foresee legal protection in the workplace and the Labour Protection Act extends protection against sexual abuse, harassment or nuisance by superiors to cover all workers, both female and male, with penalties up to THB 20 000 baht (Thailand, 2008). Sexual harassment continues to be an issue in the workplace, and women may be verbally and physically abused in the workplace, pressured to trade sexual favours for career advancement opportunities, and face risks of violence when traveling between their home and workplace.⁴³

Intimate Partner Violence (IPV) is perceived as a private or a family matter, and is considered to be underreported in Thailand. Statistics show that the numbers of reported domestic violence cases have risen from 25,767 to 31,866 between 2010 and 2013 respectively (Thai Health Promotion Foundation 2011), (The Women's Affairs and Family Development 2013). In addition, the report from the OSCC, Ministry of Public Health (MOPH) found that, in 2015, there were 23,977 women who used their services. Recent region-specific data is lacking, however, a report 2011 report indicates a higher proportion in the North and Central regions, and there was a correlation with limited education⁴⁴.

Graph 1: Proportion of married women 15-19 years old who experienced physical violence in the past year, 2009 (Report on Gender-based Violence Against Women and Girl (VAW/G) Indicators; Office of Women's

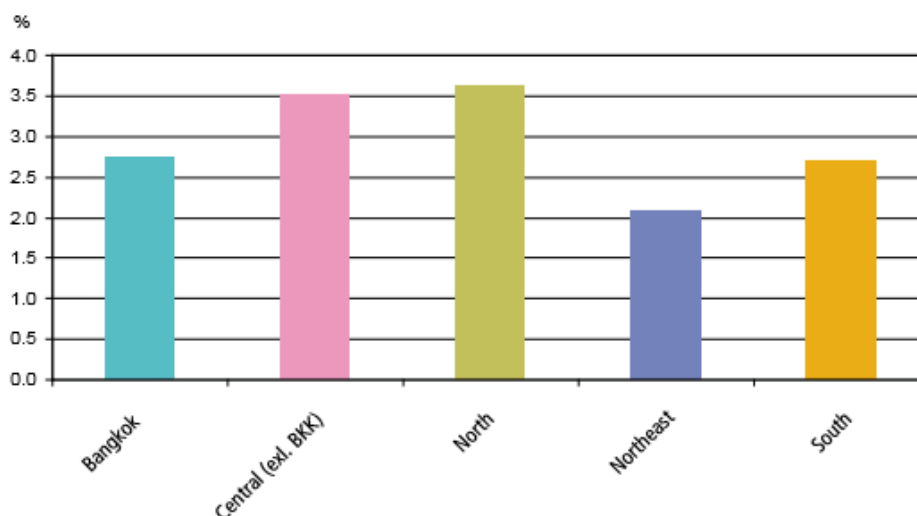
⁴² OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

⁴³ OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

⁴⁴ Report on Gender-based Violence Against Women and Girl (VAW/G) Indicators; Office of Women's Affairs and Family Development, Ministry of Social Development and Human Security, with support from UNDP Thailand, 2011

Affairs and Family Development, Ministry of Social Development and Human Security, with support from UNDP Thailand, 2011)

Proportion of married women 15–49 years old who experienced physical violence in the past year, 2009



The Domestic Violence Act enables government officials to make available medical, social and psychological assistance to survivors of domestic violence. The Thai government had implemented preventative measures including the establishment of shelters and half-way homes for children and families in crisis, One Stop Crisis Centres to provide physical and psychosocial health services to victims of sexual and domestic violence, and a 24-hour hotline. Police officers are said to be trained in gender perspectives and the psychological nature of sexual violence victims, and certain courts have integrated women-friendly measures such as the provision of counselling services and the separation of victims and perpetrators.⁴⁵ The Department of Women's Affair and Family Development Learning Centres hears complaints of gender discrimination and the Committee on Consideration of Gender Discrimination ensures that victims of gender discrimination receive legal remedies and compensation.⁴⁶ However, Thailand has been criticized under CEDAW for the lack of access to justice for survivors of gender-based violence. The criticism includes issues such as law enforcement officials' negative attitudes resulting in failure of register and investigate complaints.⁴⁷

⁴⁵ OECD. 2019. Social Institutions and Gender Index, 2019: Thailand.

⁴⁶ Thailand's Voluntary National Review on the Implementation of the 2030 Agenda for Sustainable Development, June 2018
<http://www.mfa.go.th/sep4sdgs/contents/filemanager/images/sep/VNR%202018%20English%2010.07.18.pdf>

⁴⁷ CEDAW/C/THA/CO/6-7

IV. Legal and Administrative Framework Protecting Women and Gender Equality

From the perspective of key international commitments protecting women and gender equality, in 2000, Thailand ratified the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) in 1985 and its Optional Protocol and endorsed the Beijing Platform for Action (BPFA), the Millennium Development Goals (MDGs), and finally the Sustainable Development Goals (SDGs).

Thailand has made significant efforts to integrate the international principles and instruments into the policy and programming framework, evident in the Constitution B.E. 2550 (2007), which has anti sex discrimination and gender equality provisions.

A Working Group for the Implementation of SDG 5 supports the implementation of the Goal. Representatives from government agencies and civil society such as the Thailand Sustainable Development Foundation, the Association to Empower the Potential of Women with Disabilities (AEPWWDs), the Foundation for Women, Homenet Thailand Foundation for Labour and Employment Promotion and WeMove are all active in the Working Group. The Working Group allows government agencies and civil society to exchange information, as well as discuss future direction in the implementation of SDG 5.⁴⁸

The Strategy for Women's Development Plan (2017 – 2021) is Thailand's main mechanism for promoting women's development; it aligns with SDG 5 (Gender Equality). The Strategy has the vision of "[creating] social equality without discrimination, for women to have good quality of life, be safe and secure." The Government also designed an action plan to implement the Strategy. The action plan focuses on community development, for example, the Vocational Training Program for Women and Families which provides training for the unemployed, educationally disadvantaged, and people at risk of sexual exploitation and human trafficking.

The Department of Women's Affairs and Family Development, of the Ministry of Social Development and Human Security, is the national women's machinery responsible for promoting gender equality and the empowerment of women, and social inclusion. It serves as a secretariat of the National Inter-ministerial Committee on Policy and Strategy for the Improvement of the Status of Women, chaired by the Prime Minister and develops. The Office implements a five-year cycled National Women's Development Plans, which currently falls under the period of 2012-2016.⁴⁹

In 2015, Thailand passed its first law to promote gender equality, namely the Gender Equality Act B.E. 2558, which aims to provide protection to all people from gender discrimination. It was Thailand's first legal instrument to specifically mention transgender people. Further, the Department of Women's Affairs and Family Development recognizes the different challenges faced by people whose gender identity do

⁴⁸ Thailand's Voluntary National Review on the Implementation of the 2030 Agenda for Sustainable Development, June 2018 <http://www.mfa.go.th/sep4sdgs/contents/filemanager/images/sep/VNR%202018%20English%2010.07.18.pdf>

⁴⁹ UN Women: Thailand <http://asiapacific.unwomen.org/en/countries/thailand>

not match their sex assigned at birth.⁵⁰ However, the Act has been criticized by the CEDAW Committee for making exceptions on issues related to religious principles and national security.⁵¹

Thailand's 2017 Voluntary National Review on the 2030 Agenda for Sustainable Development reported that women's status in Thailand has drastically improved in Thailand over the past several decades. The country has achieved gender parity in basic education with women doing slightly better in higher education and the proportion of women in poverty than men overall. While noting that Thailand is recognized as having among the most favourable conditions for women entrepreneurship in Asia and the Pacific,⁵² the Review also recognized the low representation of women in the public sector, in particular in national politics, in government leadership positions and in local administration. The 2017 Constitution calls for gender responsive budgeting, stipulating that the government should take into account the different necessities and needs with respect to gender, age, and conditions of persons to ensure fairness. The 2017 Voluntary National Review also noted that elements of SDG 5 would be included in the Women Development Strategy (2017-2021), which recognizes that Thailand has fallen short on several international goals and measures and the country needs to step up efforts in women's development and promote a more equal society. Thailand considers the development of a gender-disaggregated database a next step in its effort to achieve gender equality. The Government reaffirmed its commitment in 2017 to work with all stakeholders at both national and international levels for the successful promotion of gender equality and sustainable development.⁵³

The weak gender dimension in Thai policy has been observed by the Committee on the Elimination of Discrimination Against Women (CEDAW). In CEDAW's 2017 Concluding Observations on the state of gender equality in Thailand, the Committee expressed a particular concern for the situation of rural Thai women, including indigenous women and those ethnic and religious minorities, remain disproportionately affected by poverty and limited economic opportunities. CEDAW spells out that Thai women, in particular rural women, are also excluded from participation in the elaboration and implementation of policies and action plans on climate change and disaster risk reduction, despite the fact they are disproportionately affected by the effects of climate change and disasters.⁵⁴

Besides lack of clear policy directives on the integration of gender equality and limited participation of women in the formulation and implementation of climate policies and plans, another key challenge in gender and social inclusion in on climate change action is lack of disaggregated data.

⁵⁰ UNDP, MSDHS. May 2018. Legal Gender Recognition in Thailand: A Legal and Policy Review

⁵¹ CEDAW/C/THA/CO/6-7

⁵² According to Mastercard Women's Entrepreneurial Index 2016.

⁵³ Thailand's Voluntary National Review on the Implementation of the 2030 Agenda for Sustainable Development, June 2017, p. 21.

⁵⁴ CEDAW Concluding Observations on state of gender equality – Thailand 2017, p. 11.

V. Gender Analysis and Recommendations

This gender analysis acts as an entry point for addressing gender equality and women's empowerment through: i.) strategic approaches, i.e. with interventions targeted specifically at women and girls and ii.) mainstreamed approaches, i.e. ensuring that all areas of the project promote the transformation of gender inequalities to strengthen climate resilience and provide greater opportunities for all rural people to adopt adaptation options that reduce their vulnerability to the impacts of climate change and strengthen the climate resilience of households and communities in the areas directly and indirectly targeted by the project.

Over 50% of project beneficiaries are expected to be women, please see table below. The project should ensure that the particular climate challenges and needs faced by women in water management and agriculture are fully captured in project implementation. These include disadvantages related to:

- Access to information about climate change and related impacts on water and agriculture, and related ability to disseminate information
- Decision making related water use/management and agriculture
- Access to technology and related training to advance climate resilient agriculture
- Access to finance to invest in climate resilient agriculture

Table 4: Project beneficiaries disaggregated by gender

Beneficiary areas			Number of household and population			
Province	District	Sub-district	Household	Population	Male	Female
Uttaradit	Phichai	Thamafuang	1,286	3,987	1,920	2,067
		Korrum	695	2,155	1,034	1,121
		Phrayaman	1,361	4,219	2,025	2,194
Sukhothai	Srinakorn	Klongmaplup	23	71	34	37
		Nongbuo	105	325	156	169
	Khongkairas	Kokrat	1,320	4,092	1,967	2,125
		Krainok	1,744	5,407	2,595	2,812
		Kraiklang	723	2,241	1,070	1,171
		Krainai	317	983	472	511
		Baanmaisukasem	1,272	3,943	1,899	2,044
		donydoi	44	136	65	71
		Sawankhalok	431	1,336	645	691
Phitsanulok	Promphiram	Nongkham	852	2,641	1,268	1,373
		Wanywoon	2,453	7,604	3,670	3,934
		Sriphrom	1,568	4,861	2,385	2,476
		Thaloktiem	1,260	3,906	1,875	2,031
		Matong	121	375	180	195
		Thachang	485	1,504	722	782
	Bangrakham	Thananyangam	1,834	5,685	2,749	2,936
		Choomsaeng songklam	1,381	4,281	2,085	2,196
		Bangrakham	99	307	147	160
	Muang	Baankrang	626	1,941	932	1,009
Total			20,000	62,000	29,896	32,104

The project is aligned with national gender equality and women's empowerment related policies and programmes as well as international commitments under the Paris Agreement, CEDAW, SDGs, etc. The project will aim to support transformational change towards gender equality and women's empowerment and will:

- Seek to understand in greater depth the potentially gender-differentiated knowledge, experience, and skills of women and men at sub-district, district and provincial levels as well as their needs, challenges, and vulnerabilities (e.g. around access to resources, decision-making, etc.) in order to strengthen project interventions including information sharing between implementers and beneficiaries, decision-making around adaptation at community and household levels, production, post-harvest, and market development decisions, etc. This can be done through introducing/piloting approaches such as gender-responsive participatory Integrated Climate

Services for Agriculture (PICSA) and piloting and scaling up household methodologies (HHM).^{55 56} Several organizations have championed these approaches which have been applied in several in different regions to strengthen agricultural livelihoods, production, and value chain/market development, etc.

- Formulate a working group on gender. The gender focal point of each ministries (national level) will be informed about the project implementation. The project management team, led by the Royal Irrigation Department will invite a member from a provincial representative of the Women Empowerment Fund, Krungsri Bank, BAAC, female head of villages, Tambon Administrative Office, water user groups, FFS in the project provinces, Yom-Nan river basin committee, as well as a co-operatives in the project provinces. The Ministry of Social Development and Human Security (MSDHS) is mandated to promote gender equality and is therefore important to engage on issues related to gender. MSDHS has staff working on issues such as gender responsive budgeting and planning, which would be useful to engage in the project. Equally important will it be to use and try to strengthen the system om gender focal points appointed within each ministry.
- Due to the low level of women in decision-making roles in Thailand, which also applies to water management, efforts to promote and support women in these roles will be done throughout the project. This might also include ensuring that there are enabling environments, including working with men to alter harmful stereotypes.
- Ensure capacity building includes the active participation of women and men and incorporates, where relevant, issues around gender and climate information services, market development, farmer engagement, irrigation, use of innovative technology, etc.
- Given the lower percentage of land owned by women, the project needs to ensure their equal participation by providing an enabling environment for participation. The project also needs to have a broader inclusion of households members, not only focusing on the person who the land is registered on.
- Mitigate any negative impacts on rural women and girls (e.g. in terms of benefit sharing, labour division of labour, access to resources, access to technology and skills development, etc.) Where possible, enhance role of new generation, both female and male youth (aged 18-25) to build a

⁵⁵ For more about PICSA approaches, see, for example <https://ccafs.cgiar.org/participatory-integrated-climate-services-agriculture-picsa#.XOlq5Y9IDIU>

⁵⁶ For more about Household Methodologies (HHM), see, for example <https://www.ifad.org/documents/38714170/40198517/Household+methodologie+-+harnessing+the+family%27s+potential+for+change.pdf/cb0ab278-bfb4-4b4c-a237-e7841bc9e9aa>
For more on Household Methodologies, particularly the Gender Action Learning Systems (GALS) approach, See Mayoux, L. 2017. Gender Action Learning for Sustainability at Scale:: Pilot in Kyrgyzstan, May – Nov 2016 https://gamechangenetwork.org/wp-content/uploads/2016/05/GALS-Process-for-RWEE-Naryn-Pilot_-_ConsolidatedReport_Feb2017_ed.pdf

For more on adapting GALS in agricultural extension, Value Chain (VC) development and Village Savings and Loans Associations (VSLA), see PELUM. 2016. Models of Integrating GALS in Functional Adult Literacy, Agricultural Extension, Advocacy, Village Savings and Lending Associations https://uganda.oxfam.org/sites/uganda.oxfam.org/files/file_attachments/Adapting-GALS-in-development-programmes-FINAL.pdf

new generation of farmers equipped with knowledge and skills to build climate resilience in their community.

- Ensure that the Grievance Redress Mechanism is gender- and age inclusive and responsive, addressing potential barriers to women and marginalized groups. Raise awareness of the grievance mechanism among all impacted by project activities. For the grievance mechanism to be able to adequately respond to complaints raised by women or marginalized groups, staff members will be trained on gender sensitivity. Furthermore, a protocol on prevention of sexual exploitation and abuse will be adopted to ensure that any complaints related to such issues are adequately managed.
- Although an increased risk of GBV is not considered to be high for the planned project activities, strategies will be put in place to monitor and mitigate GBV. The measures will follow CARE's Guidance for Gender Based Violence (GBV) Monitoring and Mitigation within Non-GBV Focused Sectoral Programming.⁵⁷ This includes activities such as capacity building of project staff and implementing partners, development of protocols on GBV, and putting a referral list in place.
- Include a Gender Specialist to ensure that the needs and challenges faced by women are appropriately reflected in interventions. The Gender Specialist will support community consultations, data collection and mainstreaming of gender issues (and where relevant, strategic approaches targeted specifically at women/girls across all areas of the project);
- Link with institutions who can provide training to agricultural extension workers on uptake of household methodologies (e.g. Gender Action Learning Systems or GALS) in their work with women and men in vulnerable rural households and communities to complement, strengthen, and leverage farmer to farmer sharing of climate information, Value Chain (VC) and market access opportunities
- Build/strengthen linkages with institutions/networks that can strengthen gender approaches throughout the project (e.g. academic institute, agriculture and other relevant research centres, etc.). The project should also consider linking to existing women's groups, housewife groups, credit groups.
- Contributions to disaggregated data should be an integral part of any project for future sustainability. Build data collection and management in the process of capacity development at all levels. Technical support with integrated gender and social inclusion for each project/program.
- Ensure that good practices and lessons learned from the project around integrating participatory, gender transformative approaches in climate adaptation are documented and shared among stakeholders including line ministries, policymakers, and government/non-government decision-makers/planners/implementers at national and sub-national level and beyond (e.g.

⁵⁷ CARE (2014), [Guidance for Gender Based Violence \(GBV\) Monitoring and Mitigation within Non-GBV Focused Sectoral Programming](#)

regional/international). This may also contribute to reporting on international commitments such as NDCs, etc.

- Women generally carry out the same activities in agricultural cultivation as men do, and perhaps even more. For water-usage managing groups, the leaders are usually men. However, women in these groups hold roles and responsibilities including vice president, financing and accounting and various administrative positions. It is therefore important to ensure that women are encouraged and supported in leadership positions, and not exclusively to administrative roles.
- Thailand creates a supportive mechanism to promote female business start-up for agricultural business. The mechanism builds on an existing agricultural cooperative and a community enterprise that female and LGBT always active in leading the cooperative management in their hometown. It also creates job for local resident.
- Ensure complementarity with, and learn from best practices and lessons learned from “Gender Bond” - Krungsri Bank has prepared a gender bond. The bank will provide loans for female entrepreneurs. Women in the project location can access this loan via Krungsri Branch Offices nationwide. <https://www.nationthailand.com/business/30377667>
- The project will make a case of women leadership to the existing mechanism such as business start-up, social enterprise, community enterprise, co-operative of rice, fruit, flower, etc.

The gender analysis and action plan should be reviewed and updated regularly. This is particularly relevant in the context of COVID-19, as the impacts on health and livelihoods are not yet fully known or understood.

Proposed Gender Action Plan

Thailand: Enhancing climate resilience in Thailand through effective water management and sustainable agriculture

Activities	Indicators and Targets	Timeline	Responsibilities	
Impact Statement: Improved adaptive capacity and climate resilience (including food and water security and overall wellbeing) of resource dependent agricultural households and communities, including women and girls, in Phitsanulok, Uttaradit, and Sukhothai Provinces. Outcome Statement: Improved women's (consider role of youth of both genders, for future resilience and sustainability) participation and decision-making in, and benefit-sharing from, greater access to climate information services, water management infrastructure and practices, and livelihood strategies and opportunities.				
Output 1: Enhance climate and risk informed planning in the water and agricultural sectors through improved climate information and cross sectoral coordination				
Activity 1.1 Strengthening capacity to generate tailored climate information to inform water management and agriculture planning Development of climate information products informed by gender-sensitive community consultations and gender analysis Technical training for staff from TMD, RID, and relevant agencies identifies and includes active participation of women as well as men. Training includes sensitization of the needs of women of water management and agriculture	Indicator: Climate products include information highlighted as needed by women Target: Gender-informed climate information Indicator: Number of trainees, disaggregated by gender, age, health/disability, income, type of household Target: at least 30% women	By 2023 and regular thereafter	100,000	RID
Activity 1.2	Indicator: Active participation of gender	By 2025 and regular thereafter	50,000	RID/MOAC

Facilitating inter-ministerial coordination for climate-informed and integrated planning Formulate a working group on gender. The gender focal point of each ministries (national level) will be informed about the project implementation. The project management team, led by the Royal Irrigation Department will invite a member from a provincial representative of the Women Empowerment Fund, Krungsri Bank, BAAC, female head of villages, Tambon Administrative Office, water user groups, FFS in the project provinces, Yom-Nan river basin committee, as well as a co-operatives in the project provinces. Integrate a gender focus in inter-ministerial coordination for climate-informed and integrated planning (e.g. gender and agriculture/cc experts/focal points/researchers from relevant government/NGO bodies	focal point/expertise in inter-ministerial stakeholder committees, meetings, conferences, NAP process, etc. Target: Key line ministries/government bodies working on gender equality/women's empowerment. Indicator: Number of inter-agency reports, briefs, submissions including integration of gender issues (need more specifics, e.g., gender-specific, socially-inclusive activities, gender-responsive indicators) Target: 100% of relevant documentation produced by stakeholder committees, etc..		Note: Should not be much added cost (if any) as means including ministries/other stakeholders where relevant who should participate in any case. Also, documentation should not require extra funding. Gender advisor to the project and government gender focal points should ensure inclusion of gender.	ONEP/MoNRE
Activity 1.3 Expanding access to climate information for application at the household level Pilot complementary gender-responsive Participatory Integrated	Indicator: Number of farmers participating in PICSA pilot areas disaggregated by gender (and age if possible)	By 2025	100,000	RID, with MOAC and ARDA ONEP/MoNRE

Climate Services for Agriculture (PICSA) ⁵⁸ approach that involves extension workers working with rural women and men farmers to analyse together climate information and use participatory tools to develop and choose adaptation options best suited to women and men's potentially different circumstances and challenges. (Consider 'moh-din' 'moh-naam' – 'soil doctor', 'water doctor' extension and strengthening of capacity. There should be more specific effort to enhance active role of women and youth in developing knowledge and skills, and access to new technology.)	Target: 50% females (25-30% youth aged 18-25) Indicator: Number of agricultural extension workers (disaggregated by sex) in pilot sites who can apply PICSA approach with farmers. Target: 50% of agricultural extension workers in pilot sites.			
Output 2: Improve water management through strengthened infrastructure complemented by EbA measures, for greater resilience to climate change impacts				
Activity 2.1 Climate-informed engineering designs for the 13 schemes of the Yom-Nan river basin and upgrade of 2 water infrastructure Activity 2.2. Complementing of grey infrastructure with EbA measures and integration of EbA approaches into water management policy and planning	Indicator: Vulnerability and risk assessment reports capture and analyse (where relevant), data disaggregated by sex (and age if possible). Need to disaggregate by gender, age, health/disability, income, type of household from the start.	By 2023	100,000 Note: This budget is to ensure coverage of well-designed collection, analysis, and documentation of gender-disaggregated data in planned vulnerability and risk assessment through inclusion of a gender/social inclusion expert on team (or linkage to research	RID/MoAC GIZ ONEP/MoNRE

⁵⁸ For more information on PICSA, see CGIAR's Climate Change, Agriculture, and Food Security website <https://ccafs.cgiar.org/participatory-integrated-climate-services-agriculture-picsa#.XOhst49IDIV>

Integrate gender responsiveness into vulnerability and risk assessment Participatory assessment, with training of community members, especially youth/students to conduct data collection (with technical support by project/academic/research institution) Participatory assessment results should be shared and validated in the community in workshop format. Promote role of women in water user groups	Target: Number of youth volunteers trained in data collection (field interviews, data collection) Target: 40% participation of women in water user groups		institute/university who can provide guidance on gender.	
Output 3: Reduce volatility of agriculture livelihoods in drought and flood prone areas through strengthened extension support and local planning, investment in on-farm adaptation measures and greater access to finance and markets				
Activity 3.1 Application of climate information in household agriculture planning and strengthening of related support through extension services Training of trainers (ToT) for Lead Extension Workers and cascaded Training of extension workers in CSA-complementary, gender transformative household methodologies (HHM) that strengthen gender equitable intra-household work and decision-making on production, value chains, market access, and benefit-sharing.	Indicator: ToT extension workers supported by project trained on methodology Target: 100%	<i>After start of implementation and thereafter</i>	100,000 Note: This would allow for initial training of “Lead Extension Workers” in each project area/province” who would then cascade the training to other extension workers in project areas for greater coverage. These extension workers would incorporate learnings into their work with communities/households	MoAC

Activity 3.2 Implementation of on-farm climate resilient measures to improve drought and flood resilience, and improved access to finance for sustainable agriculture Integrate household methodology approach (HHM) to complement EbA approach and support agricultural extension efforts to strengthen climate-resilient farm planning including agricultural livelihoods, market access, VC development, etc.	Indicator: Extension workers applying methodology in project areas incorporating HHM approach in agricultural extension for strengthening agricultural livelihoods Target: 100%	<i>By 2025</i>	50,000 Note: This would support gender expertise advising/working with district level adaptation plan team(s) to integrate/institutionalize HHM approaches in agricultural extension to strengthen household/community resilience	MoAC
Activity 3.3 Capacity building for farmers to support market access for climate resilient agriculture products Promote participation of women in training, as agents of change towards climate resilience Make available through training materials that may be of help to women in the area (e.g. financial incentives targeting women and female entrepreneurs, information related to available help related to domestic violence)	Indicator: women participating in training activities; use of mobile application for climate change information for crop planning; adoption of climate resilient practices; access to finance for investment in climate resilient agriculture resilience Target: 50% of beneficiaries are women	<i>By 2025</i>	50,000	MoAC